

Measuring Unfair Inequality: Reconciling Equality of Opportunity and Freedom from Poverty

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Empirical evidence on distributional preferences shows that people do not judge inequality as problematic per se but that they take into account the fairness or unfairness of the outcome. This article conceptualizes a view of unfair inequality and introduces a new measure of inequality based on two widely held fairness principles: equality of opportunity and freedom from poverty. It develops a method for decomposing inequality and its trends into an unfair and a fair component. We provide two empirical applications of our measure. First, we analyse the development of inequality in the US from 1969 to 2014 from a fairness perspective. Second, we conduct a corresponding international comparison between the US and 31 European countries in 2010. Our results document that unfair inequality matches the well-documented inequality growth in the US since 1980. This trend is driven by decreases in social mobility, i.e., increasing importance of parental education and occupation for the income of their children. Among the 32 countries of our international comparison, the land of opportunity ranks among the most unfair societies in 2010.

Key words: Inequality, Equality of opportunity, Poverty, Fairness, Measurement.

JEL Codes: D31, D63, I32

1. INTRODUCTION

Rising income inequality in many countries around the world has led to intense debates—both in academia and in the public. Calls for more redistribution are often countered by pointing out that inequality is (1) necessary to incentivize individuals and (2) may reflect just deserts in a market economy. However, standard measures of inequality are inappropriate to inform the fairness

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debate because they neither correspond to standard principles of distributive justice, nor to the distributional preferences upheld by the larger public. In this article, we propose a new measure of (unfair) inequality based on two widely held normative principles, namely equality of opportunity and freedom from poverty. Bringing this new measure to the data, we provide important insights about the fairness of inequality, both over time (in the US) and across countries (in 2010).

Following the seminal work by [Piketty and Saez \(2003\)](#), the literature has documented a continued increase of income inequality since the beginning of the 1980s in many Western societies.¹ This evidence has strongly influenced public discourse. For example, based on the slogan “*We are the 99%*” the Occupy Wall Street movement has fiercely advocated for more redistribution. To the contrary, free-market pundits emphasize that through trickle-down effects everybody benefits from growth among the job creators at the top. More redistribution would disincentivize those individuals and lead to lower welfare for everybody in the long-run. While the equity-efficiency trade-off dominates public discourse on inequality, an explicit discussion of what we understand by an equitable distribution of income is mostly absent. To the contrary, the implicit assumption in much of public discourse, as well as in the recent economics literature, seems to be that less inequality by necessity implies a more equitable distribution. However, it is highly questionable whether equity is adequately represented by inequality measures that invoke perfect equality as the normative benchmark. For instance, is it really the case that everybody receiving the same income, i.e., a Gini coefficient of zero, represents the most equitable distribution when people exert different levels of effort?

Most theories of distributive justice argue that we should not be concerned by inequality per se, but that we should focus on the sources and structure of inequality. In general, these theories differentiate between fair (justifiable) and unfair (unjustifiable) inequality. Unfair inequality shall be eliminated completely while fair inequalities ought to persist.² For example, according to responsibility-sensitive egalitarian theories of justice, outcome inequalities are unfair if they are rooted in factors beyond individual control. These factors could not have been influenced by individual choice and therefore people should not be held responsible for the (dis)advantages that follow from them.³ In line with this reasoning, individuals are more willing to accept income differences which are due to effort and preferences rather than exogenous circumstances ([Fong, 2001](#); [Cappelen, Hole, Sørensen, and Tungodden, 2007](#); [Alesina and Giuliano, 2011](#); [Alesina, Stantcheva, and Teso, 2018](#)).⁴ Yet, in spite of its wide acceptance, the notion of individual responsibility is insufficient to define fairness (e.g. [Konow, 2003](#); [Konow and Schwettmann, 2016](#)). For example, when an outcome is such that it brings deep deprivation to an individual, questions of how it came about seem secondary to the moral imperative of addressing the extremity of the outcome, be it hunger, homelessness, violence or insecurity ([Bourguignon, Ferreira, and Walton, 2006](#)).⁵ Hence, while outcome differences based

1. Among others: [Atkinson and Piketty \(2007\)](#), [Leigh \(2007\)](#), [Roine and Waldenström \(2015\)](#), [Güvenen and Kaplan \(2017\)](#), [Piketty, Saez, and Zucman \(2018\)](#).

2. In the social choice literature these two intuitions are formally represented by *compensation* and *reward* principles ([Fleurbaey, 2008](#); [Fleurbaey and Maniquet, 2011](#)).

3. This distinction is emphasized in [Rawls \(1971\)](#), [Sen \(1980\)](#), [Dworkin \(1981a\)](#), [Dworkin \(1981b\)](#), [Arneson \(1989\)](#), [Cohen \(1989\)](#), [Roemer \(1993\)](#), [Roemer \(1998\)](#), [Lippert-Rasmussen \(2001\)](#), and [Lippert-Rasmussen \(2011\)](#).

4. Moreover, the literature branches on intergenerational mobility (e.g. [Solon, 1992](#); [Björklund and Jäntti, 1997](#); [Black and Devereux, 2011](#); [Corak, 2013](#); [Chetty, Hendren, Kline, and Saez, 2014a](#)), the gender pay gap (e.g. [Blau and Kahn, 2017](#); [Kleven, Landa, Sogaard, and Egholt, 2018](#)) and also on racial disparities (e.g. [Lang and Lehmann, 2012](#); [Kreisman and Rangel, 2015](#)) are concerned with inequalities that are rooted in one specific factor beyond individual control. The volume of academic research on these topics is a further indication that circumstance-based inequalities are of foremost public interest.

5. To illustrate this point, [Kanbur and Wagstaff \(2016\)](#) suggest the following thought experiment: Imagine yourself serving on a soup line. The indigents move forwards and you hand out hot soup. But in one case a new piece of information

on exogenous circumstances imply violations of fairness, the reverse statement does not hold true. In addition to the responsibility criterion there are many reasons why a given outcome distribution could be considered unfair—one of them being that not everybody has enough to make ends meet.

In this article, we propose the first family of measures for unfair inequality that incorporate the principles of equality of opportunity (EOP) and freedom from poverty (FfP) in a co-equal fashion. In line with the previous discussion, we take seriously the idea that equity is not represented by equality in outcomes, but that it requires life success to be orthogonal to exogenous circumstances (EOP) *and* that everybody should have enough to make ends meet (FfP).

We build on the norm-based approach towards inequality measurement (Cowell, 1985; Magdalou and Nock, 2011). In a first step, we construct a fair income distribution that complies with the principles of EOP and FfP.⁶ In a second step, we measure unfair inequality as the divergence between this norm distribution and the observed income distribution. We show that our proposed measure is easily interpretable and exhibits desirable properties identified in the measurement literature. It furthermore nests standard measures of equality of opportunity and poverty.

Our article makes two main contributions. First, we develop the first measure of unfair inequality that reconciles EOP and FfP in a co-equal fashion. Both EOP and FfP have a vast theoretical and empirical literature. Yet, characterizations of unfairness that have relied on separate application of either principle have been criticized concerning their theoretical scope, as well as their policy implications (Kanbur and Wagstaff, 2016). Moreover, previous attempts to reconcile the two principles are scant and subject to important drawbacks. For example, existing works give priority to either EOP or FfP, while treating the second principle as a mere weighting factor (Brunori, Ferreira, Lugo, and Peragine, 2013). We address these shortcomings by treating EOP and FfP as co-equal principles conveying different grounds for compensation. That is, we develop an inequality measure that detects unfairness emanating from unequal opportunities or poverty even if one of the two guiding principles is satisfied.

Second, we provide two empirical applications of our measure yielding important insights for the inequality debate and the design of appropriate policy responses. First, we analyse the development of inequality in disposable household income in the US over the time period 1969–2014. Our results show that the US trend in unfair inequality has mirrored the marked increase of total inequality since 1980. While total inequality in the US has more than doubled in the time period 1980–2014, so has unfair inequality. The underlying relative share of unfair inequality has increased from 15.2% to 18.9%. This trend is especially driven by increasing inequality across individuals with different socio-economic background characteristics as described by their parental education and occupation. Hence, increasing unfairness in the US results from increased violations of the EOP principle; and decreases in social mobility across generations in particular. Second, we compare inequality in disposable household income between the US and 31 European countries in 2010. In absolute terms, the US have the second highest level of unfair inequality after Greece. Furthermore, we show that unfairness in the US has a remarkably different structure than in European societies with comparable levels of unfairness. While unfair inequality in these European countries is especially driven by violations of the FfP principle in the aftermath of

is given to you. You are told that the outcome of the person in front of you was not due to circumstances but a lack of effort. Would you withdraw your soup holding hand because her outcome is morally justifiable according to the responsibility criterion? If not, clearly some other principle is cutting across the power of the responsibility-sensitive egalitarian argument.

6. Note that standard measures of inequality, such as the Gini index, can also be understood as norm-based measures, in which the norm vector requires perfect equality. The explicit construction of a norm distribution lays bare the normative assumptions that underpin the respective inequality measure.

the 2008 financial crisis, unfairness in the US is predominantly driven by violations of the EOp principle.

We emphasize that these empirical findings are contingent on normative choices and hence open to debate. Our measurement approach provides a blueprint for how to decompose total inequality into its *fair* and *unfair* components based on the principles of EOp and FfP. To implement these measures one must take a stance on the following questions: What are the individual (non-)responsibility characteristics that warrant compensation (EOp)? What is the minimal basic income necessary to make ends meet (FfP)? We do not attempt to settle these questions in this article. Instead, in the spirit of Foster (1998), we provide extensive sensitivity analysis based on (1) different circumstance sets, (2) different treatments of the correlation between individual circumstances and efforts, and (3) different basic income thresholds. We therefore provide a menu of different normative assumptions based on which the reader may draw her own conclusion about the development of unfairness in the US over time and differences in unfairness across countries.

The remainder of this article is organized as follows. In Section 2, we clarify the underlying normative principles of EOp and FfP. In Section 3, we develop our measure of unfair inequality. Section 4 provides two empirical applications describing unfair inequality in the US from 1969 to 2014, as well as an international comparison in 2010. Sensitivity analyses with respect to alternative normative assumptions are provided in Section 5. Lastly, Section 6 concludes.

2. NORMATIVE PRINCIPLES

2.1. Equality of opportunity

Equality of opportunity (EOp) is a popular concept of fairness that is used to evaluate distributions of various outcomes, including health, education and income. Following the seminal contributions by Van de gaer (1993), Fleurbaey (1995), and Roemer (1993, 1998), a vivid theoretical and empirical literature evolved that weaves the idea of personal responsibility into inequality research.⁷ Opportunity egalitarians deem inequalities ethically acceptable to the extent that they are rooted in factors of individual responsibility. To the contrary, they condemn inequalities that follow from factors beyond individual control. Prominent examples of the latter are biological sex, race, or the socio-economic status of parents. If individual responsibility factors were the sole determinants of the outcome distribution, EOp would be realized to its full extent.

To operationalize the opportunity-egalitarian idea, the literature draws on the concepts of *circumstances* and *efforts*. Circumstances are those outcome determinants for which individuals shall not be held responsible whereas efforts belong to the realm of personal responsibility. To the extent that the former rather than the latter are stronger (weaker) determinants of the outcome distribution, a society is considered less (more) fair than otherwise. Measures of EOp are underpinned by two fundamental ideas. First, people should be compensated for unequal circumstances. A prominent formulation of this idea is the principle of *ex ante compensation* which postulates that opportunity sets ought to be equalized across people with different circumstances. The principle is *ex ante* because opportunity sets are evaluated prospectively without regard for the individual level of effort exertion. Second, people should be appropriately rewarded for their efforts. While there are again different formulations of this idea, one prominent version is the principle of *utilitarian reward*. Utilitarian reward states that effort should be rewarded in a way that maximizes the aggregate outcome of people with the same circumstances. It entails that outcome

7. See Roemer and Trannoy (2016), Ferreira and Peragine (2016), and Ramos and Van de gaer (2016) for recent overviews.

differences between individuals with the same circumstances are a matter of indifference. *Ex ante utilitarian* measures of EOp therefore boil down to measures of between-group inequality where groups are defined by their circumstance characteristics.⁸ The precise cut between circumstances and efforts is normatively contentious. For example, some argue in favour of including genetic endowments into the set of circumstances (Lefranc, Pistolesi, and Trannoy, 2009) while others deny that natural endowments provide ground for compensation (Miller, 1996). Similarly, it is widely debated whether the correlation between effort levels and circumstances constitutes a ground for compensation or not. While some argue in favour of holding people responsible for their preferences regardless of how they are formed (Barry, 2005), others allocate such correlation to the circumstances that demand compensation (Roemer, 1998). In our empirical baseline, we draw on commonly accepted circumstance characteristics and allocate the correlation between circumstances and efforts to the unfair determinants of inequality. However, we provide sensitivity analyses for different responsibility cuts in Section 5.

Beyond theoretical reasoning, there is compelling empirical evidence that people indeed disapprove of inequalities that are rooted in factors beyond individual control.⁹ Alesina et al. (2018) use information treatments to show that policy preferences with respect to taxation and spending on opportunity-equalizing policies are robustly correlated with perceptions of social mobility. The lower social mobility within a society, the more people are willing to remedy existing inequalities by policy interventions. Faravelli (2007) demonstrates that perceptions of justice tend to more equal distributions when income differences originate from contextual factors that could not have been influenced by individuals. The works of Cappelen et al. (2007) and Krawczyk (2010) confirm that people uphold the equal-opportunity ideal even if it adversely affects their own material interests.

2.2. Freedom from poverty

Poverty is an important focal point in public debates about the distribution of material resources. In the philosophical literature, the focus on the least advantaged has been defended by reference to sufficientarian conceptions of justice (Frankfurt, 1987) and arguments that consider material deprivation as a violation of the rights we have in virtue of being humans (Fleurbaey, 2007).¹⁰ Akin to the literature on EOp, the normative concern for poverty operates on a principle of compensation: poor people are entitled to be compensated so as to attain the material conditions to live a life of reasonable comfort.

While there is wide-spread appreciation for the multi-dimensionality of poverty (Aaberge and Brandolini, 2015), much of the empirical poverty literature focuses on income. In general, poverty measurement follows a two-step process. First, set a threshold that partitions

8. See Fleurbaey and Peragine (2013) and Ramos and Van de gaer (2016) for a comprehensive discussion of different compensation and reward principles.

9. A more nuanced distinction separates factors beyond individual control into “brute luck,” i.e., lotteries that cannot be avoided, and “option luck,” i.e., lotteries with voluntary participation. Existing literature agrees that people have a strong tendency to compensate brute luck, while there is more heterogeneity in the treatment of option luck (e.g. Cappelen, Konow, Sørensen, and Tungodden, 2013a). Additionally, Møllerstrom, Reme, and Sørensen (2015) provide evidence that compensation for brute luck may be influenced by people’s tendency to expose themselves to option luck. Lefranc et al. (2009) and Lefranc and Trannoy (2017) discuss the treatment of luck within the EOp framework.

10. Some object that freedom from poverty does not belong to the theoretical realm of fairness or even justice although it is morally objectionable. Such moral objections could be raised from a humanitarian or human rights perspective. In this article, we use the term “unfair” in a colloquial sense to indicate that a distribution of some good is unfair if it raises moral objection. This practice is also consistent with a *generic* instead of a *specific* interpretation of justice. See Konow (2001) for a thorough discussion of this distinction.

the population into its deprived and non-deprived factions. All else equal, the more lenient the definition of the poverty line, the larger the group to which compensation is owed. Second, choose a function to aggregate the gaps between observed incomes and the poverty line for those whose income falls below the threshold. In analogy to the cut between circumstances and effort, the appropriate setting of the poverty line is widely debated in the literature (among others [Foster, 1998](#); [Decerf, 2017](#)). In our baseline empirical application, we draw on an internationally comparable absolute poverty line. However, we provide sensitivity analyses for different thresholds in Section 5.

The concern for poverty alleviation is strongly reflected in the distributional preferences of the general public. The evidence summarized in [Konow \(2003\)](#) and [Konow and Schwettmann \(2016\)](#) indicates that fairness preferences are sensitive to individual needs, and reflect a concern for everybody having enough to make ends meet. [Cappelen, Moene, Sørensen, and Tungodden \(2013b\)](#) use an international dictator game to show that transfers increase if the recipient comes from a poorer country. [Fisman, Kuziemko, and Vannutelli \(2020\)](#) show that inequality aversion goes hand in hand with a preference for increasing the incomes of the worst-off in society.

2.3. *Reconciling EOp and FfP*

In this work, we treat EOp and FfP as co-equal principles conveying different grounds for compensation. Our approach is philosophically inspired by the recognition that EOp and FfP are individually insufficient to characterize what a fair distribution of resources requires ([Anderson, 1999](#); [da Vita, 2007](#)). These theoretical insights are bolstered by empirical evidence that distributional preferences are sensitive to (1) *ex ante* inequalities that are determined by exogenous circumstances and (2) *ex post* inequalities that are insensitive to responsibility considerations. For example, [Cappelen et al. \(2013a\)](#) show that people endorse an *ex ante* equal-opportunity ethic, however, they also correct for *ex post* inequalities that are the result of luck. [Andreoni, Aydin, Barton, Bernheim, and Naecker \(2020\)](#) suggest that social preferences are a mix of *ex ante* and *ex post* considerations where the latter gain in importance once the outcome is observed. Consistent with these findings [Gaertner and Schwettmann \(2007\)](#) show that people tend to compensate extreme outcomes irrespective of whether they are the result of individual responsibility factors or not. In [Supplementary Figure S.5](#), we furthermore show survey evidence on public support for different principles of justice in 18 European countries that are part of our empirical application. A consistent pattern emerges: people are not perfect outcome egalitarians. Instead, they most strongly endorse a distribution of income that is sensitive to individual need (FfP) and rewards individual effort but not family background characteristics (EOp).

In spite of this evidence, previous attempts to reconcile the (*ex ante*) EOp principle with the (*ex post*) FfP principle are scant. First, [Brunori et al. \(2013\)](#) propose an “opportunity-sensitive poverty measure” according to which identical incomes below the poverty line receive less weight the more advantageous the circumstances of the poor individual. However, since EOp serves as a mere weighting factor in the evaluation of incomes below the poverty line, their measure does not detect any unfairness in societies that are free from poverty but that are characterized by severe violations of EOp. The measure is therefore informative if one aims to prioritize poor individuals based on the responsibility criterion. However, it does not allow to quantify the overall level of fairness in an income distribution. Second, [Ferreira and Peragine \(2016\)](#) suggest the construction of “opportunity-deprivation profiles” where members of circumstance types are considered opportunity-deprived if their average outcome falls below a pre-specified poverty line. Effectively, they apply standard poverty measures to circumstance types instead of individuals.

As a consequence, this measure is informative for the identification of particularly opportunity-deprived types. However, just as the “opportunity-sensitive poverty measure” it does not allow to quantify the overall level of fairness in an income distribution.

3. MEASURING UNFAIR INEQUALITY

In this section, we describe the construction of unfair inequality measures that treat EOp and FfP as co-equal grounds for compensation.

3.1. Notation

Consider the society $\mathcal{N} = \{1, 2, \dots, N\}$ and an associated vector of non-negative incomes $y = (y_1, y_2, \dots, y_N)$. y corresponds to the observed income distribution. Let us furthermore define a basic income $y_{\min}$. It is a fixed income threshold defining what is required to make ends meet in a given society at a given time. Based on y and $y_{\min}$, we can partition the population into a poor and a non-poor faction:

$$\mathcal{P} = \{i \in \mathcal{N} \mid y_i \leq y_{\min}\}; \quad \mathcal{R} = \mathcal{N} \setminus \mathcal{P}. \quad (3.1)$$

Individual incomes are a function of two sets of factors: first, *circumstances* beyond individual control $\Omega \subseteq \mathbb{R}^C$. Second, individual *efforts* $\Theta \subseteq \mathbb{R}^E$. We define the vector $\omega_i \in \Omega$ as a comprehensive description of the circumstances with which $i \in \mathcal{N}$ is endowed. Analogously, we define the vector $\theta_i \in \Theta$ as a comprehensive description of the efforts that are exerted by $i \in \mathcal{N}$. Based on the realizations of circumstances, we can partition the population into T (S) *circumstance types* (*effort tranches*) that are defined as follows:

$$\mathcal{T}(\omega) = \{i \in \mathcal{N} : \omega_i = \omega\}; \quad \mathcal{S}(\theta) = \{i \in \mathcal{N} : \theta_i = \theta\}. \quad (3.2)$$

For any subgroup $\mathcal{X} \subseteq \mathcal{N}$, we denote by $N_{\mathcal{X}} = \text{card}(\mathcal{X})$ the number of individuals in this subgroup and by $\mu_{\mathcal{X}} = \frac{1}{N_{\mathcal{X}}} \sum_{i \in \mathcal{X}} y_i$ their average income. For ease of notation, we let hereafter $N = N_{\mathcal{N}}$ and $\mu = \mu_{\mathcal{N}}$.

Next to the empirical income distribution y , consider a fair norm distribution $y^* = (y_1^*, y_2^*, \dots, y_N^*)$. It is the normative bliss distribution for which society should strive in absence of incentive constraints and behavioural responses to redistribution. While y is given in the data, y^* must be constructed based on explicit normative principles.¹¹

3.2. Measuring divergence

Endowed with y and y^* one must decide how to aggregate the discrepancies between both vectors into a scalar measure of unfair inequality. Prominent divergence measures include the works by Cowell (1985), Magdalou and Nock (2011), and Almås, Cappelen, Lind, Sørensen, and Tungodden (2011), each of which generalizes standard measures of inequality. While Cowell (1985) and Magdalou and Nock (2011) build on the entropy class of inequality measures, Almås et al. (2011) generalize the Gini index. In contrast to standard measures of inequality, these generalized measures do not decrease (increase) with progressive

11. Standard measures of inequality such as the Gini coefficient or top income shares adhere to the norm of outcome egalitarianism, i.e., the norm distribution corresponds to the perfect equality distribution where each individual is assigned the mean income of the empirical distribution: $y_i^* = \mu$, $\forall i \in \mathcal{N}$.

(regressive) transfers from rich (poor) to poor (rich) but rather with transfers that reduce (increase) the distance between y and y^* . This requirement is equivalent to the standard Pigou-Dalton principle of transfers if and only if $y^* = \mu$. Otherwise, transfers from poor to rich can be desirable if the income of the poor exceeds its norm value, while the income of the rich falls short of it.

In our baseline, we use the measure proposed by [Magdalou and Nock \(2011\)](#) yielding the following aggregator for the divergence between y and y^* .¹²

$$D(y||y^*) = \sum_{i \in \mathcal{N}} [\phi(y_i) - \phi(y_i^*) - (y_i - y_i^*)\phi'(y_i^*)],$$

$$\text{where } \phi(z) = \begin{cases} -\ln z, & \text{if } \alpha = 0, \\ z \ln z, & \text{if } \alpha = 1, \\ \frac{1}{\alpha(\alpha-1)} z^\alpha, & \text{otherwise.} \end{cases} \quad (3.3)$$

α is indicative of different value judgments: the higher α , the more weight is attached to positive divergences of empirical income y_i from its respective norm income y_i^* . The lower α , the more weight is attached to shortfalls from y_i^* . In the baseline, we choose $\alpha = 0$. This choice is guided by the fact that the MN-measure with $\alpha = 0$ nests the mean log deviation (MLD) if we set $y_i^* = \mu$, $\forall i \in \mathcal{N}$. As such we ensure close proximity to the empirical literature on EOp, in which the MLD is prevalent (among others [Ferreira and Gignoux, 2011](#); [Hufe, Peichl, Roemer, and Ungerer, 2017](#)). Furthermore, attaching higher weight to shortfalls from y_i^* is consistent with experimental evidence showing a preference for overcompensating the undeserving instead of failing to compensate the deserving ([Cappelen, Cappelen, and Tungodden, 2018](#)).¹³ Thus, our baseline measure of unfair inequality aggregates divergences between y and y^* as follows:

$$D(y||y^*) = \frac{1}{N} \sum_{i \in \mathcal{N}} \left[\ln \frac{y_i^*}{y_i} - \frac{y_i^* - y_i}{y_i^*} \right].^{14} \quad (3.4)$$

We will now turn to the construction of a norm vector y^* that accords with the principles of EOp and FfP.

3.3. Baseline measure

3.3.1. Norm vector. Let $\mathcal{D} \subseteq \mathbb{R}_+^N$ be the set containing all possible norm distributions y^* . In the following, we will define subsets of eligible distributions $\mathcal{D}^h \in \mathcal{D}$ that are consistent with the normative intuitions of EOp and FfP.

First, we characterize EOp by reference to the principles of *ex ante* compensation and utilitarian reward ([Fleurbaey and Peragine, 2013](#); [Ramos and Van de gaer, 2016](#)). These principles state that the expected income of an individual should not correlate with her circumstance type. Thus, we are infinitely inequality averse with respect to inequalities *between* circumstance types. The ideal

12. We abbreviate this class with “MN” in the following. The MN-family of divergence measures is characterized by the properties of *scale invariance*, the *principle of population*, and *subgroup decomposability*. These properties directly carry over to our measures of unfair inequality. Robustness checks using the measures of [Cowell \(1985\)](#) and [Almås et al. \(2011\)](#) are provided in Section 5.4.

13. Robustness checks using alternative specifications of α are provided in Section 5.4.

14. We can scale the measure by $1/N$ to satisfy the *principle of population* without further adjustment since we will constrain y^* such that $\mu^* = \mu$ ([Magdalou and Nock, 2011](#)).

of an equal-opportunity society is realized if there is equality in average incomes across types. $\mathcal{D}^1$ is the subset of distributions for which this criterion is satisfied:

$$\mathcal{D}^1 = \left\{ y^* \in \mathcal{D} \mid \mu_{T(\omega)}^* = \frac{1}{N_{T(\omega)}} \sum_{i \in T(\omega)} y_i^* = \frac{1}{N} \sum_{j \in \mathcal{N}} y_j = \mu, \forall \omega \in \Omega \right\}. \quad (3.5)$$

Note that $\mathcal{D}^1$ implies $\mu^* = \mu$. By fixing the volume of resources, we let the distribution of resources be the only margin of difference between y and y^* .¹⁵ Furthermore, by invoking $\mathcal{D}^1$ we treat the correlation between Ω and Θ as morally objectionable. This assumption is in line with the normative account of Roemer (1998). However, we provide sensitivity checks to this assumption in Section 5.1.

Second, according to FfP people have a claim for basic income $y_{\min}$ even if their low income follows from factors within their control. Therefore, we want to identify those who are poor due to a lack of effort exertion instead of exogenous circumstances and compensate them such that they are able to make ends meet. We follow extant literature and let effort tranches $\mathcal{S}(\theta)$ be represented by relative incomes within $T(\omega)$. Within their types, we hold individuals fully responsible for their income y_i , i.e., within $T(\omega)$ relative differences in y are proportional to relative differences in θ .¹⁶ Hence, we define a partition according to which people are labelled (non-)poor after considering their counterfactual gains from opportunity equalization while maintaining relative income (effort) differences within types:

$$\mathcal{P}(\omega) = \left\{ i \in T(\omega) \mid y_i \frac{\mu}{\mu_{T(\omega)}} \leq y_{\min} \right\}; \quad \mathcal{R}(\omega) = T(\omega) \setminus \mathcal{P}(\omega), \forall \omega \in \Omega. \quad (3.6)$$

Based on the definition of $\mathcal{P}(\omega)$, we formulate the FfP requirement:

$$\mathcal{D}^2 = \left\{ y^* \in \mathcal{D} \mid y_i^* = y_{\min}, \forall i \in \mathcal{P}(\omega), \forall \omega \in \Omega \right\}. \quad (3.7)$$

The FfP requirement consists of two components: $y_i^* = \frac{1}{N_{\mathcal{P}(\omega)}} \sum_{j \in \mathcal{P}(\omega)} y_j^* = \mu_{\mathcal{P}(\omega)}^*$ and $\mu_{\mathcal{P}(\omega)}^* = y_{\min}$. The first component states infinite inequality aversion with respect to income differences among the poor—they all have an *equal* claim to a certain level of resources. The second component states infinite inequality aversion with respect to the average shortfall of the poor population from the poverty line—they all have an equal claim to nothing less (but also nothing more) than the *basic income* $y_{\min}$.

15. Cappelen and Tungodden (2017) call this restriction the “no-waste-condition.” It is standard in the literature on inequality measurement which abstracts from the efficiency costs to reach a norm distribution. Even in (optimal) policy analysis abstracting from behavioural responses often is a useful benchmark (see the discussion in Bierbrauer, Boyer, and Peichl, 2021). Accounting for efficiency costs, however, could be part of further analysis. Assuming the joint minimization of EOp and FfP to be a goal of public policy, our framework could be integrated into models of fair taxation (Fleurbaey and Maniquet, 2006; Weinzierl, 2014; Ooghe and Peichl, 2015; Saez and Stantcheva, 2016). In such a framework, the planner seeks to realize a specific notion of fairness while taking behavioural responses to taxation into account. See Fleurbaey and Maniquet (2018) for a recent overview.

16. In our baseline, we rely on a relative conception of effort where absolute effort exertion of individuals is evaluated relative to the average behaviour within their circumstance type; i.e., while the propensity to study or to work long hours may vary across circumstance types, one does not hold individuals responsible average differences in these behaviours. An alternative to this identification approach is presented in Roemer (1998) who identifies effort tranches by the quantiles of type-specific income distributions. This identification approach also relies on a relative conception of effort. However, it is stronger since the type-specific distribution of efforts (and not just their average) is effectively treated as a circumstance worthy of compensation. Robustness checks using this alternative specification are provided in Section 5.3. See Appendix B for further theoretical details.

Third, there is no inequality aversion with respect to the share of income that exceeds $y_{\min}$. Therefore, we impose a proportionality requirement.¹⁷ $\mathcal{D}^3$ denotes the subset of distributions that respect relative income differences in excess income above $y_{\min}$ after accounting for opportunity equalization:

$$\mathcal{D}^3 = \left\{ y^* \in \mathcal{D} \mid \frac{y_i^* - y_{\min}}{y_j^* - y_{\min}} = \frac{y_i \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}}{y_j \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}}, \forall i, j \in \mathcal{R}(\omega), \forall \omega \in \Omega \right\}. \quad (3.8)$$

The intersection $\cap_{h=1}^3 \mathcal{D}^h$ characterizes our baseline norm distribution:

Proposition 1. Suppose $\mu > y_{\min}$. Then, the intersection $\cap_{h=1}^3 \mathcal{D}^h$ yields a singleton which defines the norm distribution y^* :

$$y_i^* = \begin{cases} y_{\min}, & \forall i \in \mathcal{P}(\omega), \forall \omega \in \Omega, \\ y_{\min} + \tilde{y}_i \times \delta_{\mathcal{T}(\omega)}, & \forall i \in \mathcal{R}(\omega), \forall \omega \in \Omega, \end{cases} \quad (3.9)$$

where $\tilde{y}_i = y_i \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}$ and $\delta_{\mathcal{T}(\omega)} = \frac{\mu - y_{\min}}{\frac{N_{\mathcal{R}(\omega)}}{N_{\mathcal{T}(\omega)}} \left(\frac{\mu_{\mathcal{R}(\omega)}}{\mu_{\mathcal{T}(\omega)}} \mu - y_{\min} \right)}$.

While the formal proof for Proposition 1 is disclosed in Appendix A, we describe its intuition in the following.¹⁸ Norm incomes of individuals in $\mathcal{P}(\omega)$ are pinned down by basic income $y_{\min}$. This prescription follows from the FfP requirement (3.7): those who are poor due to factors other than exogenous circumstances are owed compensation to make ends meet but nothing more.

Norm incomes of individuals in $\mathcal{R}(\omega)$ are pinned down by basic income $y_{\min}$ plus a type-specific proportional transfer rate $\delta_{\mathcal{T}(\omega)}$ that is applied to $\tilde{y}_i$, i.e., individual income in excess of $y_{\min}$. The focus on $\tilde{y}_i$ follows from the proportionality requirement (3.8): above $y_{\min}$ fair incomes must remain proportional to the counterfactual equal-opportunity distribution.

The type-specific transfer rate $\delta_{\mathcal{T}(\omega)}$ is chosen to ensure the satisfaction of the EOp requirement (3.5) while accounting for income gains of those who are lifted to $y_{\min}$ through the FfP requirement (3.7). To understand its mechanics, let us reformulate $\delta_{\mathcal{T}(\omega)}$ as follows:

$$\delta_{\mathcal{T}(\omega)} = \frac{\mu - y_{\min}}{\mu - \frac{N_{\mathcal{P}(\omega)}}{N_{\mathcal{T}(\omega)}} \frac{\mu_{\mathcal{P}(\omega)}}{\mu_{\mathcal{T}(\omega)}} \mu - \frac{N_{\mathcal{R}(\omega)}}{N_{\mathcal{T}(\omega)}} y_{\min}}. \quad (3.10)$$

Note $\frac{\mu_{\mathcal{P}(\omega)}}{\mu_{\mathcal{T}(\omega)}} = \frac{y_{\min}}{\mu} \implies \delta_{\mathcal{T}(\omega)} = 1$: there is no tax/subsidy on $\tilde{y}_i$ in type $\mathcal{T}(\omega)$ if the ratio of average incomes of the poor and average incomes of the type correspond to the ratio of the target situation where both FfP and EOp are satisfied. Similarly, $\frac{\mu_{\mathcal{P}(\omega)}}{\mu_{\mathcal{T}(\omega)}} < (>) \frac{y_{\min}}{\mu} \implies \delta_{\mathcal{T}(\omega)} < (>) 1$: there is a tax (subsidy) on $\tilde{y}_i$ in type $\mathcal{T}(\omega)$ if the relative average incomes of the poor fall short of (exceed) their

17. Instead of a “flat tax” that preserves relative income differences one could also formulate the measure in terms of a “lump-sum tax” preserving absolute income differences (e.g. Bossert and Fleurbaey, 1995). However, such an absolute version can only be calculated under stronger feasibility conditions that are rarely satisfied in our empirical application. In cases where it is satisfied, the results are very close to our baseline measure. As a consequence, we forego a detailed analysis of such an absolute norm in this article.

18. We derive norm distributions from axioms that describe normative principles. Alternatively, one may understand the derivation as an optimization under constraints. For example, we could minimize inequality of opportunity ($\mathcal{D}^1$) while considering freedom from poverty ($\mathcal{D}^2$) and the proportionality requirement ($\mathcal{D}^3$) as optimization constraints. Such alternative derivation, however, would not change the substantial result. See also Appendix A for further detail.

relative average income in the target situation. Hence, if $\mathcal{P}(\omega)$ receive too little of the type-specific resources, $\mathcal{R}(\omega)$ are urged to compensate and vice versa.

The fair income distribution y^* is a function of simple summary statistics of the empirical income distribution y . Some may argue that y^* should be independent of y . First, we note that the underlying principles that inform the construction of y^* are always independent of y . Second, we note that the dependence of y^* on y in the implementation of these principles is not particular to our measurement approach. To the contrary, such dependence characterizes many standard measures of inequality, poverty and inequality of opportunity.¹⁹ In fact, whether and to what extent an insulation of y^* from y is desirable, depends on the normative intuitions one strives to capture. For example, $y_{\min}$ can be set in absolute terms, or in relative terms as some functional of y . The former is preferable if one thinks that the poverty concept applies to basic human needs. The latter is preferable if one aims to capture aspects of social deprivation as well (Foster, 1998). Within our measurement approach, the extent of such dependence can be strengthened or loosened in several ways. First, in the baseline analysis, we choose an absolute poverty threshold and therefore insulate $y_{\min}$ from changes in y . However, we provide sensitivity analyses based on relative poverty thresholds in Section 5.2.²⁰ Second, in the baseline analysis y^* is dependent on changes in the intra-type variance of incomes. Such dependence follows from the proportionality requirement (3.8) that proposes to honour relative income differences within types by interpreting them as indicators of differential effort exertion. However, in Section 3.4, we introduce an alternative norm distribution that insulates y^* from such dependence by harmonizing intra-type variances across circumstance types.

3.3.2. Measure and comparative statics. Substituting the norm distribution given in (3.9) into the divergence measure given in (3.4), we obtain our baseline measure of unfair inequality:

$$D(y||y^*) = \frac{1}{N} \sum_{i \in \mathcal{P}(\omega)} \left\{ \ln \frac{y_{\min}}{y_i} - \left(\frac{y_{\min} - y_i}{y_{\min}} \right) \right\} + \frac{1}{N} \sum_{i \in \mathcal{R}(\omega)} \left\{ \ln \left(\frac{y_{\min} + \tilde{y}_i \delta_{\mathcal{T}(\omega)}}{y_i} \right) - \left(\frac{y_{\min} + \tilde{y}_i \delta_{\mathcal{T}(\omega)} - y_i}{y_{\min} + \tilde{y}_i \delta_{\mathcal{T}(\omega)}} \right) \right\}, \quad (3.11)$$

where $\delta_{\mathcal{T}(\omega)}$ represents the type-specific scaling factor that is applied to $\tilde{y}_i$ —the share of counterfactual income above $y_{\min}$. To further illustrate the properties of this measure, we provide comparative statics in the following.

Assume $y_{\min} \rightarrow 0$. The limiting case of $y_{\min} = 0$ is equivalent to abstracting from the concern for FfP altogether. EOp remains the only normative foundation for inequality aversion. At the limit, $\mathcal{P}(\omega) = \emptyset$, $\mu_{\mathcal{R}(\omega)} = \mu_{\mathcal{T}(\omega)}$, and $N_{\mathcal{R}(\omega)} = N_{\mathcal{T}(\omega)}$. As a consequence, $\delta_{\mathcal{T}(\omega)} = 1$, $\forall \omega \in \Omega$. The resulting norm vector and the measure of unfair inequality read as follows:

$$y_i^* = y_i \frac{\mu}{\mu_{\mathcal{T}(\omega)}}, \quad \forall i \in \mathcal{T}(\omega), \quad \forall \omega \in \Omega, \quad (3.12)$$

19. For example, the standard approach to inequality measurement can be characterized as finding a suitable distance measure between y and a norm vector where every individual has the mean of the distribution, i.e., $y_i^* = \mu$, $\forall i \in \mathcal{N}$. The properties of the distance measure can be further specified (e.g. the Pigou-Dalton property, scale independence, decomposability, etc.). But as the empirical vector changes, the norm vector changes as well.

20. Such poverty lines are relative in that they appeal to the concept of relative deprivation by calculating $y_{\min}$ as a function of the observed distribution y . Once these relative thresholds are determined we hold them fixed for our calculations—i.e., they are not relative to any counterfactual distribution in which either EOp or FfP is satisfied.

$$D(y||y^*) = \frac{1}{N} \sum_{i \in \mathcal{N}} \ln \frac{\mu}{\mu_{\mathcal{T}(\omega)}}. \quad (3.13)$$

With $y_{\min} = 0$, unfair inequality collapses to inequality in the distribution of average outcomes among circumstance types. Hence, as $y_{\min} \rightarrow 0$, the measure converges to the standard *ex ante* utilitarian measure of inequality of opportunity in which the MLD is applied to a smoothed distribution of type-specific mean incomes.

Assume $N_{\mathcal{P}(\omega)} \rightarrow 0$, $\forall \omega \in \Omega$. Note the difference to our previous thought experiment where we abstracted from the concern for FfP altogether. The limiting case of $N_{\mathcal{P}(\omega)} = 0$ corresponds to a society that values FfP below $y_{\min}$ but happens to be in the fortunate position of having zero poverty incidence once incomes are corrected for unequal opportunities. At the limit, $\mathcal{P}(\omega) = \emptyset$, $\mu_{\mathcal{R}(\omega)} = \mu_{\mathcal{T}(\omega)}$, and $N_{\mathcal{R}(\omega)} = N_{\mathcal{T}(\omega)}$. As a consequence, $\delta_{\mathcal{T}(\omega)} = 1$, $\forall \omega \in \Omega$. The resulting norm vector and the measure of unfair inequality read as follows:

$$y_i^* = y_i \frac{\mu}{\mu_{\mathcal{T}(\omega)}}, \quad \forall i \in \mathcal{T}(\omega), \quad \forall \omega \in \Omega, \quad (3.14)$$

$$D(y||y^*) = \frac{1}{N} \sum_{i \in \mathcal{N}} \ln \frac{\mu}{\mu_{\mathcal{T}(\omega)}}. \quad (3.15)$$

With $N_{\mathcal{P}(\omega)} = 0$, $\forall \omega \in \Omega$, opportunity equalization is sufficient to satisfy the criteria of both EOP and FfP. Hence, as $N_{\mathcal{P}(\omega)} \rightarrow 0$, $\forall \omega \in \Omega$, the measure of unfair inequality again converges to the standard *ex ante* utilitarian measure of inequality of opportunity. The limiting case of $N_{\mathcal{P}(\omega)} = 0$, $\forall \omega \in \Omega$ thus illustrates that our measure continues to detect unfairness through violations of EOP even if FfP is perfectly satisfied.

Assume we reduce the set of non-responsibility characteristics that constitute unfair outcome determinants from an opportunity-egalitarian perspective. This reduction can be represented by letting the number of circumstance types travel to one, i.e. $T \rightarrow 1$. At the limit, the entire population would be considered a single circumstance type. FfP remains the only normative foundation for inequality aversion. $T = 1$ leads to $\mathcal{T}(\omega) = \mathcal{N}$, $\mathcal{P}(\omega) = \mathcal{P}$, and $\mathcal{R}(\omega) = \mathcal{R}$. Furthermore, $N_{\mathcal{P}(\omega)} = N_{\mathcal{P}}$, $\mu_{\mathcal{T}(\omega)} = \mu$, and $\mu_{\mathcal{P}(\omega)} = \mu_{\mathcal{P}}$. As a consequence, $\tilde{y}_i = y_i - y_{\min}$ and $\delta_{\mathcal{T}(\omega)} = \delta = \left(1 - \frac{PG}{RG}\right)$, where PG (RG) denotes the poverty (richness) gap.²¹ The resulting norm vector and the measure of unfair inequality read as follows:

$$y_i^* = \begin{cases} y_{\min}, & \forall i \in \mathcal{P}, \\ y_{\min} + (y_i - y_{\min}) \left(1 - \frac{PG}{RG}\right), & \forall i \in \mathcal{R}, \end{cases} \quad (3.16)$$

$$D(y||y^*) = \underbrace{\frac{1}{N} \sum_{i \in \mathcal{P}} \ln \left(\frac{y_{\min}}{y_i} \right)}_{\text{Watts Index}} - \underbrace{\frac{1}{N} \sum_{i \in \mathcal{P}} \left(\frac{y_{\min} - y_i}{y_{\min}} \right)}_{\text{Poverty Gap Ratio}} + \frac{1}{N} \sum_{i \in \mathcal{R}} \left\{ \ln \left(\frac{y_{\min} + (y_i - y_{\min})\delta}{y_i} \right) - \left(\frac{(y_i - y_{\min})(\delta - 1)}{y_{\min} + (y_i - y_{\min})\delta} \right) \right\}. \quad (3.17)$$

21. Following existing literature we define the poverty (richness) gap as $PG = N_{\mathcal{P}}/N(y_{\min} - \mu_{\mathcal{P}})$ and $RG = N_{\mathcal{R}}/N(\mu_{\mathcal{R}} - y_{\min})$, respectively (Peichl, Schaefer, and Scheicher, 2010).

$T = 1$ leads to a scaling factor δ that is uniform across all $i \in \mathcal{R}$. δ is determined by the ratio of the poverty gap and the total volume of excess income above $y_{\min}$. The decomposability property of the MN-measures allows us evaluate unfairness in the truncated distribution $y_{\mathcal{P}} = (y_1, y_2, \dots, y_{\min})$. Up to $y_{\min}$, unfair inequality is characterized by the difference between the Watts index (Zheng, 1993) and the poverty gap ratio. Individually, these are well-known measures of poverty. However, also their combination bears a number of desirable properties that have been identified in the literature on poverty measurement (e.g. Ravallion and Chen, 2003). These include *monotonicity* (as opposed to the headcount ratio), the *principle of transfers* (as opposed to the poverty gap taken as a stand-alone measure) and *additive decomposability*. Note that we do not obtain a measure of poverty that satisfies the *focus axiom*. Our approach frames poverty as an aspect of inequality and thus imposes requirements on how the funds to eradicate poverty should be raised—see the proportionality condition (3.8). Therefore, it is not indifferent to transfers between individuals with incomes above $y_{\min}$.

Assume $\mu_{\mathcal{T}(\omega)} \rightarrow \mu, \forall \omega \in \Omega$. Note the difference to our previous thought experiment where we abstracted from the concern for EOp altogether. In contrast to the previous case, the normative concern for EOp remains intact, however, the EOp principle is increasingly satisfied as $\mu_{\mathcal{T}(\omega)} \rightarrow \mu, \forall \omega \in \Omega$. The limiting case corresponds to an equal-opportunity society without disparities in average outcomes across circumstance types. At the limit, $\tilde{y}_i = y_i - y_{\min}$, $\delta_{\mathcal{T}(\omega)} = \left(1 - \frac{PG(\omega)}{RG(\omega)}\right)$, where $PG(\omega)$ ($RG(\omega)$) denotes the type-specific poverty (richness) gap. The resulting norm vector and the measure of unfair inequality read as follows:

$$y_i^* = \begin{cases} y_{\min}, & \forall i \in \mathcal{P}, \forall \omega \in \Omega, \\ y_{\min} + (y_i - y_{\min}) \left(1 - \frac{PG(\omega)}{RG(\omega)}\right), & \forall i \in \mathcal{R}, \forall \omega \in \Omega, \end{cases} \quad (3.18)$$

$$\begin{aligned} D(y||y^*) &= \underbrace{\frac{1}{N} \sum_{i \in \mathcal{P}} \ln \left(\frac{y_{\min}}{y_i} \right)}_{\text{Watts Index}} - \underbrace{\frac{1}{N} \sum_{i \in \mathcal{P}} \left(\frac{y_{\min} - y_i}{y_{\min}} \right)}_{\text{Poverty Gap Ratio}} \\ &+ \frac{1}{N} \sum_{i \in \mathcal{R}} \left\{ \ln \left(\frac{y_{\min} + (y_i - y_{\min}) \delta_{\mathcal{T}(\omega)}}{y_i} \right) - \left(\frac{(y_i - y_{\min}) (\delta_{\mathcal{T}(\omega)} - 1)}{y_{\min} + (y_i - y_{\min}) \delta_{\mathcal{T}(\omega)}} \right) \right\}. \end{aligned} \quad (3.19)$$

With $\mu_{\mathcal{T}(\omega)} = \mu, \forall \omega \in \Omega$, we calculate poverty-eradicating transfers across types by reference to the *type-specific* poverty gap and the *type-specific* income share that exceeds $y_{\min}$. The limiting case shows that our measure continues to detect unfairness through violations of FfP even if EOp is perfectly satisfied.

The previous comparative statics illustrate the advantages of our measure of unfair inequality. First, it is easily interpretable since it nests well-known measures of both EOp and FfP. If we abstract from the concern for FfP ($y_{\min} = 0$), we obtain a standard measure for inequality of opportunity. If we abstract from the concern for EOp ($T = 1$), we obtain a combination of the Watts index and the poverty gap ratio, both of which are well-established measures of poverty.

Second, the proposed measure treats EOp and FfP as co-equal principles and therefore detects unfair inequality even if either of the two principles is perfectly satisfied. If there is zero poverty incidence ($N_{\mathcal{P}(\omega)} = 0, \forall \omega \in \Omega$), it still detects unfair inequality based on average outcome differences across circumstance types. If the income distribution is perfectly opportunity-egalitarian ($\mu_{\mathcal{T}(\omega)} = \mu, \forall \omega \in \Omega$), it still requires type-specific transfers from rich to poor in order to assure the satisfaction of both FfP and EOp.

3.4. *Alternative conceptualizations*

There are different ways of conceptualizing EOp (Roemer and Trannoy, 2016). In this section, we briefly discuss two alternations to the EOp concept.²²

First, the baseline norm applies a criterion of *weak equality of opportunity*. It only requires the expectation of outcomes to be identically distributed across circumstance types (Lefranc *et al.*, 2009). To the contrary, *strong equality of opportunity* would require equality of outcomes conditional on exerting similar levels of effort.

Second, the baseline norm treats EOp and FfP as *non-separable* in their scope of application. For example, it evaluates type-specific opportunity sets by reference to $\mu_{\mathcal{T}(\omega)}$ —the average incomes of all $i \in \mathcal{T}(\omega)$. To the contrary, under a *separability assumption*, EOp and FfP would operate on disjunct supports of the income distribution y . While FfP characterizes the normative requirement for $\mathcal{P}$, the distributional ideal of EOp only applies to $\mathcal{R}$.

In Appendix B, we derive alternative versions of our measure that are based on *strong equality of opportunity*, and *separability*, respectively. The empirical implications of these alternations are discussed in Section 5.3.

4. EMPIRICAL APPLICATION

To illustrate the proposed measure of unfair inequality, we provide two empirical applications. First, we use the Panel Study of Income Dynamics (PSID) to analyse the development of unfair inequality in the US over the time period 1969–2014. Second, we combine the PSID and the EU Statistics on Income and Living Conditions (EU-SILC) to conduct a cross-sectional analysis in which we benchmark unfair inequality in the US against unfair inequality in 31 European countries in 2010.²³

We re-iterate that the implementation of our measurement approach affords normative assumptions that are open to debate. In our baseline analysis, we choose circumstances Ω and basic income $y_{\min}$ based on standard choices in the empirical literature on equality of opportunity and poverty. These choices may not find unanimous support. Therefore, we interpret the empirical results presented in this section as one plausible description of unfair inequality in the US and Europe. Differences based on alternating normative assumptions are presented in Section 5.

4.1. *Unfair inequality in the US over time*

4.1.1. Data source. The PSID is a main resource for the study of inequality, poverty and intergenerational transmission processes in the US (see Johnson, McGonagle, Freedman, and Sastry, 2018; Smeeding, 2018, and the overview articles in the same issue). At its inception in 1968, the PSID consisted of a nationally representative sample of 2,930 families and an oversample of 1,872 low-income families that are tracked until the present day. All individuals who leave their original households automatically

22. In addition, we provide the following extensions: In [Supplementary material F.1](#), we illustrate how additional inequality aversion may be introduced into our framework. In [Supplementary material F.2](#), we illustrate how heterogeneity in individual needs may be integrated based on individual-specific deprivation thresholds.

23. Note that much of the recent literature on inequality trends draws on administrative data sources (Burkhauser *et al.*, 2012). However, in the context of this study survey data such as the PSID or EU-SILC provide important advantages since the operationalization of EOp and FfP requires detailed information on individual background characteristics and an accurate representation of the lower tail of the income distribution. Administrative data are restricted in both dimensions since tax returns collect only basic demographic information and because the bottom half of the distribution pays little personal income tax.

become independent units in the PSID sampling frame. To match compositional changes of the US population through post-1968 immigration flows, the PSID added a Latino sample and an immigrant sample in its 1990 and 1997 waves, respectively.²⁴ Starting in 1997, it has switched from an annual to a biennial survey rhythm. In its most recent waves, the PSID covers the members of more than 9,000 families and provides rich information on their incomes, family background characteristics and living practices.

In this study we focus on individuals aged 25–60 over the survey (income reference) periods 1970–2015 (1969–2014).²⁵ We will now briefly outline the construction of the inputs to our inequality measure: y , Ω , Θ , and $y_{\min}$. Further detail on the construction of all relevant variables, as well as descriptive statistics are disclosed in [Supplementary materials B](#) and [D](#).

4.1.2. Outcome variable. To assess the distribution of economic resources from a fairness perspective, we use the income components created by the PSID Cross-National Equivalence File (CNEF) to define annual disposable household income as the sum of total household income from labour, asset flows, windfall gains, private transfers, public transfers, private retirement income, and social security pensions.²⁶ We deduct total household taxes as calculated by the NBER TAXSIM calculator ([Butrica and Burkhauser, 1997](#)).

Our measure of unfairness puts a strong emphasis on the lower end of the income distribution. It is well-known that poverty estimates based on survey data tend to be upward biased due to the under-reporting of government benefit receipts ([Meyer and Mok, 2019](#); [Mittag, 2019](#)). Furthermore, it has been shown that households with extremely low reported incomes tend to misreport their income from earnings ([Brewer, Etheridge, and O'Dea, 2017](#); [Meyer, Wu, Mooers, and Medalia, 2021](#)). To mitigate distortions from benefit under-reporting, we use the time series provided in [Meyer, Mok, and Sullivan \(2015\)](#) to scale reported public transfers by a year-specific under-reporting factor that is calculated based on a comparison between the aggregate level of benefits receipts reported in the PSID and the aggregate expenditure levels from administrative program data. To cushion distortions from the under-reporting of labour incomes we identify individuals that report zero earnings but non-zero working hours. We replace their reported earnings level by a prediction from a Mincer wage regression and adjust household labour income by the sum of these correction values over all household members. In total only about 1% of our person-year observations are affected by this imputation procedure.

To account for differences in need and standard of living by household composition, we scale all household incomes by the modified OECD equivalence scale.²⁷ For the sake of inter-period and between-country comparisons, we deflate all income figures with the purchasing power

24. We exclude the Latino sample from our investigation as it was dropped in 1995 and did not reflect the full range of post-1968 immigrants.

25. We employ cross-sectional sample weights for all calculations. However, one may worry that infrequent PSID updates for compositional changes in the US population distort comparisons over time. To address such concerns, we calculate population weights for 48 age-sex-race-cells ($8 \times 2 \times 3$) in the Annual Social and Economic Supplement of the Current Population Survey (CPS-ASEC) and rescale the provided PSID individual weights to match their CPS-ASEC counterparts. This rescaling has a negligible effect on our results suggesting that the standard PSID weights do a good job in representing the underlying US population.

26. In [Supplementary Table S.1](#), we provide a detailed breakdown of these income aggregates into their single components. We note that public transfers mostly include non-medical cash assistance. Of the most important in-kind transfers it includes food stamps but not housing subsidies.

27. The modified OECD equivalence scale assigns a value of 1 to the household head, of 0.5 to each additional adult member, and of 0.3 to each child below age 14. Throughout the article, we deflate observed incomes by this household-specific factor in order to acknowledge differences in needs across households of different size and age structure while accounting for economies of scale in consumption.

parity (PPP) adjustment factors for household consumption provided by the Penn World Tables (Feenstra *et al.*, 2015). Lastly, we curb the influence of outliers by winsorizing at the 1st and the 99.5th-percentile of the year-specific income distribution.

4.1.3. Circumstance types and effort tranches. In an equal-opportunity society, there are no differences in outcomes across individuals with different circumstance characteristics but comparable levels of effort. Our measure of unfairness therefore requires to partition the population into circumstance types. Thereby a tension arises. On the one hand, the more parsimonious the type partition, the more we underestimate the influence of individual circumstances on life outcomes (Ferreira and Gignoux, 2011). On the other hand, limited degrees of freedom suggest restrictions on the granularity of the type partition to avoid noisy estimates of the relevant type parameters. In this work, we use four circumstance variables to partition the population into a maximum of 36 circumstance types. First, we include the biological sex of the respondent. Second, we include a binary indicator differentiating among non-Hispanic white individuals and the remaining population. Third, we construct a categorical variable based on whether the highest educated parent (1) dropped out of secondary education, (2) attained a secondary school degree, or (3) acquired at least some tertiary education. Lastly, we proxy the occupational status of parents by grouping them in (1) elementary occupations, (2) semi-skilled occupations, or (3) skilled occupations. These are standard circumstances used in the empirical literature on inequality of opportunity. However, we present sensitivity analyses based on alternative type partitions in Section 5.1.

Replacing our baseline notion of weak EOp with strong EOp additionally requires the identification of effort tranches. To this end, we further partition each type-specific income distribution into 20 quantiles and replace individual incomes with the within-type average of their respective effort tranche. Hence, for each year, we perform our calculations on a maximum population of 36×20 cells, where each cell represents a particular circumstance-effort combination. In Supplementary Figure S.3, we show that this standardization of income distributions has a negligible impact on conventional inequality and poverty measures in the time period of interest.

4.1.4. Basic income threshold. The specification of poverty thresholds that allow for meaningful comparisons over time and across countries is a topic of widespread academic debate. For example, the official US poverty line is based on expenditure data from the 1950s that reflects three times the cost of a well-balanced diet. Since then it has been updated only by inflation adjustments without taking account of potential changes in the needs of different family types (Meyer and Sullivan, 2012). The international poverty line of the World Bank is currently set at \$1.90 per capita and day in PPP-adjusted dollars. In view of its low value, it is criticized for being irrelevant in countries outside of the developing world (Allen, 2017). Lastly, both EU and OECD define relative poverty lines as a fraction of median equivalized disposable household income. Poverty measurement based on relative lines, however, takes us beyond the focus on basic needs, since such lines may react to changes in the upper percentiles of the distribution irrespective of income changes for those in need (Foster, 1998).

For our baseline estimates we rely on a revised set of international absolute poverty lines as calculated by Jolliffe and Prydz (2016) in a two-step procedure. First, they match official national poverty headcounts to the PovcalNet expenditure data of the World Bank and calculate the implied poverty thresholds. Second, they group the resulting range of national poverty lines according to indicators of economic development and take the group median as an internationally comparable poverty line for the respective class of countries. Their procedure recovers the \$1.90 line for

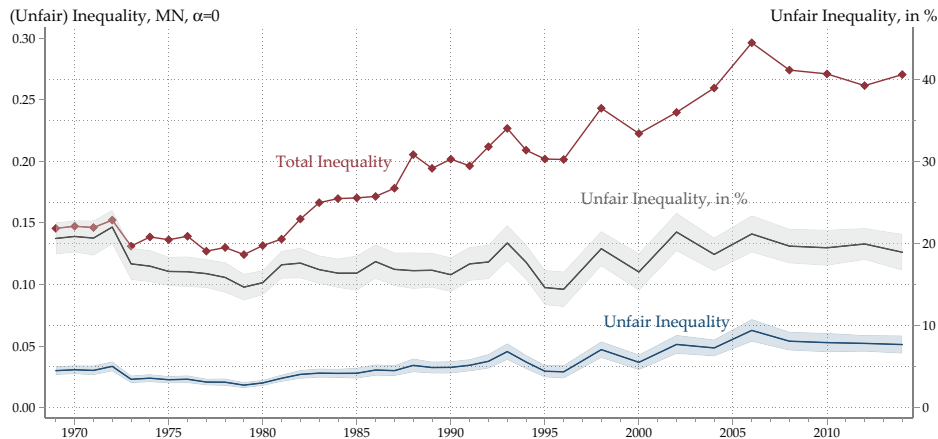


FIGURE 1
Unfair inequality in the US, 1969–2014
Baseline results

Data: PSID.

Notes: Own calculations. This figure displays the development of (unfair) inequality in the US over the period 1969–2014. (Unfair) inequality is calculated based on the divergence measure proposed by Magdalou and Nock (2011) with $\alpha = 0$ (MN, $\alpha = 0$) which corresponds to the MLD for total inequality. Relative measures of unfair inequality are expressed in percent (in %) of total inequality. The shaded areas show bootstrapped 95% confidence intervals based on 500 draws.

the least developed economies but yields more relevant poverty thresholds for economically advanced countries. In our baseline estimate, we take their set of national poverty lines and group countries in quintiles of PPP-adjusted household final consumption expenditure per capita. For single households in the US, this procedure yields a PPP-adjusted poverty line of \$12,874 annually that we hold constant (in real terms) over the period of our analysis. Sensitivity analyses based on alternative poverty thresholds are presented in Section 5.2.

4.1.5. Baseline results. Figure 1 displays the development of (unfair) inequality in the US over the time period 1969–2014. The upper line shows the development of total inequality as measured by the divergence of the empirical income distribution from a perfectly outcome egalitarian distribution in which $y^* = \mu$, $\forall i \in \mathcal{N}$. The time series replicates the well-documented pattern of inequality development in the US (among others Heathcote, Perri, and Violante, 2010a; Burkhauser et al., 2012; Piketty et al., 2018): Slight inequality decreases throughout the 1970s are followed by strong inequality increases in the 1980s. This trend continues until the present day, most notably interrupted by the economic crises following the burst of the dot-com bubble at the turn of the century and the global financial crisis in the late 2000s.

The lower blue line displays the development of unfair inequality as measured by the divergence of the empirical income distribution from a norm distribution in which the ideals of EOp and FfP are realized to their full extent (see equation 3.11). Unfair inequality remains at a lower level than total inequality as the latter provides an upper bound for the former in any given country at any given point in time. However, it is noteworthy that unfair inequality follows a similar time trend as total inequality. Starting with decreases of unfair inequality until 1980, we observe a steady increase of unfairness until the present day and downward movements that are largely coincidental with economic downturns.

The intermediate black line shows the share of total inequality that is in violation of EOp and FfP. It is calculated as the ratio between unfair inequality and total inequality and converted into percentage terms. Starting from a level of 20.6% in 1969, unfair inequality drops to a share of

15.2% in 1980. This development suggests that the observed decreases in inequality over the 1970s were accompanied by an even stronger reduction of unfair inequality. In spite of 50% inequality growth in the 1980s, the share of inequality attributable to violations of EO_p and FfP remained roughly stable at this level until 1990. While the subsequent two decades are characterized by a more erratic pattern, unfair inequality follows a slightly steeper growth curve than total inequality. Starting at a level 16.2% in 1990, the unfair share of inequality climbs to 21.2% in 2006 and stalls at a level of 18.9% in the latest period of observation. Some may be surprised by the low relative share of unfair inequality. However, we emphasize that our measures are based on disposable household income. Therefore, they evaluate remaining unfairness after taking transfers through existing welfare state institutions and redistribution within households into account.²⁸

4.1.6. Decomposition. To develop a better understanding for the observed inequality trends, we conduct a Shapley value decomposition to identify the contributions of different components that underpin our normative principles. That is, we quantify the contributions of FfP and EO_p, respectively. Furthermore, we decompose the latter into the contributions from the circumstance characteristics biological sex, race, parental education, and parental occupation. This decomposition furthermore allows us to embed our measure of unfairness into the larger literature branches on US trends in poverty, gender income gaps, racial disparities, and social mobility.

The Shapley value procedure quantifies the contribution of each of the aforementioned factors by calculating the average marginal decline in unfair inequality once we eliminate it from our calculations. For example, one could quantify the marginal impact of FfP on unfair inequality by decreasing $y_{\min}$ from our baseline threshold of \$12,874 to \$0. Analogously, one could quantify the marginal impact of biological sex by excluding it from the list of variables that define our type partition. However, in both steps the estimate of the marginal impact depends on the specification of the remaining normative criteria. To avoid such path-dependencies, we estimate the individual contribution of each factor by averaging their marginal impacts on unfair inequality across all possible elimination sequences (Shorrocks, 2012). The results of this decomposition are shown in Figure 2.

In 1969, approximately half of unfair inequality, that is 9.7% of total inequality, was associated with violations of the FfP principle. The previously described attenuation of relative unfairness in the 1970s is almost fully explained by decreased violations of the FfP principle. While EO_p shows only a slightly decreasing trend over the 1970s, the contribution of FfP to total inequality is halved, dropping from 0.013 points (9.0%) in 1970 to 0.006 points (5.0%) in 1979. Following the decrease of the 1970s, the contribution of FfP bounces back to its initial levels in the 1980s and subsequently follows a flat time trend that persists until the present day.²⁹ In 2014, violations of FfP contribute 0.014 points to our measure of unfairness and explain 5.1% of total inequality.

At first glance, our results on poverty are in line with official statistics that also show a flat time trend in poverty rates across the period of investigation (U.S. Bureau of the Census, 2019). However, the official poverty concept in the US differs from ours in important aspects such that this analogy only holds superficially. Official poverty statistics rely on the poverty headcount ratio applied to an annually adjusted poverty line that is based on the pre-government income

28. Moreover, it is well understood in the empirical literature that standard estimates of inequality of opportunity provide only lower bounds of their true value (Ferreira and Gignoux, 2011; Hufe *et al.*, 2017).

29. While the absolute contribution of FfP is rather stable between 1969 and 2014, its relative contribution is halved from 9.7% to 5.1%. This decrease in the relative contribution follows mechanically from the increase in total inequality. For further illustration, see also Supplementary Figure S.6 in which we fit locally smoothed time trends for the relative contributions of both EO_p and FfP.

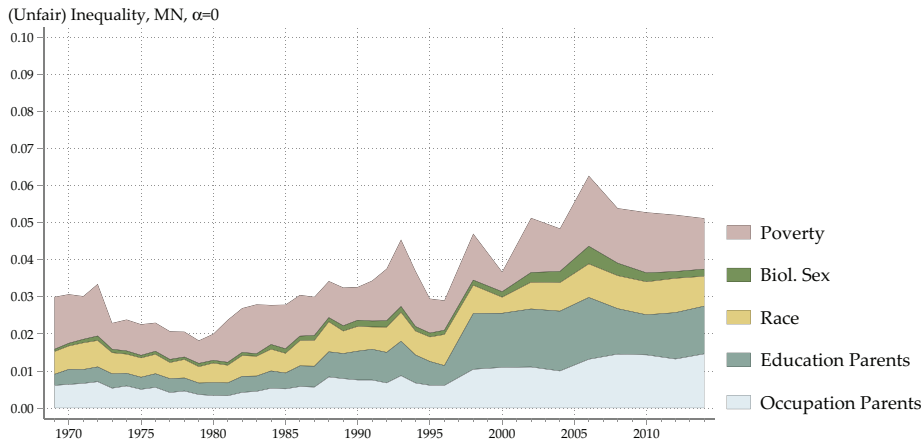


FIGURE 2
Unfair inequality in the US, 1969–2014
Decomposition

Data: PSID.

Notes: Own calculations. This figure displays a decomposition of unfair inequality in the US over the period 1969–2014. (Unfair) inequality is calculated based on the divergence measure proposed by Magdalou and Nock (2011) with $\alpha=0$ (MN, $\alpha=0$) which corresponds to the MLD for total inequality. The decomposition is based on the Shapley value procedure proposed in Shorrocks (2012).

of families. To the contrary, we apply a time-constant absolute poverty threshold to disposable household income after taxes and transfers and measure poverty as a linear combination of the poverty gap ratio and the Watts index (Section 3). In fact, applying the headcount ratio to our income concept and the time-constant poverty line, we find that the share of poor individuals drops by more than 40% over time (Supplementary Figure S.7 and Supplementary Table S.5).³⁰ However, while the share of poor households has constantly decreased over time the intensity of poverty as measured by the poverty gap ratio and the Watts index has first decreased in the 1970s and then rebounded since the mid-1990s. As a consequence, we also find a relatively constant poverty trend over time, but for different reasons than official US government statistics.

The stable poverty trend, however, is superseded by marked increases in the violations of EOp. After slight decreases in the 1970s, the EOp contribution to total inequality increases from 0.013 points (9.8%) in 1980, over 0.024 points (11.7%) and 0.031 points (14.1%) in 1990 and 2000, to 0.037 points (13.9%) in the latest period of observation.

Analysing the EOp component in further detail, we note that the contribution of biological sex to overall inequality is negligible and hovers around the 1%-mark in relative terms. Hence, our measure does not reflect the well-documented decrease in earnings differences between males and females (Blau and Kahn, 2017). This deviation is not unexpected and follows from our focus on disposable household income. Accounting for resource sharing at the household level evens out any intra-household inequality among males and females. As such, all our results on biological sex are driven by single-headed households. Within this group the flat time trend in the contribution of sex-based differences to total inequality can be rationalized by two countervailing forces that are displayed in Supplementary Figure S.8. First, income differences among male and female-headed single households have been decreasing over the time period 1969–2014. Second, the prevalence of single-headed households has been increasing for both males and females. While

30. See Wimer, Fox, Garfinkel, Kaushal, and Waldfogel (2016) for similar results.

the first trend depresses the contribution of sex-based differences to total inequality, the second trend magnifies the remaining differences leading to relatively time-constant contributions of this component to unfairness in the US.³¹

In analogy to biological sex, the contribution of race to unfairness in the US is largely stagnant at approximately 0.007 points across the time period of observation. In relative terms, the contribution of race decreases slightly from 4.2% to 3.0%, again reflecting the marked increase of total inequality. This flat trend echoes previous findings that there has been little progress in closing the black–white earnings gap since the 1970s (Bayer and Charles, 2018; Derenoncourt and Montialoux, 2021).³²

With the contributions from sex- and race-based differences rather constant over time, the witnessed increase of the EOp component is entirely driven by the increased importance of parental background variables—namely parental education and occupation. While these factors jointly contributed 0.009 points (6.3%) in 1969, their importance has tripled to 0.028 points (10.2%) in 2014. Interpreting the covariances between parental education and occupation and individual income as a proxy for social mobility, our findings suggest that the US has become increasingly immobile in the time period from 1969 to 2014. This finding is in line with Aaronson and Mazumder (2008); Davis and Mazumder (2019) who find that the intergenerational elasticity of income has declined for cohorts entering the labour market after 1980, as well as Hilger (2019) who documents a similar time trend for educational mobility. However, we note that the assessment of intergenerational mobility trends in the US is contentious. In contrast to the previously cited works, Lee and Solon (2009), Chetty, Hendren, Kline, Saez, and Turner (2014b), and Song, Massey, Rolf, Ferrie, Rothbaum, and Xie (2020) conclude that inter-generational mobility has stayed constant over the time period of investigation. The disparity of results is explained by various drawbacks of the underlying data sources, as well as different measurement choices. While our measurement approach is not strictly comparable to either of these papers, our results are in line with the first set of works.³³

To summarize, unfair inequality in the US by-and-large replicates the development of total inequality. In the 1970s, unfairness decreased due to decreases in poverty. To the contrary, the growth of unfair inequality since the 1980s is almost exclusively attributable to increased violations of the EOp principle, and the growing importance of parental education and education for the income of their offspring in particular.

31. See also Lundberg, Pollak, and Stearns (2016) on the interaction between changing gender gaps, family structures, and the intergenerational transmission of advantages.

32. See also Supplementary Figure S.9 for complementary evidence on the stability of non-white disposable income gaps in our data.

33. Mobility measures can be decomposed into (1) the copula of parental background characteristics and child outcomes and (2) the marginal distributions of child outcomes and parental background characteristics, respectively (Chetty *et al.*, 2014b). Rank-mobility measures such as intergenerational correlations (IGC) and rank–rank correlations depend on (1) while holding (2) constant. To the contrary, mobility measures like the intergenerational elasticity (IGE) allow for changes in (2). Our measurement approach is closer to the second class as we compare different marginal distributions in the parent and the child generation that we allow to change over time. However, our measure differs from a typical IGE estimate in at least three important dimensions. (1) We model child income as a function of parental education and occupation instead of parental income. (2) We summarize persistence by calculating inequality in a predicted distribution instead of interpreting regression parameters. (3) Child outcomes refer to annual incomes at various points of the life-cycle instead of modelling them so as to mimic lifetime income (Nyblom and Stuhler, 2016). To provide a closer analogy to standard IGE estimates, we re-estimate our measure of unfairness for different age groups at different points in time while excluding all determinants of unfairness except for parental background characteristics. The results, displayed in Supplementary Figure S.10, suggest that relative mobility has decreased at all points of the individual life-cycle with more pronounced changes at older ages. This pattern is consistent with earnings profiles that fan out over the life-cycle.

4.2. Cross-country differences in unfair inequality

4.2.1. Data source. For the purpose of an international comparison we combine the PSID with the 2011 wave of EU-SILC. EU-SILC serves as the official database for monitoring inequality, poverty and social exclusion in the EU and covers a total of 31 countries (see e.g. Atkinson, Guio, and Marlier (2017) and the references cited therein). We use the 2011 cross-sectional wave as it contains a special survey module on parental background information that allows us to construct types from a broad range of circumstance variables.³⁴ As in the PSID, incomes are reported for the year preceding the survey leading to 2010 as the year of our cross-sectional comparison. The data preparation closely follows the procedures outlined for the PSID. Further detail on variable construction, as well as descriptive statistics are provided in [Supplementary materials B](#) and [D](#).

4.2.2. Outcome variable. We construct disposable household income as the sum of total household income from labour, asset flows, private transfers, public transfers, private retirement income and social security pensions, and deduct taxes on wealth (if applicable), income and social security contributions.³⁵ In analogy to the PSID, we scale reported public transfers by a country-specific under-reporting factor and adjust labour incomes by imputing individual labour incomes of respondents with zero labour incomes but non-zero working hours. Only about 1% of respondents are affected by the latter imputation. Furthermore, we deflate household incomes by the modified OECD equivalence scale, adjust for purchasing power parities and winsorize country-specific income distributions at the 1st and 99.5th percentiles.

4.2.3. Circumstance types and effort tranches. For each country we partition the population based on the following circumstance characteristics: (1) biological sex, (2) migration background, (3) educational achievement of the highest educated parent, and (4) the highest occupation category of either parent. While circumstances (1), (3), and (4) mirror the PSID specification, we replace the binary race variable of the PSID with a binary indicator for whether respondents were born in their current country of residence. In total, we partition the population into 36 circumstance types which we again subdivide into 20 quantiles to identify effort tranches. As evidenced in [Supplementary Figure S.3](#), this transformation is innocuous with respect to cross-country comparisons of inequality and poverty statistics.

4.2.4. Basic income threshold. Internationally comparable absolute poverty thresholds are again constructed based on the procedure suggested by Jolliffe and Prydz (2016). Twenty-one out of the 31 European countries belong to the highest quintile of countries in terms of PPP-adjusted household final consumption expenditures per capita. Hence, they are characterized by the same poverty threshold as the US: \$12,874 per annum (PPP-adj.). Ten Eastern European countries only belong to the second highest quintile and are therefore characterized by a lower poverty threshold of \$3,957 per annum (PPP-adj.).

4.2.5. Baseline results. Figure 3 replicates Figure 1 for the cross-country comparison. The red diamonds indicate total inequality, the blue squares unfair inequality. The

34. In contrast to the PSID, EU-SILC consists of rotating panels and each household stays in the data for only 4 years. Hence, one cannot use the panel dimension to construct circumstance variables.

35. In [Supplementary Table S.2](#), we provide a detailed breakdown of these income aggregates into their single components.

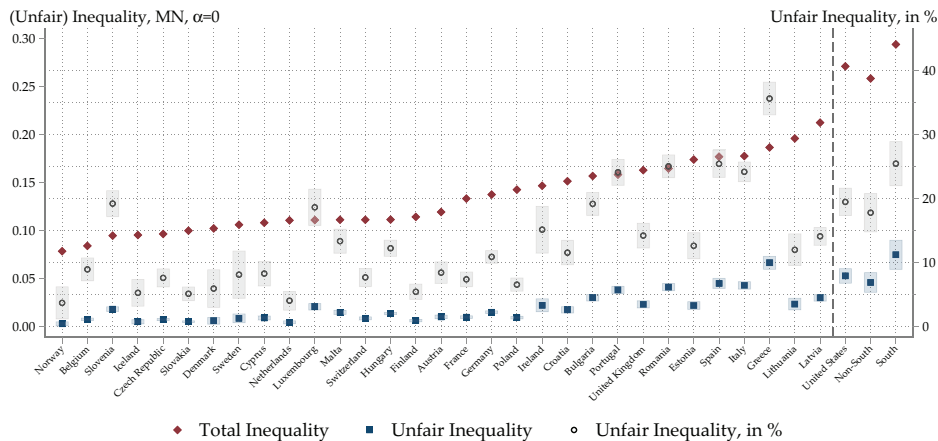


FIGURE 3
Unfair inequality across countries, 2010
Baseline results

Data: PSID and EU-SILC.

Notes: Own calculations. This figure displays cross-country differences in (unfair) inequality in 2010. Data points to the left of the vertical dashed line refer to the European country sample. Data points to the right of the vertical dashed line refer to the US and its census regions. (Unfair) inequality is calculated based on the divergence measure proposed by Magdalou and Nock (2011) with $\alpha=0$ (MN, $\alpha=0$) which corresponds to the MLD for total inequality. Relative measures of unfair inequality are expressed in percent (in %) of total inequality. The shaded areas show bootstrapped 95% confidence intervals based on 500 draws.

black hollow circles show the relative share of unfair inequality. Countries are ordered from left to right by their level of total inequality. The dashed vertical line separates the European countries from the US sample. Acknowledging the special role of the Southern states in terms of intergenerational transmission processes (Chetty *et al.*, 2014a; Bratberg, Davis, Mazumder, Nybom, Schnitzlein, and Vaage, 2017) and poverty prevalence (Ziliak, 2006), we also provide results separating the South of the US from the rest of the country based on the census region groupings of the US Census Bureau.

The US are by far the most unequal society in our country sample with inequality figures about 25% higher than the most unequal European societies. At the other end of the spectrum we find Norway, Belgium and Slovenia. The most unfair societies in 2010 are Greece, the US, Spain, Italy, and Romania closely followed by Portugal. Treating the South of the US as a separate country, it would attain the highest level of unfairness of all countries. In relative terms, EOP and FfP explain on average 26.6% of total inequality in the European countries of this group. The US attains an unfair share of 19.5%. The lower unfairness share of the US follows mechanically from its higher levels of total inequality. The group of countries with the least extent of unfair inequality consists of Nordic countries plus the Netherlands. Country rankings differ depending on whether we analyse total inequality or unfairness. While for example the Netherlands ranks 10th in terms of total inequality, it ranks second in terms of least unfair inequality.

4.2.6. Decomposition. The US differs markedly from its European counterparts in terms the processes that determine unfair inequality. Figure 4 shows the results of a Shapley value decomposition of unfair inequality into its different components.

In the European group of countries with the highest unfairness (Greece, Portugal, Romania, Spain, Italy), violations of the FfP principle consistently explain more than half of the detected unfair inequality. 2010 marks a peak year of the European sovereign debt crisis, and Greece, Portugal, Spain, and Italy were among the countries most affected by it. To highlight the

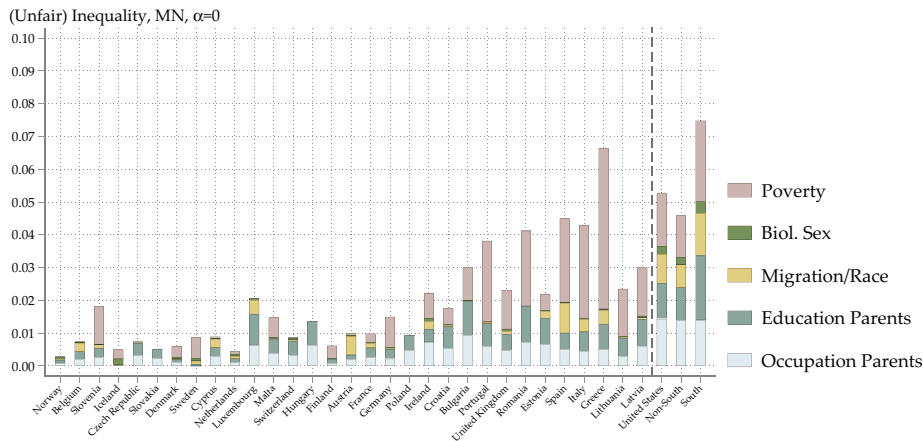


FIGURE 4
Unfair inequality across countries, 2010
Decomposition

Data: PSID and EU-SILC.

Notes: Own calculations. This figure displays a decomposition of cross-country differences in unfair inequality in 2010. Data points to the left of the vertical dashed line refer to the European country sample. Data points to the right of the vertical dashed line refer to the US and its census regions. (Unfair) inequality is calculated based on the divergence measure proposed by [Magdalou and Nock \(2011\)](#) with $\alpha = 0$ (MN, $\alpha = 0$) which corresponds to the MLD for total inequality. The decomposition is based on the Shapley value procedure proposed in [Shorrocks \(2012\)](#).

differential impact of the economic crisis on unfairness in Europe and the US, we calculate the difference between the Watts index and the poverty gap ratio for the six most unfair societies in our country sample (Greece, US, Spain, Italy, Romania, and Portugal) from 2006 to 2014. Since the FfP component nests the difference between these two poverty measures, it can be interpreted as a proxy statistic for the longitudinal development of FfP in these countries. The results are displayed in [Supplementary Figure S.11](#). Romania is the least economically developed country in the considered country group. In Romania, the financial crisis ended a trend of decreasing poverty and led to increased violations of the FfP principle in its aftermath. Similarly, in the group of Southern European countries the FfP proxy increases markedly after 2008. This evidence suggests that high levels of unfair inequality among the European countries in 2010 followed from the economic downturn that accompanied the financial crisis, and which in turn led to increased violations of the FfP principle.

In contrast to the European group, the difference between Watts index and poverty gap ratio is completely flat in the US over the crisis years. Instead, [Figure 4](#) shows that unfairness in the US is strongly driven by the EOp component. This difference cannot be explained by differential importance of biological sex. Due to our focus on disposable household income, income differences across the sexes have a negligible impact on unfair inequality in Europe and the US alike. Neither is this difference a mere consequence of replacing the race indicator with the immigration background indicator. Even abstracting from the migration/race circumstance, the US would be characterized by the highest degree of unequal opportunities in our country sample. It is the contributions of parental education and occupation that are the highest among all countries under consideration and place the US among the most unfair societies in our country sample. In line with the findings of [Chetty et al. \(2014a\)](#) and [Hilger \(2019\)](#), the lack of social mobility is particularly pronounced in the Southern states of the US. However, even when focusing on the non-Southern states only, the US ranks among the countries with the highest intergenerational persistence in our country sample.

5. SENSITIVITY ANALYSIS

In this section, we investigate the sensitivity of our baseline results to alternative normative assumptions. For brevity, we only present results for the longitudinal analysis of the US in the main body of this paper. However, every sensitivity check is conducted in an analogous way for the cross-country comparison—see [Supplementary Figures S.12–S.15](#) and [Supplementary Table S.7](#).

First, in principle the measurement approach adopted in this paper takes a neutral stance on the specification of the model primitives Ω , Θ , and $y_{\min}$. Hence, it may accommodate a wide array of different views on the responsibility cut, as well as the basic income $y_{\min}$. We acknowledge that these choices may be normatively contentious. It is not our ambition to resolve such disagreement. Instead we provide results for alternative choices of Ω , Θ , and $y_{\min}$ in Sections 5.1 and 5.2, respectively.

Second, our baseline measures is based on a weak conceptualization of EOp and treats EOp and FfP as non-separable in their scope of application. We provide empirical results for unfair inequality based on strong equality of opportunity and separability in Section 5.3.³⁶

Third, differences between y and y^* may be aggregated by different divergence measures that put different weights on positive and negative divergences from norm incomes, respectively. We therefore provide robustness analyses with respect to the use of different divergence measures in Section 5.4.

5.1. *Alternative responsibility cuts*

Any measurement of responsibility-sensitive egalitarianism requires a stance on the features of life for which people should be held responsible. In our baseline estimates we assume that people should not be held responsible for (1) their biological sex, (2) their race, (3) the occupation of their parents, and (4) the education of their parents. However, there may be further characteristics beyond individual control that evoke normative concern. Examples could be the quality of neighbourhoods in which people grew up ([Chetty, Hendren, and Katz, 2016](#)), parenting practices ([Doepke, Sorrenti, and Zilibotti, 2019](#)), or genetic endowments ([Papageorge and Thom, 2020](#)).

To be sure, the PSID puts strong constraints on testing the influence of different circumstance characteristics.³⁷ We therefore proceed as follows: first, we extract two additional circumstances that are consistently measured across the period of our analysis: (1) the census region in which respondents grew up and (2) the migration background of parents. We convert both variables into a vector of binary indicators and add them to our set of circumstances. Second, we repeat our analysis for all circumstance combinations that yield the same number of types as in our baseline analysis (36 types).³⁸ Hence, we repeat our analysis for 210 different specifications of Ω . The results are presented in Figure 5. The upper line again marks the development of total inequality. The lower crosses mark unfair inequality under each of the different specifications of Ω in any

36. See discussion in Section 3.4 and Appendix B for a formal derivation.

37. The PSID has introduced the Child and Development Supplement (CDS) in 1997 with follow-up waves in 2002/03 and 2007/08. The CDS provides very detailed information on the living environments of 3,563 children aged 0–12 in the initial wave. However, even the oldest children from the 1997 CDS cohort are only now in their early 30s—an age that is commonly believed to be the minimum threshold to approximate long-term earnings potential. Respecting sensible age thresholds and due to sample attrition over time, the CDS sample is too small to exploit its richer circumstance information for the income decompositions that underlie our empirical analysis—see also our discussion in Section 4.1.

38. We keep the granularity of the type partition constant to ensure the comparability to our baseline results and to balance the concerns for underestimating the influence of circumstances and noisy estimates of the relevant type parameters—see also our discussion in Section 4.1.

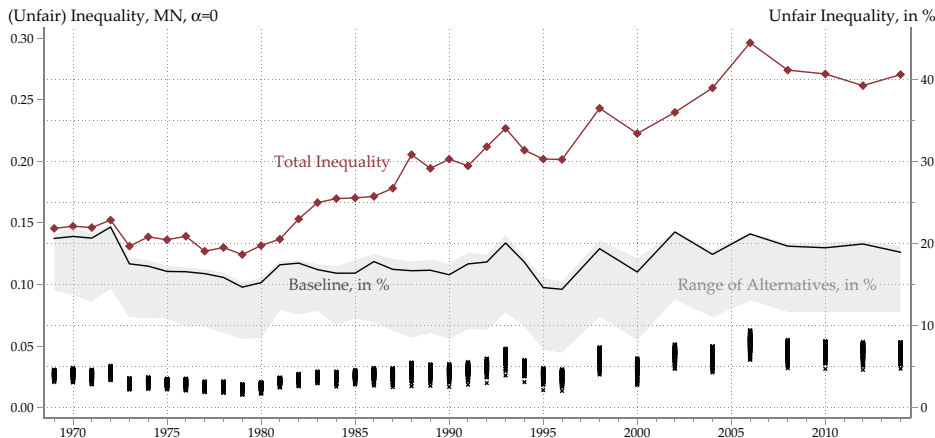


FIGURE 5
Unfair inequality in the US, 1969–2014
Alternative circumstance sets

Data: PSID.

Notes: Own calculations. This figure displays the development of (unfair) inequality in the US over the period 1969–2014 according to alternative specifications of the circumstance set Ω . (Unfair) inequality is calculated based on the divergence measure proposed by Magdalou and Nock (2011) with $\alpha=0$ (MN, $\alpha=0$) which corresponds to the MLD for total inequality. Relative measures of unfair inequality are expressed in percent (in %) of total inequality. The grey area shows the range of unfair inequality in percent (in %) of total inequality depending on the alternative measurement specifications.

given year. The black line marks the relative share of unfair inequality from our baseline estimate. The grey area shows the range between the lower and the upper envelope of the relative share of unfairness according to the alternative measurement specifications.

Our conclusions with respect to the time trend of unfair inequality in the US remain unaffected by the specification of Ω . However, we register level differences depending on the factors for which we hold people responsible. According to the most conservative specification of Ω , unfair inequality in the US amounts to 11.6% of total inequality in 2014. The upper bound is 19.6%. We acknowledge that the alternative circumstance information in the PSID remains limited to geographical and migration background information. EU-SILC avails a broader range of circumstance characteristics that are consistently elicited across all sample countries. These include (1) the relationship status of parents, (2) the number of siblings, (3) the financial situation of the parental household, as well as (4) property ownership of parents. We again test 210 different specifications of Ω for the EU-SILC countries holding the maximum number of types constant at 36. [Supplementary Figure S.12](#) reveals that in spite of level differences the general conclusions from our cross-country comparison remain robust to this broader set of alternative circumstance characteristics.

Another normative assumption relates to the correlation between circumstances Ω and efforts Θ . In our baseline measure, we treat the correlation between both components as morally objectionable. For example, part of the income gap between whites and non-whites can be explained by differences in educational attainment which itself is at least partially under the control of individuals (Gelbach, 2016). Circumstances thus exert a direct and an indirect effect on life outcomes. While in our baseline we follow Roemer (1998) and consider both effects as normatively objectionable, others have suggested to hold people responsible for effort and preference variables regardless of how they are formed (Barry, 2005). To test the sensitivity of our baseline results to this alternative normative stance, we repeat our analysis while partialling out the indirect effect that circumstances exert through individual efforts. To this end, we consider

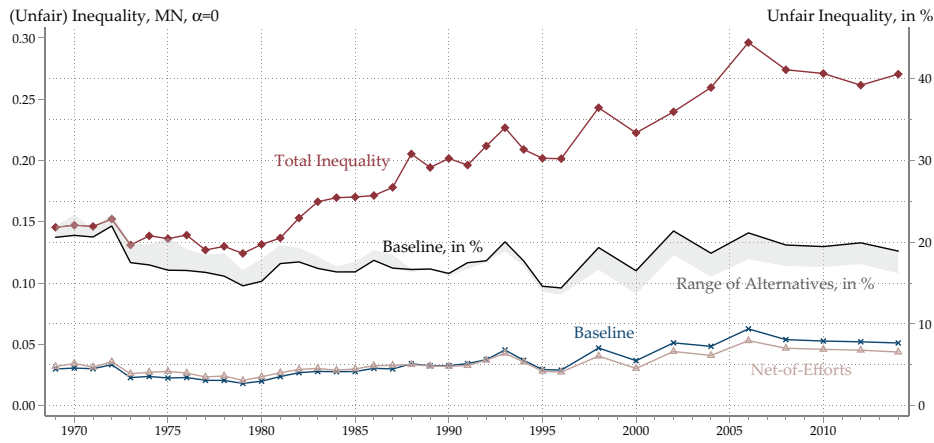


FIGURE 6
Unfair inequality in the US, 1969-2014
Accounting for preferences

Data: PSID.

Notes: Own calculations. This figure displays the development of (unfair) inequality in the US over the period 1969-2014 according to alternative treatments of the correlation between the effort set Θ and the circumstance set Ω . (Unfair) inequality is calculated based on the divergence measure proposed by Magdalou and Nock (2011) with $\alpha = 0$ (MN, $\alpha = 0$) which corresponds to the MLD for total inequality. Relative measures of unfair inequality are expressed in percent (in %) of total inequality. The grey area shows the range of unfair inequality in percent (in %) of total inequality depending on the alternative measurement specifications.

two variables that are partially under the control of individuals and highly predictive of incomes—(1) educational attainment and (2) annual working hours—and clean circumstances from their correlation with these effort variables before repeating our analysis.³⁹ If circumstances had no impact independent of the considered efforts, we would see a sharp drop of unfair inequality in comparison to our baseline results.

Figure 6 shows the differences between our baseline and the alternative responsibility cut. We note a moderation of the previously described time trend when holding people responsible for the correlation between circumstances Ω and efforts Θ . In contrast to our baseline, unfair inequality starts at higher levels in 1969 and increases more moderately in the 1990s. Combining this moderation of the time trend in absolute unfair inequality with the increasing slope of total inequality, the relative share of unfairness decreases over time and attains 16.2% in 2014. The differential development of our baseline and the alternative measure is consistent with evidence on the increasing stratification of college completion by parental background characteristics (Davis and Mazumder, 2019; Hilger, 2019), increasing college wage premia (Heathcote, Storesletten, and Violante, 2010b), and longer working hours among the highly educated (Fuentes and Leamer, 2019). Once we shut down educational attainment and working hours as channels of circumstance influence, unfairness does no longer reflect the growing importance of these factors for the determination of incomes over time.

5.2. Alternative basic income thresholds

There is no clear consensus on how to set a basic income threshold $y_{\min}$ that captures the material requirements to make ends meet. Acknowledging the arbitrariness of any threshold, Foster (1998) suggests to move beyond normative and empirical disagreements on the “correct” value of $y_{\min}$

39. We describe the exact steps of this procedure in [Supplementary material E](#).

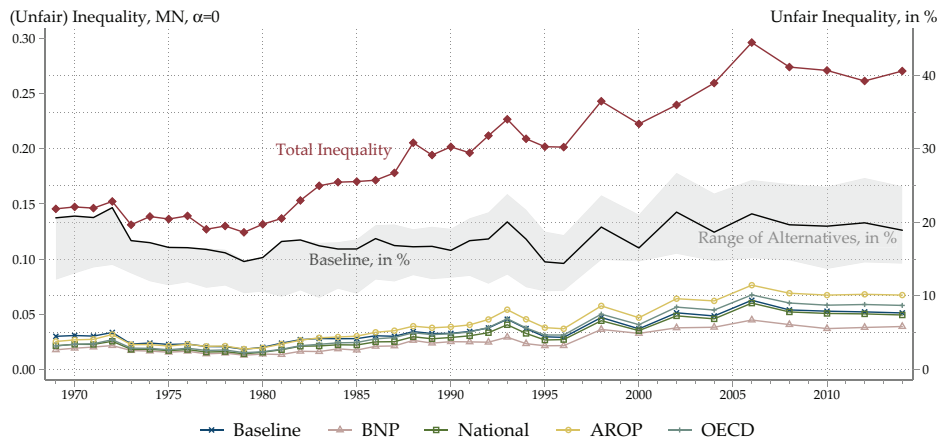


FIGURE 7
Unfair inequality in the US, 1969–2014
Alternative minimum thresholds

Data: PSID.

Notes: Own calculations. This figure displays the development of (unfair) inequality in the US over the period 1969–2014 according to alternative specifications of the poverty threshold $y_{\min}$. (Unfair) inequality is calculated based on the divergence measure proposed by Magdalou and Nock (2011) with $\alpha=0$ (MN, $\alpha=0$) which corresponds to the MLD for total inequality. Relative measures of unfair inequality are expressed in percent (in %) of total inequality. The grey area shows the range of unfair inequality in percent (in %) of total inequality depending on the alternative measurement specifications. The construction of the alternative minimum thresholds is discussed in Supplementary material B.

and to show the robustness of the main conclusions based on different plausible specifications of $y_{\min}$ instead. In this spirit, we provide alternative measures of unfair inequality based on four different poverty lines. First, Allen (2017) uses a linear programming approach to calculate the PPP-adjusted minimal cost of a basic needs consumption basket for different climatic regions of the world. For the four countries overlapping with our sample (US, Lithuania, UK, France) he calculates an average basic needs poverty (BNP) line of \$3.96 (PPP-adj.) per capita and day which we apply to all countries and years in our sample. Second, we repeat our analysis by using the official country-year-specific national poverty lines of the US Census Bureau and EUROSTAT. Third, we calculate relative poverty lines based on the suggestions of the OECD and EUROSTAT. While the OECD proposes a poverty line at 50% of the median equivalized disposable household income, EUROSTAT proposes an at-risk-of-poverty (AROP) line at 60% of the median of the same distribution.⁴⁰ The results for these different poverty thresholds are shown in Figure 7.

Our general conclusions with respect to the trend of unfairness in the US are insensitive to the specification of $y_{\min}$. If anything, the relative poverty thresholds of the OECD and AROP tend to magnify the relative increase of unfairness since the 1990s. However, we observe sharp level differences in unfair inequality depending on the stringency of $y_{\min}$. Proponents of the AROP threshold (\$18,737) would conclude that unfairness explained 24.9% of total inequality in 2014, while proponents of the BNP (\$1,445) threshold would detect a relative share of 14.4%.

40. Note that the official poverty statistics of EUROSTAT are also calculated by reference to the AROP threshold. The AROP lines presented in this work differ nevertheless from the national poverty lines provided by EUROSTAT since we calculate them by observing the sample restrictions and variable definitions used in this article.

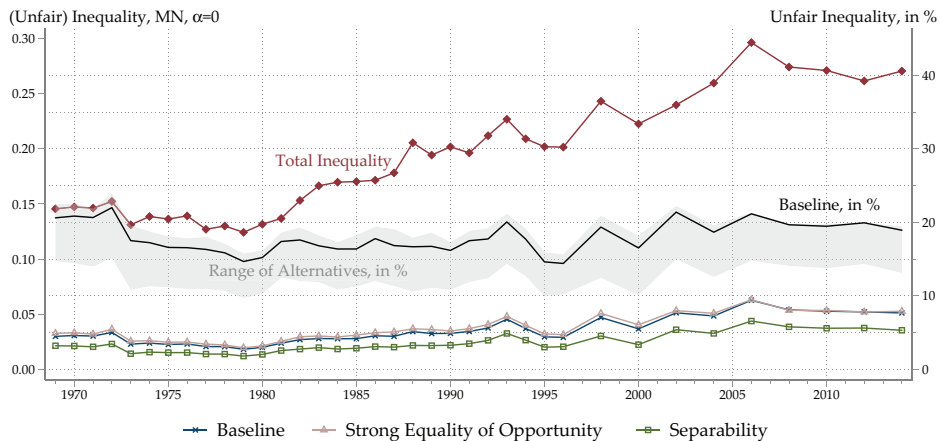


FIGURE 8
Unfair inequality in the US, 1969–2014
Alternative norm distributions

Data: PSID.

Notes: Own calculations. This figure displays the development of (unfair) inequality in the US over the period 1969–2014 according to the alternative norm distributions outlined in Section 3.4. (Unfair) inequality is calculated based on the divergence measure proposed by Magdalou and Nock (2011) with $\alpha=0$ (MN, $\alpha=0$) which corresponds to the MLD for total inequality. Relative measures of unfair inequality are expressed in percent (in %) of total inequality. The grey area shows the range of unfair inequality in percent (in %) of total inequality depending on the alternative measurement specifications.

5.3. Alternative norm distributions

Our baseline estimates of unfair inequality are based on weak EOp and reconcile EOp and FfP in a non-separable way. In Section 3.4, we have presented alternative norm distributions that divert from the baseline by operating on a strong notion of EOp, or assume separability between EOp and FfP. Figure 8 presents the development of (unfair) inequality in the US under each of these different conceptualizations. The black line marks the relative share of unfair inequality from our baseline estimate. The grey area shows the range between the lower and the upper envelope of the relative share of unfairness according to the alternative measurement specifications.

Our conclusions with respect to the time trend of unfair inequality in the US are robust to the different conceptualizations: a decrease in the relative share of unfair inequality until 1980 is followed by a stagnation throughout the following decade and increases throughout the 1990s until the present day. However, level differences exist. Invoking strong equality of opportunity yields results that are congruent to our baseline. Invoking separability consistently leads to lower levels of unfair inequality. Separability entails that (1) opportunity sets of circumstance types are evaluated by excess incomes above $y_{\min}$ only and (2) empirically poor individuals are excluded from compensation through opportunity-equalizing transfers beyond $y_{\min}$. Both features make the distribution of type-specific advantages more homogeneous and therefore require less transfers across types to attain the normative bliss distribution y^* . If one prefers the separability assumption over our baseline measure, one would conclude that unfairness amounts to 13.1% instead of 18.9% of total inequality in 2014.

5.4. Alternative divergence measures

Our baseline measure of unfair inequality employs the divergence measure proposed by Magdalou and Nock (2011) with $\alpha=0$. In addition to alternations in the weighting parameter α , we present results based on the measures of Cowell (1985) and Almås *et al.* (2011). The

TABLE 1
Rank Correlation across Years, US

	Magdalou and Nock			Cowell			Almås <i>et al.</i>
	$\alpha = 0$	$\alpha = 1$	$\alpha = 2$	$\alpha = 0$	$\alpha = 1$	$\alpha = 2$	
Magdalou and Nock							
$\alpha = 0$	1.00	.	.	.	.	.	.
$\alpha = 1$	0.99	1.00	.	.	.	.	.
$\alpha = 2$	0.97	0.99	1.00	.	.	.	.
Cowell							
$\alpha = 0$	0.99	1.00	0.98	1.00	.	.	.
$\alpha = 1$	0.99	1.00	0.99	1.00	1.00	.	.
$\alpha = 2$	0.98	1.00	0.99	1.00	1.00	1.00	.
Almås <i>et al.</i>							
	0.96	0.98	1.00	0.98	0.98	0.98	1.00

Data: PSID.

Notes: Own calculations. This table displays rank correlations for unfair inequality across years based on different divergence measures. Unfair inequality is calculated based on the divergence measures proposed by [Magdalou and Nock \(2011\)](#), [Cowell \(1985\)](#), and [Almås *et al.* \(2011\)](#).

[Cowell](#)-family is another generalization of the entropy class of inequality indexes that varies with an inequality aversion parameter α . The [Cowell](#)-family and the MN-family coincide exactly for $\alpha = 1$. Moreover, we employ the unfairness Gini proposed by [Almås *et al.* \(2011\)](#) which tends to put relatively less weight on negative divergences from the reference distribution.

In spite of their differences, all measures yield highly comparable results in terms of cross-period comparisons of unfair inequality. Table 1 shows rank-correlations for the different measures and their parameterizations for the US sample. All correlation coefficients are at a level of at least 0.96. Hence, our conclusions are robust to using alternative divergence measures.

6. CONCLUSION

In this article, we propose a novel measure of unfair inequality that reconciles the ideals of EOP and FfP. In fact, we provide the first work that combines these widely endorsed principles of justice into a joint measure of unfair inequality by treating both as co-equal grounds for compensation.

Next to illustrating our measurement approach and showcasing its flexibility to various normative alternations, we provide two empirical applications. First, we analyse the development of inequality in the US over the time period 1969–2014 through the lens of our unfairness measure. Second, we provide a corresponding international comparison between the US and 31 European countries in 2010. In combination, both analyses yield important insights for current debates on inequality. First, the US trend in unfair inequality has traced the marked increase of total inequality since 1980. Second, this trend is mainly driven by a less equal distribution of opportunities across people with different parental education and occupation characteristics. Third, unfairness in the US shows a remarkably different structure than in comparable European societies. In 2010, unfairness in Europe is largely driven by the consequences of the 2008 financial crisis; unfairness in the US is driven by the intergenerational transmission of disadvantages. The underlying determinants of the latter are arguably more persistent than income shortfalls due to economic downturns illustrating the challenge presented to policymakers willing to address unfairness in the US.

While we provide comprehensive robustness checks for our findings, there are shortcomings which suggest a wide avenue for further research. At the empirical level, it includes addressing the well-known drawbacks of survey data by the use of suitable administrative datasets. Furthermore, we have shown in this work that our measurement approach lends itself to various refinements and

extensions with respect to the conceptualization of unfairness. While we were careful to choose our guiding principles to broadly match the fairness perceptions of a larger public, we look forward to tailor our approach even stronger to forthcoming empirical evidence on the normative preferences upheld by individuals.

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Supplementary Data

Supplementary data are available at *Review of Economic Studies* online.

Data Availability Statement

The data and code underlying this research is available on Zenodo at <https://doi.org/10.5281/zenodo.5772718>.

APPENDIX

A. PROOF OF PROPOSITION 1

Proof of Proposition 1. The proof proceeds in two steps. First, we show that $y^* \in \cap_{h=1}^3 \mathcal{D}^h$. Second, we show that $\cap_{h=1}^3 \mathcal{D}^h$ is a singleton. The second step is established by contradiction.

Step 1. Define y^* as in equation (3.9):

$$y_i^* = \begin{cases} y_{\min}, & \forall i \in \mathcal{P}(\omega), \forall \omega \in \Omega, \\ y_{\min} + (y_i \frac{\mu}{\mu_{T(\omega)}} - y_{\min}) \times \frac{\frac{\mu - y_{\min}}{N_{\mathcal{R}(\omega)} (\frac{\mu_{\mathcal{R}(\omega)}}{\mu_{T(\omega)}} \mu - y_{\min})}}{\frac{\mu_{\mathcal{R}(\omega)}}{\mu_{T(\omega)}} \mu - y_{\min}}, & \forall i \in \mathcal{R}(\omega), \forall \omega \in \Omega. \end{cases}$$

We reshape the second term of equation (3.9) as follows:

$$\begin{aligned} \mu &= y_{\min} + \frac{y_i^* - y_{\min}}{y_i \frac{\mu}{\mu_{T(\omega)}} - y_{\min}} \times \frac{N_{\mathcal{R}(\omega)} (\frac{\mu_{\mathcal{R}(\omega)}}{\mu_{T(\omega)}} \mu - y_{\min})}{\frac{\mu_{\mathcal{R}(\omega)}}{\mu_{T(\omega)}} \mu - y_{\min}} \\ &= y_{\min} + \frac{1}{N_{T(\omega)}} \sum_{j \in \mathcal{R}(\omega)} (y_j^* - y_{\min}) \frac{\mu_{\mathcal{R}(\omega)} \frac{\mu}{\mu_{T(\omega)}} - y_{\min}}{y_j \frac{\mu}{\mu_{T(\omega)}} - y_{\min}} \\ &= \frac{1}{N_{T(\omega)}} \left[\underbrace{\sum_{j \in \mathcal{P}(\omega)} y_{\min}}_{=y_j^*, \forall j \in \mathcal{P}(\omega)} + \underbrace{\sum_{j \in \mathcal{R}(\omega)} y_{\min} + (y_j^* - y_{\min}) \frac{y_j \frac{\mu}{\mu_{T(\omega)}} - y_{\min}}{y_j \frac{\mu}{\mu_{T(\omega)}} - y_{\min}}}_{=y_j^*, \forall j \in \mathcal{R}(\omega)} \right]. \end{aligned}$$

The first term corresponds to the norm income of any $j \in \mathcal{P}(\omega)$ as prescribed by $\mathcal{D}^2$; the second term corresponds to the norm income of any $j \in \mathcal{R}(\omega)$ as prescribed by $\mathcal{D}^3$. Simplifying the previous equation we obtain the EO requirement $\mathcal{D}^1$:

$$\mu = \frac{1}{N_{T(\omega)}} \sum_{j \in T(\omega)} y_j^* = \mu_{T(\omega)}^*.$$

We conclude $y^* \in \cap_{h=1}^3 \mathcal{D}^h$.

Step 2. Now let's assume there is $\hat{y}^* \neq y^*$ such that $\hat{y}^* \in \cap_{h=1}^3 \mathcal{D}^h$.

By $\mathcal{D}^2$ we know that for any $j \in \mathcal{P}(\omega)$:

$$\hat{y}_i^* = y_{\min}.$$

By $\mathcal{D}^3$, we know that for any $j \in \mathcal{R}(\omega)$:

$$\hat{y}_i^* = y_{\min} + \frac{y_i \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}}{y_j \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}} (\hat{y}_j^* - y_{\min}).$$

Using both requirements in $\mathcal{D}^1$ and simplifying, we get:

$$\begin{aligned} \mu &= \frac{1}{N_{\mathcal{T}(\omega)}} \left[\sum_{i \in \mathcal{P}(\omega)} y_{\min} + \sum_{i \in \mathcal{R}(\omega)} \left(y_{\min} + \frac{y_i \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}}{y_j \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}} (\hat{y}_j^* - y_{\min}) \right) \right] \\ &= y_{\min} + \frac{1}{N_{\mathcal{T}(\omega)}} \sum_{i \in \mathcal{R}(\omega)} \frac{\hat{y}_j^* - y_{\min}}{y_j \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}} (y_i \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}) \\ &= y_{\min} + \frac{\hat{y}_j^* - y_{\min}}{y_j \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}} \times \frac{1}{N_{\mathcal{T}(\omega)}} \sum_{i \in \mathcal{R}(\omega)} (y_i \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}) \\ &= y_{\min} + \frac{\hat{y}_j^* - y_{\min}}{y_j \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}} \times \frac{N_{\mathcal{R}(\omega)}}{N_{\mathcal{T}(\omega)}} \left(\frac{\mu_{\mathcal{R}(\omega)}}{\mu_{\mathcal{T}(\omega)}} \mu - y_{\min} \right). \end{aligned}$$

Solving for $\hat{y}_j^*$ we obtain:

$$\hat{y}_j^* = \begin{cases} y_{\min}, & \forall j \in \mathcal{P}(\omega), \forall \omega \in \Omega, \\ y_{\min} + (y_j \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}) \times \frac{\frac{\mu - y_{\min}}{N_{\mathcal{R}(\omega)} (\frac{\mu_{\mathcal{R}(\omega)}}{\mu_{\mathcal{T}(\omega)}} \mu - y_{\min})}}{1}, & \forall j \in \mathcal{R}(\omega), \forall \omega \in \Omega. \end{cases}$$

Hence, in contrast to the initial assumption of $\hat{y}^* \neq y^*$, we obtain $\hat{y}^* = y^*$ which is a contradiction. We conclude $\cap_{h=1}^3 \mathcal{D}^h$ is a singleton.⁴¹ $\square$

B. ALTERNATIVE CONCEPTUALIZATIONS

In this appendix, we provide formal derivations of the alternative norm distributions discussed in Section 3.4. Proofs for the propositions proceed analogously to the proof of Proposition 1 and are collected in [Supplementary material A](#).

B.1. Strong equality of opportunity.

We divert from the baseline by replacing *weak equality of opportunity* with *strong equality of opportunity*. The satisfaction of strong EOp requires the equalization of all moments of the type-specific income distribution. We therefore reformulate (3.5) as follows:

$$\mathcal{D}^{1a} = \left\{ y^* \in \mathcal{D} \mid y_i^* = \frac{1}{N_{\mathcal{S}(\theta)}} \sum_{j \in \mathcal{S}(\theta)} y_j^* = \mu_{\mathcal{S}(\theta)}^*, \forall i \in \mathcal{S}(\theta), \forall \theta \in \Theta \right\}. \quad (\text{B.1})$$

Note that $\mathcal{D}^{1a}$ implies $\mu^* = \mu$. Invoking strong EOp requires a subsequent redefinition of the poor and the non-poor fraction of the population, as well as of the FFP and the proportionality requirement:

$$\mathcal{P}(\theta) = \left\{ i \in \mathcal{S}(\theta) \mid y_i \frac{\mu_{\mathcal{S}(\theta)}}{y_i} \leq y_{\min} \right\}; \quad \mathcal{R}(\theta) = \mathcal{S}(\theta) \setminus \mathcal{P}(\theta), \forall \theta \in \Theta \quad (\text{B.2})$$

41. In footnote 18, we highlight the possibility to derive the same norm distribution from an optimization under constraints. The formal set-up for such an optimization problem could look as follows:

$$\begin{aligned} &\min \sum_T (\mu_{\mathcal{T}(\omega)}^* - \mu) \text{ subject to} \\ &y_i^* = y_{\min}, \forall i \in \mathcal{P}(\omega), \quad \frac{y_i^* - y_{\min}}{y_j^* - y_{\min}} = \frac{y_i \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}}{y_j \frac{\mu}{\mu_{\mathcal{T}(\omega)}} - y_{\min}}, \forall i, j \in \mathcal{R}(\omega). \end{aligned}$$

$$\mathcal{D}^{2a} = \left\{ y^* \in \mathcal{D} \mid y_i^* = y_{\min}, \forall i \in \mathcal{P}(\theta), \forall \theta \in \Theta \right\}, \quad (\text{B.3})$$

$$\mathcal{D}^{3a} = \left\{ y^* \in \mathcal{D} \mid \frac{y_i^* - y_{\min}}{y_j^* - y_{\min}} = \frac{\mu \mathcal{S}(\theta) - y_{\min}}{\mu \mathcal{S}(\theta') - y_{\min}}, \forall i \in \mathcal{R}(\theta), \forall j \in \mathcal{R}(\theta'), \forall \theta \in \Theta \right\}. \quad (\text{B.4})$$

Proposition B.1. Suppose $\mu > y_{\min}$. Then, the intersection $\mathcal{D}^{1a} \cap \mathcal{D}^{2a} \cap \mathcal{D}^{3a}$ yields a singleton which defines the norm distribution y^* :

$$y_i^* = \begin{cases} y_{\min}, & \forall i \in \mathcal{P}(\theta), \forall \theta \in \Theta, \\ y_{\min} + (\mu \mathcal{S}(\theta) - y_{\min}) \frac{\frac{\mu - y_{\min}}{N \mathcal{R}(\Theta)}}{(\mu \mathcal{R}(\Theta) - y_{\min})}, & \forall i \in \mathcal{R}(\theta), \forall \theta \in \Theta, \end{cases} \quad (\text{B.5})$$

where $\mathcal{R}(\Theta) = \cup \mathcal{R}(\theta)$.

B.2. Separability.

We divert from the baseline by replacing *non-separability* with *separability*. According to separability FfP applies for any $i \in \mathcal{P}$, whereas EOp applies for any $i \in \mathcal{R}$. We therefore reformulate (3.5) as follows:

$$\mathcal{D}^{1b} = \left\{ y^* \in \mathcal{D} \mid \mu_{\mathcal{T}(\omega) \cap \mathcal{R}}^* = \frac{1}{N_{\mathcal{T}(\omega) \cap \mathcal{R}}} \sum_{i \in \mathcal{T}(\omega) \cap \mathcal{R}} y_i^* = \mu_{\mathcal{R}}^*, \forall \omega \in \Omega \right\}. \quad (\text{B.6})$$

Invoking separability requires a subsequent redefinition of FfP and the proportionality requirement:

$$\mathcal{D}^{2b} = \left\{ y^* \in \mathcal{D} \mid y_i^* = y_{\min}, \forall i \in \mathcal{P} \right\}, \quad (\text{B.7})$$

$$\mathcal{D}^{3b} = \left\{ y^* \in \mathcal{D} \mid \frac{y_i^* - y_{\min}}{y_j^* - y_{\min}} = \frac{y_i - y_{\min}}{y_j - y_{\min}}, \forall i, j \in \mathcal{T}(\omega) \cap \mathcal{R}, \forall \omega \in \Omega \right\}. \quad (\text{B.8})$$

Note that $\mathcal{D}^{1b} \cap \mathcal{D}^{2b} \cap \mathcal{D}^{3b}$ does not imply the satisfaction of constant resources. Therefore, we additionally impose:

$$\mathcal{D}^{4b} = \left\{ y^* \in \mathcal{D} \mid \mu = \frac{1}{N} \sum_{i \in \mathcal{N}} y_i = \frac{1}{N} \sum_{i \in \mathcal{N}} y_i^* = \mu^* \right\}. \quad (\text{B.9})$$

Proposition B.2. Suppose $\mu > y_{\min}$. Then, the intersection $\mathcal{D}^{1b} \cap \mathcal{D}^{2b} \cap \mathcal{D}^{3b} \cap \mathcal{D}^{4b}$ yields a singleton which defines the norm distribution y^* :

$$y_i^* = \begin{cases} y_{\min}, & \forall i \in \mathcal{T}(\omega) \cap \mathcal{P}, \forall \omega \in \Omega, \\ y_{\min} + (y_i - y_{\min}) \times \frac{(\mu - y_{\min})}{\frac{N \mathcal{R}}{N} (\mu \mathcal{T}(\omega) \cap \mathcal{R} - y_{\min})}, & \forall i \in \mathcal{T}(\omega) \cap \mathcal{R}, \forall \omega \in \Omega. \end{cases} \quad (\text{B.10})$$

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Genetic Endowments, Educational Outcomes, and the Mediating Influence of School Quality*

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Abstract

Genetic endowments are fixed at conception and matter for the educational attainment of individuals. Do better schools mitigate or magnify the outcomes of this genetic lottery? We analyze the interplay of genetic endowments and school quality for educational attainment in the United States. Our results suggest that higher-quality schools are substitutes for genetic endowments: a 1 SD increase in school quality reduces the positive effect of a 1 SD increase in the relevant polygenic index (PGI) by 17%. This substitutability is underpinned by relative gains in health, language ability, patience, and risk aversion of low-PGI students.

JEL-Codes: I29; I21; J24

Keywords: Polygenic indexes, School resources, Skill formation

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1 INTRODUCTION

Education is a key determinant of life outcomes, both for individuals and for societies as a whole (Acemoglu and Autor, 2011; Hanushek and Woessmann, 2008; Krueger and Lindahl, 2001). At the same time, a broad literature has shown that genetic endowments are crucial for educational success. In heritability studies, genetic endowments account for 40% of the variation in years of education (Branigan et al., 2013; Lee et al., 2018). High heritability, however, does not imply that the impact of genes on education is immutable. On the contrary, growing evidence shows that the impact of genetic endowments on life outcomes varies with social environments like families, neighborhoods, and schools (Cesarini and Visscher, 2017; Koellinger and Harden, 2018). This observation raises important questions for researchers and policy-makers alike: can schools moderate the link between genetic endowments and educational outcomes; and if yes, do they magnify or reduce genetic inequality in educational attainment?

Answers to these questions are important to address concerns about educational inequality. Since genetic endowments are fixed at conception and remain constant over the life course of individuals, genetic inequality contradicts the widespread goal of providing equal educational opportunities to all members of society (Alesina et al., 2018; Lergetporer et al., 2020). In line with this policy goal, there is a long-standing literature in economics that inquires “the question of how well schools reduce the inequity of birth” (Coleman et al., 1966, p.36). This literature has focused on inequalities by socioeconomic status (SES), race, and gender. However, detailed evidence of how well schools address inequality due to genetic endowments is scant due to the long-term unavailability of genetic endowments at the individual level.

In this paper, we address this gap. We use novel, individual-level data on genetic endowments to study the interplay of genes and school quality in the production of educational attainment. That is, we investigate whether high-quality schools magnify or reduce genetic inequality in educational success. Notably, the effect sign is a priori unclear, since different branches of related literature provide support for effects in both directions. On the one hand, behavioral geneticists have found that advantaged family environments and genetic endowments complement each other (Turkheimer et al., 2003; Woodley of Menie et al., 2021)—a result often referred to as the Scarr-Rowe interaction. Hence, if school environments worked similarly to family environments, high-quality schools would magnify genetic inequality. On the other hand, recent evidence on heterogeneity in school effects suggests that school quality and student socioeconomic background substitute each other (Jackson et al., forthcoming). Hence, if disadvantages based on genetic endowments were similar to disadvantages based on students’ socioeconomic backgrounds, high-quality schools would reduce genetic inequality.

To study the interplay of genetic endowments and school quality in the production of educational attainment, we use data from the National Longitudinal Study of Adolescent to Adult Health (Add Health). Add Health is a 5-wave panel study that follows a representative sample

of US high school students from 1994/95 to the present. To measure genetic endowments, we leverage recent advances in molecular biology and use a polygenic index for educational attainment (PGI^{EA} , Becker et al., 2021; Dudbridge, 2013; Lee et al., 2018). PGI^{EA} is an individual measure of the genetic propensity to attain education.¹ It offers important advantages over traditional proxies for "innate ability", such as student test scores and IQ tests (Brinch and Galloway, 2012; Hanushek and Woessmann, 2008, 2012; Heckman et al., 2010). More specifically, it is a fine-grained DNA-based measure that is fixed at conception and cannot be modified thereafter. To measure the quality of school environments, we use information from headmaster surveys and construct school quality indicators based on the following observable characteristics: teacher experience, teacher turnover, teacher education, and class size. To the best of our knowledge, Add Health is the only (publicly available) data set that offers detailed information on schooling environments from both survey and administrative sources for a genotyped sample of reasonable size.

For identification, we combine a between-family comparison with the control function approach suggested by Altonji and Mansfield (2018). We discuss the underlying identification assumptions in detail and provide tests for their satisfaction. A first identification challenge arises from the fact that the genetic endowments of children are a function of the genetic endowments of their parents. Therefore, our parameters of interest may be confounded by *genetic nurture effects*, i.e., genetic endowments of children may be correlated with other family characteristics that co-determine educational attainment. In response, we show that the relevant point estimates from the between-family design can be replicated in a smaller sibling sample that allows us to control for genetic nurture by including family-fixed effects. This result suggests that any remaining confounding from genetic nurture effects is small and unlikely to overturn our main results. A second identification challenge arises from the fact that children sort into schools based on family background. Therefore, our parameters of interest may be confounded by *selection effects*, i.e., school quality may be correlated with other family characteristics that co-determine educational attainment. In response, we follow the control function approach suggested in Altonji and Mansfield (2018), i.e., we use group-level averages of observable characteristics to remove all cross-group variation that results from sorting into school catchment areas. Furthermore, we calculate different summary statistics to quantify the potential magnitude of confounding by unobservables (Cinelli and Hazlett, 2020; Oster, 2019). The results suggest that any residual confounding from selection effects is small and unlikely to overturn our main results. Lastly, gene-environment interactions can only be identified if genetic endowments and the environmental variable of interest are distributed independently of each other. This prerequisite may be violated if children sort into schools based on their genetic endowments. In response, we show that we cannot reject the equality of PGI^{EA} distributions at different levels of school quality. This result rules out selection into schools based on ge-

¹In addition, PGI^{EA} is highly predictive of several life outcomes that are closely related to educational attainment. These outcomes include earnings, wealth and (non-)cognitive skills (Barth et al., 2020; Buser et al., 2021a; Demange et al., 2021; Houmark et al., 2020; Lee et al., 2018; Muslimova et al., 2020; Papageorge and Thom, 2020).

netic endowments. In summary, all our tests point to the satisfaction of crucial identification assumptions and provide support for the between-family design adopted in this paper.

Our results can be summarized as follows. First, genetic endowments and school quality are highly predictive of years of education: a one-standard-deviation increase in PGI^{EA} (school quality) increases educational attainment by ≈ 0.36 (0.14) years.² Second, genetic endowments and school quality are substitutes in the production of educational attainment: a one-standard-deviation increase in school quality reduces the positive association between educational attainment and PGI^{EA} from 0.36 to 0.30 years—a decrease of $\approx 17\%$. This result implies that improvements in school quality may reduce genetic inequality in educational attainment.

We perform a series of robustness checks to evaluate whether our results are conflated by competing mechanisms. We first show that our measures for school quality do not pick up the effects of other school characteristics that may correlate with student outcomes. These characteristics comprise school-level policies such as the prevalence of ability grouping and retention as well as the demographic composition of teachers. Next, we demonstrate that our results are not driven by gene-environment interactions that reflect family instead of school environments. To that end, we estimate flexible models including higher-order polynomials of PGI^{EA} and our measures for school quality and account for all possible interactions with a broad set of family background characteristics (Biroli et al., 2022; Keller, 2014). In all these robustness analyses, our main results remain unaffected.

We also analyze mechanisms that underpin the substitutability of genetic endowments and school quality. First, to uncover which types of skills drive our results, we analyze the associations of PGI^{EA} and school quality with a range of intermediate outcomes that are highly predictive of educational attainment. These intermediate outcomes include cognitive skills, economic preferences, personality measures, and health. We find substitutability of genetic endowments and school quality for the formation of verbal intelligence, risk-aversion, patience, and subjective health. Hence, we consider these intermediate outcomes plausible transmission channels for the overall result regarding educational attainment. Second, educational attainment summarizes information from various educational stages, where each stage requires a different mix of skills (Cunha et al., 2006, 2010). Therefore, we repeat our analysis by replacing total educational attainment with binary variables indicating whether respondents have obtained a given educational degree. We find substitutability of genetic endowments and school quality for high school and college graduation, but there is no substitutability for post-graduate degrees. This pattern stands in notable contrast to the gene-environment interaction for family SES. Consistent with evidence from recent studies (Buser et al., 2021a; Papageorge and Thom, 2020), we show that the interaction of genetic endowments and family socioeconomic status switches its sign as individuals progress through the educational system and that there is a substantial complementarity for the attainment of post-graduate degrees. The contrast between these pat-

²These increases correspond to 16% (6%) of a standard deviation.

terns indicates that gene-environment interplay varies across different sources of investments in children: relative to their high PGI^{EA} peers, genetically disadvantaged children consistently gain more from having attended a high-quality school than from having a high-SES family background.

Our study contributes to two strands of literature. First, we contribute to the literature on gene-environment interactions. There is a large literature showing that the association between genetic endowments and life outcomes varies with family SES (Figlio et al., 2017; Houmark et al., 2020; Papageorge and Thom, 2020; Ronda et al., 2022). However, evidence on gene-environment interactions in the school context is scant. Houmark et al. (2022) show that genetic gaps in achievement among Danish students increase across grades 2–8, particularly among children from low SES families. In contrast to their study, we do not focus on the dynamics of genetic gradients across the educational biography of students but investigate whether school quality can moderate these gradients. Barcellos et al. (2021) use a regression discontinuity design to show relative gains in educational attainment for low PGI^{EA} students after a compulsory schooling reform in the UK.³ In contrast to their paper, we do not focus on variation in the length of schooling but on variation in the quality of schools. Trejo et al. (2018) investigate whether the socioeconomic composition of schools moderates the strength of genetic gradients in educational and occupational attainment. Using data from Add Health and the Wisconsin Longitudinal Study, they find inconclusive evidence for the presence of gene-environment interplay. In contrast to their study, we do not focus on school contexts, e.g., the socioeconomic composition of students, but on quality indicators that reflect school practices and policies: teacher experience, teacher turnover, teacher education, and class size.⁴ This shift of focus also has methodological implications. The socioeconomic composition of schools directly reflects endogenous sorting that leads to gene-environment correlations. Since the independent distribution of genetic endowments and environmental variables is a prerequisite for the identification of their interaction, the interpretation of gene-environment interactions with school composition variables is not straightforward. To the contrary, we show that our measure of school quality is independently distributed of PGI^{EA} and we strengthen identification by using the control function approach of Altonji and Mansfield (2018) to account for cross-group variation that results from endogenous sorting into schools.

Second, we contribute to the literature on school quality. A broad literature demonstrates that school quality raises educational attainment, wages, and health, and reduces crime (Beuermann and Jackson, 2022; Beuermann et al., 2022; Deming, 2011; Deming et al., 2014). Specifically, teacher quality, which is a core component of our school quality measure, has been shown to improve student outcomes in the short- and long-term (Chetty et al., 2014a,b; Jackson, 2019; Rivkin et al., 2005; Rockoff, 2004). However, it is less known which student groups benefit the

³However, these relative gains in educational attainment did not translate in corresponding relative income gains as high PGI^{EA} students experienced higher returns to education.

⁴See also Jennings et al. (2015) and Raudenbush and Willms (1995) for the conceptual distinction between school context and school practice when estimating school effects.

most from school quality. On the one hand, disadvantaged students may benefit more, as they have more room for improvement; on the other hand, they may benefit less, as they do not have the means to take advantage of better schools (Cunha et al., 2010). In line with the ambiguity of theoretical predictions, the empirical evidence is mixed. For example, Walters (2018) show that disadvantaged students benefit more from admission to charter schools in Boston, while Dustan et al. (2017) show that disadvantaged students benefit less from admission to elite public high schools in Mexico City. Many studies in this literature, analyze samples of applicants to charter or elite schools, which are not representative of disadvantaged students in general. A notable exception is a study by Jackson et al. (forthcoming) who use a sample of all schools and students in Chicago’s public school district to demonstrate that disadvantaged students gain relatively more from higher-quality schools. In contrast to these previous studies, which have evaluated heterogeneity by socioeconomic status, we focus on genetic endowments as the source of disadvantage. The focus on genetic endowments is highly relevant since they are fixed at conception and cannot be influenced by individual choice. Therefore, they constitute a source of inequality that is perceived as unfair by many people in the United States and other Western societies (Almås et al., 2020). Furthermore, they have substantial explanatory power for educational attainment that is not captured by standard measures of disadvantage like family SES.⁵ Using a representative sample of schools from all over the United States, we show that genetically disadvantaged students gain relatively more from high-quality schools than their advantaged peers.

Furthermore, our paper relates to the literature on educational inequality and intergenerational educational mobility (Blanden et al., 2023; Chetty et al., 2014c,d; Hanushek et al., 2021; Rossin-Slater and Wüst, 2020). This literature shows that educational inequality is strongly persistent across generations, part of which can be explained by the intra-family transmission of genetic endowments. Our results show that investments in school quality can address this channel of intergenerational persistence in the United States.

The remainder of this paper is structured as follows. In section 2, we provide an introduction to the measurement of genetic endowments. In section 3, we detail our empirical strategy. After introducing our data sources in section 4, we present results in section 5. Section 6 concludes the paper.

2 MEASURING GENETIC ENDOWMENT

The "First Law of Behavior Genetics" states that all human traits are heritable; that is, genetic endowments explain the expression of each trait, at least to some extent (Turkheimer, 2000).

⁵See Supplementary Figure S.1 where we show that PGI^{EA} is correlated with family SES but that PGI^{EA} distributions have significant overlap at different levels of family SES. This pattern suggests that genetic endowments are a distinct dimension of student advantage that cannot be captured by family SES alone.

The empirical challenge is to identify specific sequences in the genome that are related to the traits of interest.⁶ Recent advances in molecular genetics have enabled a novel method of genetic discovery: genome-wide association studies (GWAS). GWAS exploit the most common type of genetic variation between humans, so-called single-nucleotide polymorphisms (SNPs). SNPs occur when a single nucleotide—the basic building block of DNA molecules—differs at a specific position in the genome. Humans have around ten million of these SNPs. GWAS estimate separate linear regressions that relate the SNP of individual i at genome location j to an outcome of interest y :

$$y_i = \psi_j^y \text{SNP}_{ij} + \delta C_i + \varepsilon_i. \quad (1)$$

$\text{SNP}_{ij} \in \{0, 1, 2\}$ is a count variable and indicates the number of minor alleles that individual i possesses at location j . Minor alleles are the less frequent genetic variation within a population. As humans inherit one of each chromosome from each parent, they possess either zero, one, or two minor alleles at each location j . C_i is a vector of control variables to filter out spurious correlations due to non-biological differences across population groups. A particular SNP coefficient ψ_j^y is considered genome-wide significant if the null hypothesis of non-association is rejected at a level of $p < 5 \times 10^{-8}$ (Chanock et al., 2007). The p -value is set low to account for multiple hypothesis testing.

The association of single SNPs with y is small, but jointly they can explain a substantial share of observed outcome differences between individuals (Lee et al., 2018). In particular, the estimated SNP coefficients can be used to construct polygenic indexes (PGI). PGIs are scalar measures of an individual's genetic predisposition for an outcome y relative to the population. Formally, PGI_i^y are constructed as a linear aggregation of all SNP_{ij} using the GWAS coefficients $\hat{\psi}_j^y$ as weights:

$$\text{PGI}_i^y = \sum_j \hat{\psi}_j^y \text{SNP}_{ij}. \quad (2)$$

To avoid overfitting, equation (1) is estimated in a discovery sample, whereas the PGI_i^y is constructed in a hold-out sample (Wray et al., 2014).

PGIs are now available for a variety of outcomes. These include, for example, the body mass index and height (Yengo et al., 2018), attention deficit hyperactivity disorder (Demontis et al., 2019), major depressive disorder (Howard et al., 2019), intelligence (Savage et al., 2018), smoking (Liu et al., 2019), and sleep duration (Jansen et al., 2019). For our analysis, we rely on the PGI for educational attainment from Lee et al. (2018) which we denote by PGI^{EA} . It is based on information from 1.1 million individuals and explains around 12.7% of the variance in educational attainment in the United States.

The interpretation of PGIs is not trivial. First, PGIs are not pure measures of biological influ-

⁶Human genetic information is stored in 23 chromosome pairs that consist of deoxyribonucleic acid (DNA) molecules. These chromosomes, in turn, contain 20,000 to 25,000 genes—specific DNA sequences that provide instructions for building proteins. More than 99% of the sequences are identical in all humans.

ence. For example, GWAS coefficients may capture environmental factors such as population stratification across geographic regions (Abdellaoui et al., 2019). To address this concern, we follow standard practice in the literature and always control for the first 20 principal components of the genetic data in our empirical analysis. Second, PGIs are noisy measures of genetic endowments.⁷ For example, GWAS coefficients are estimated in finite samples leading to measurement errors in the PGI weights. Furthermore, the explanatory power of PGIs depends on the context of their application. If a PGI is applied in one context, while the underlying GWAS was estimated in another context, the predictive power of the PGI will be attenuated.⁸ To address the concern of attenuation bias due to measurement error, we use the procedure of Becker et al. (2021) and provide robustness checks based on measurement-error-corrected estimates for the coefficients of PGI^{EA} and the gene-environment interaction. Reassuringly, these analyses suggest that our conclusions about the relative strength of gene-environment interactions are not affected by measurement error in PGI^{EA} .

3 EMPIRICAL STRATEGY

3.1 Empirical Model

Consider a model in which the skills θ of child i at age a are determined by prior skill levels θ_{ia-1} , family inputs I_{ia}^F , school inputs I_{ia}^S , and genetic endowments G_i . There are three phases of skill accumulation:

$$\theta_{ia} = \begin{cases} f_a(G_i) & , \text{ for child age } a = -1, \\ f_a(I_{ia}^F, \theta_{ia-1}, G_i) & , \text{ for child age } a = 0, \dots, 5, \\ f_a(I_{ia}^S, I_{ia}^F, \theta_{ia-1}, G_i) & , \text{ for child age } a = 6, \dots, A. \end{cases} \quad (3)$$

Skills at conception are determined by genetic endowments only. For children aged $a = 0, \dots, 5$, i.e., in the period after conception and before attending school, families are the only source of investments in children. Family inputs may include health behaviors during pregnancy, monetary investments such as buying books, and time investments such as reading to the child. For $a = 6, \dots, A$, schools are an additional source of investments in children. School-based inputs

⁷The heritability of educational attainment is estimated to be 40% whereas PGI^{EA} explains 12.7% of the variance in educational attainment (Branigan et al., 2013).

⁸For example, educational attainment in a country without compulsory schooling likely correlates with a different set of genetic endowments than in a country with high-quality compulsory schools. In our study, this concern is limited: we apply PGI^{EA} to a sample from the United States, while the underlying GWAS predominantly draws on samples from the United States and other industrialized countries with comparable education systems (Lee et al., 2018).

may include the quality of instruction by teachers and school policies to regulate behavior.

Completed education Y is a function of individual skills accumulated at age $a = A$:

$$Y_i = m(\theta_{iA}). \quad (4)$$

By recursively substituting equations (3) and (4) across ages $a = 1, \dots, A$, we obtain a model in which educational attainment is determined by initial genetic endowments, the full history of family inputs, and the full history of school inputs:

$$Y_i = h(I_{iA}^S, \dots, I_{i6}^S, I_{iA}^F, \dots, I_{i1}^F, G_i). \quad (5)$$

We are interested in the complementarity of school inputs and genetic endowments at a particular child age a :

$$\kappa = \frac{\partial^2 h(I_{ia}^S, \overline{I_{ia-1}^S}, \dots, \overline{I_{i6}^S}, \overline{I_{ia}^F}, \overline{I_{ia-1}^F}, \dots, \overline{I_{i1}^F}, G_i)}{\partial I_{ia}^S \partial G_i}. \quad (6)$$

If $\kappa < 0$, genetic endowments and school inputs at age a are *substitutes* in the production of educational attainment, i.e., high-quality schools are more productive for individuals with relatively disadvantageous genetic endowments. Reversely, if $\kappa > 0$, genetic endowments and school inputs at age a are *complements* in the production of educational attainment, i.e., high-quality schools are more productive for individuals with relatively advantageous genetic endowments.

In this study, we measure genetic endowments G using the polygenic score PGI^{EA} . Furthermore, we focus on school inputs I_{ia}^S during adolescence ($14 \leq a \leq 18$) by constructing a summary index Q^S that proxies for the quality of high schools attended by individuals. Then, we estimate κ using a linear regression model with an interaction term:

$$Y_i = \alpha PGI_i^{EA} + \beta Q_i^S + \kappa (PGI_i^{EA} \times Q_i^S) + \mathbf{X}_i(a)\gamma + \epsilon_i, \quad (7)$$

where $\mathbf{X}_i(a)$ denotes a vector of control variables to condition on the history of family and school inputs up to age $a = 14$.

3.2 Conditions for Identification

The parameter of interest κ is identified if the following conditions are met: (i) exogenous variation in PGI^{EA} , (ii) exogenous variation in Q^S , and (iii) independent variation in PGI^{EA} and Q^S (Almond and Mazumder, 2013; Biroli et al., 2022; Johnson and Jackson, 2019; Nicoletti and

Rabe, 2014). In the following, we discuss each of these conditions, potential threats to their satisfaction, and how we address these threats in this paper.

(i) Exogenous variation in PGI^{EA} . Genetic endowments are not exogenous to family characteristics as the genetic endowments of children are drawn from the genetic pool of their biological parents. As a consequence, PGI^{EA} is a function of maternal and paternal genetic endowments that may correlate with family inputs $I_{i1}^F, \dots, I_{ia}^F$. Hence, when estimating equation (7), α and κ may be confounded by *genetic nurture effects* (Kong et al., 2018).⁹ In particular, there may be a positive correlation between advantageous genetic endowments and family environments in which children receive more investments. Genetic nurture can be controlled by estimating a sibling fixed effects model that relies on within-family variation in PGI^{EA} only (Houmark et al., 2020; Kweon et al., 2020; Selzam et al., 2019); in a non-transmitted genes design, in which both maternal and paternal genetic endowments are included in control vector $\mathbf{X}_i(a)$; or in an adoption design, in which offspring are biologically unrelated to their parents.¹⁰ All approaches, however, are very data-demanding. For example, the sibling design requires a large sample of biological siblings with sequenced DNA data to construct PGI^{EA} . Therefore, it can only be applied to a limited set of existing data sets.

In this study, we estimate a between-family model using an extensive set of pre-determined family background characteristics to control for genetic nurture effects. This approach is standard in the literature and aims to approximate condition (i) while maximizing statistical power to estimate the gene-environment interaction κ (Domingue et al., 2020). We formally assess the residual potential for confounding through genetic nurture effects by comparing the estimates of α from the between-family model to a sibling fixed effects model that we estimate on a subset of our data ($N = 677$, Table 2). Reassuringly, both point estimates are precisely estimated; yet we cannot reject the null hypothesis that they are equal ($p = 0.863$). This result suggests that after conditioning on $\mathbf{X}_i(a)$, residual genetic nurture is close to zero and unlikely to overturn our main findings.

(ii) Exogenous variation in $\mathbf{Q}^{\text{S}}$. School quality is not exogenous to family characteristics as parents choose schools for their children (Altonji et al., 2005; Beuermann et al., 2022; Rothstein, 2006). As a consequence, $\mathbf{Q}^{\text{S}}$ is a function of family and child characteristics that may correlate with family inputs $I_{i1}^F, \dots, I_{ia}^F$. Hence, in estimation model (7), β and κ may be confounded by *selection effects* (Altonji et al., 2005; Altonji and Mansfield, 2018). In particular, there may be a positive correlation between school quality and family environments in which children receive more investments. Sorting into schools can be controlled in quasi-experimental settings,

⁹A related concept is passive gene-environment correlation (Plomin et al., 1977). In contrast to genetic nurture, passive gene-environment correlation is a narrower concept that focuses only on the intentional reaction of the environment to the genetic endowments of individuals. We will only refer to genetic nurture in the following.

¹⁰See Demange et al. (2020) for a detailed comparison of all three approaches.

e.g., by using variation from admission lotteries (Angrist et al., 2016; Cullen et al., 2006), or the geographic design of catchment areas (Laliberté, 2021). Existing data sets that avail such variation, however, do not contain sequenced DNA data which is needed to measure genetic endowments at the individual level.

In this study, we use an extensive set of pre-determined family background characteristics to account for selection into schools. Furthermore, we apply the control function approach proposed by Altonji and Mansfield (2018). In their paper, Altonji and Mansfield (2018) consider settings where individuals sort into treatments, e.g., when families move to neighborhoods based on school quality. They show that group-level averages of observable characteristics are correlated with unobservable characteristics. Based on this insight they suggest that controlling for a limited number of group-level averages may remove all cross-group variation in both observable and unobservable characteristics. Agrawal et al. (2019) extend this result to settings where the variable of interest is an interaction between group-level factors, such as Q^S , and observed individual characteristics, such as PGI^{EA} . We provide supporting evidence for their conjecture in our setting by showing that school-level averages for only five student characteristics remove all correlation between educational attainment and an extensive set of measures for family background, including parental education and wages (Table 2). In addition, we formally assess the sensitivity of our results to residual confounding by calculating summary statistics for selection on unobservables (Cinelli and Hazlett, 2020; Oster, 2019). Reassuringly, these summary statistics consistently point to a low potential for selection on unobservables (Supplementary Figure S.2). These results suggest that after conditioning on $X_i(a)$, residual confounding due to selection into schools is low and unlikely to overturn our main findings. We further support this assertion in Appendix C. In this appendix, we show analytically and based on simulations that any residual confounding due to positive selection into schools is likely to attenuate our estimate for the gene-environment interaction κ towards zero.

(iii) Independent variation in PGI^{EA} and Q^S . In addition to conditions (i) and (ii), PGI^{EA} and Q^S have to be distributed independently of each other. A strong correlation between PGI^{EA} and Q^S would imply little variation in PGI^{EA} at different levels of Q^S and vice versa. As a consequence, there would be insufficient variation to identify α , β , and κ separately from each other.

To verify that condition (iii) is satisfied, we present empirical evidence that PGI^{EA} and Q^S are indeed distributed independently of each other. This conclusion holds both unconditionally and conditional on $X_i(a)$ (Figure 2 and Supplementary Figure S.3.). We note that this finding does not imply that parents choose schools randomly. To wit, in Supplementary Table S.1 we show that there is a positive gene-environment correlation concerning peer characteristics like socioeconomic background and educational achievement. However, consistent with existing literature, this pattern suggests that parents select schools based on factors like proximity and peer quality but not necessarily based on the school practices that underpin our measures of school quality (Abdulkadiroğlu et al., 2020; Beuermann et al., 2022).

In summary: an ideal design to estimate the complementarity parameter κ would combine a sibling fixed effects model with experimental variation in school characteristics among children of the same biological parents. We are not aware of any data set that simultaneously includes genetic data at the individual level, a large set of siblings, and quasi-experimental within-family variation in school assignments. Therefore, we approximate the ideal-type conditions with the best data available to us. Within this setting, causal identification of α , β , and κ relies on rather strong assumptions. We provide extensive empirical evidence that supports the satisfaction of these assumptions and the validity of our research design. Yet, in the absence of clear-cut quasi-experimental variation, we choose to err on the side of caution and speak of *associations* instead *causal effects* in the remainder of the paper.

4 DATA

We use data from the National Longitudinal Study of Adolescent to Adult Health (Add Health), a 5-wave panel study that focuses on the determinants of health-related behaviors and health outcomes. Add Health is a nationally representative sample of adolescents enrolled in grades 7–12 in 1994/95. Initial information (wave 1, $N = 20,745$) was collected from a stratified sample of 80 high schools across the United States as well as from their associated feeder schools. In addition to in-depth interviews with adolescents, questionnaires were administered to school representatives, parents, and roughly 90,000 students of the sampled schools. Follow-up in-home questionnaires were collected in 1996 (wave 2, $N = 14,738$), 2001/02 (wave 3, $N = 15,179$), and 2008/09 (wave 4, $N = 15,701$). In the most recent wave (2016/18, $N = 12,300$), Add Health respondents are between 33 and 43 years old.

In the following, we describe our main variables of interest. Detailed descriptions of all variables used in our analysis are disclosed in Supplementary Material [D](#).

Outcomes. We measure educational attainment Y_i by the total number of years of education after age 27. In each wave, respondents were asked about their highest level of education at the time of the interview. For each individual, we use the most recent information and transform education levels into years of education, following the mapping suggested by Domingue et al. (2015).¹¹

To analyze the mechanisms behind our headline results, we additionally use a series of measures for (non-)cognitive skills, health, and academic degrees. First, measures for (non-)cognitive

¹¹Numeric values in parentheses: eighth grade or less (8), some high school (10), high school graduate (12), GED (12), some vocational/technical training (13), some community college (14), some college (14), completed vocational/technical training (14), associate or junior college degree (14), completed college (16), some graduate school (17), completed a master’s degree (18), some post-baccalaureate professional education (18), some graduate training beyond a master’s degree (19), completed post-baccalaureate professional education (19), completed a doctoral degree (20).

skills and health serve as proxy variables for θ_{iA} and allow us to analyze the dimensions of skill development that drive the main findings on educational attainment. We proxy cognitive skills using the Picture Vocabulary Test (PVT), a test of receptive hearing vocabulary that is a widely-used measure of verbal ability and scholastic aptitude. We proxy non-cognitive skills by self-reported measures of general risk aversion and patience (Falk et al., 2018) and self-reported information on the Big Five personality traits (Almlund et al., 2011). Regarding health, we use quality-adjusted life years (QALY), which we derive from self-assessed health measures as well as a summary index of diagnosed health conditions. Second, academic degrees allow us to investigate at which educational stage our results emerge. We focus on whether respondents finished high school, obtained a college degree, or completed a post-graduate degree.

Genetic endowments. Add Health obtained saliva samples from consenting participants in wave 4. After quality control procedures, genotyped data is available for 9,974 individuals and 609,130 SNPs. Add Health uses this data to calculate different PGIs using summary statistics from existing GWAS. We use a PGI for educational attainment, referred to as PGI^{EA} , that is based on the GWAS by Lee et al. (2018).¹²

Lee et al. (2018) perform a meta-analysis of 71 quality-controlled cohort-level GWAS. Their meta-analysis produced association statistics for around 10 million SNPs, of which 1,271 reached genome-wide significance. Genes near these genome-wide significant SNPs are relevant for the central nervous system, and many of them encode proteins that carry out neurophysiological functions such as neurotransmitter secretion or synaptic plasticity. They are relevant for brain-development processes before and after birth.

PGI^{EA} is highly predictive of educational attainment and has been widely used in existing studies. Lee et al. (2018) suggest that PGI^{EA} is a better predictor for years of education than household income. Including the score in a regression of years of education on a set of controls yields an incremental R^2 of 0.127 in the Add Health sample. Among other uses, PGI^{EA} has been used to study the formation of early childhood skills (Belsky et al., 2016; Houmark et al., 2020), educational attainment (Domingue et al., 2015), earnings (Papageorge and Thom, 2020), wealth accumulation (Barth et al., 2020), and social mobility (Belsky et al., 2018).

School quality. In waves 1 and 2, Add Health administered detailed questionnaires to headmasters of Add Health schools. The schools are also linked to administrative data from the Common Core of Data (CCD) and the Private School Survey (PSS). We use these sources to construct an indicator for Q^S using a principal component analysis that includes the following

¹²Lee et al. (2018) construct PGI^{EA} for two prediction cohorts, Add Health and the Health and Retirement Study (HRS). PGI^{EA} is based on results from the meta-analysis in which these two cohorts were excluded from the discovery sample. PGI^{EA} was generated from HapMap3 SNPs using the software LDpred—a Bayesian method that weights each SNP by the posterior mean of its conditional effect given other SNPs.

school-level inputs (component weights in parenthesis): (i) the share of teachers with a master’s degree (+0.63), (ii) the share of teachers with school-specific tenure of more than five years (+0.62), (iii) the share of new teachers in the current school year (−0.36), and (iv) the average class size (−0.31). As an alternative to the principal component analysis, we also construct an indicator for Q^S using the aggregation method proposed by Anderson (2008) and Kling et al. (2007).

We focus on school inputs for which there is strong evidence in the existing literature on how they enter the education production function. Clotfelter et al. (2010) and Jacob et al. (2018) demonstrate that academic credentials, which we proxy by the share of teachers with a master’s degree, are positively associated with teacher effectiveness. Papay and Kraft (2015) and Rockoff (2004) show that teaching experience, which we proxy by the share of teachers with more than five years of tenure, correlates with teacher performance and student achievement. Hanushek et al. (2016), Hill and Jones (2018), Hwang et al. (2021), and Ronfeldt et al. (2013) demonstrate that a high teacher turnover, which we proxy by the share of new teachers, impairs teaching quality and student achievement. Finally, the large literature on class size reductions (Angrist and Lavy, 1999; Angrist et al., 2019; Chetty et al., 2011; Fredriksson et al., 2013; Krueger, 1999; Leuven and Løkken, 2020) finds either positive or zero associations with student achievement.

Control variables. Add Health provides extensive information on the environments to which respondents were exposed during childhood and in school. We approximate the identification prerequisites discussed in section 3 by choosing a vector $X_i(a)$ that includes an extensive set of predetermined variables for child and family characteristics as well as school-level averages of observable characteristics to implement the control function approach by Altonji and Mansfield (2018).

We control for child characteristics by including variables for child age at the time of the survey (in months), biological sex, and their interaction as well as an indicator for firstborns. Furthermore, we follow standard practice in the literature and account for population stratification in genetic endowments by including the first 20 principal components of the full matrix of genetic data.

We control for family background characteristics by including maternal and paternal education (in years), the family’s religious affiliation (Christian/non-Christian), parental birthplace (US/non-US), and maternal age at birth (in years). Furthermore, we include the mean and standard deviation of potential wages for both mothers and fathers across child age 0–14.¹³ Furthermore, all estimations include a vector of state-fixed effects. Note that we focus on predeter-

¹³Note that Add Health contains information on actual income. However, actual income may be a bad control as it reflects parental responses to both PGI^{EA} and Q^S . Therefore, we follow the procedure of Shenhav (2021) and combine data from the 1970 Census and the March Current Population Survey (1975–2000) to construct potential wages for gender/education/census region/race/ethnicity cells and match these potential wages to parents of children aged $a = 1, \dots, 14$.

mined variables—variables that are fixed before the period of observation—to avoid smearing through "bad controls" (Angrist and Pischke, 2009). However, in robustness analyses, we expand the vector of controls to include potentially endogenous parental investments and family income. Our results remain unaffected.

To further account for selection into schools, we include a control function based on group characteristics that are observable at the school level. The in-school questionnaires in wave 1 of Add Health collect information about approximately 90,000 students in Add Health schools. We use these data to construct five variables for the control function: the share of white peers, the share of peers with single mothers, the average years of education of peers' mothers, and the average GPAs in English and Math (standardized by grade and federal state). To avoid mechanical relationships, all school-level averages are calculated excluding the respondent's school cohort.

Analysis sample. We apply the following sample selection criteria. First, we restrict our sample to genotyped respondents of European descent. This is common practice in the literature because GWAS are predominantly conducted on this ancestry group. As a consequence, there is a lack of statistical power to account for population stratification between ancestry groups and estimates of genetic influence would be biased without this restriction (Martin et al., 2017; Ware et al., 2017).

Second, our main subsample comprises individuals who visited an Add Health high school or an associated feeder school in wave 1. However, for a subset of individuals from the feeder schools, we have no information about whether they transferred from their feeder schools to the designated Add Health school. In a robustness check presented in Supplementary Table S.2, we drop these individuals from the sample. In another robustness check, we additionally exclude respondents for whom we do not have information on whether they graduated from an Add Health high school. We note that neither excluding potential movers before high school nor potential movers during high school overturns our main conclusions.

Third, we drop all observations with missing information in Y_i , G_i , Q^S , and $X_i(a)$ by list-wise deletion.

Applying these restrictions, we obtain a sample of 4,036 individuals from 72 high schools across the United States, for which we provide summary statistics in Table 1. 54% are female, and the average age measured at wave 1 equals ≈ 16 years (192 months). The average educational attainment in our sample is 14.7 years which exceeds the average educational attainment in the parental generation by ≈ 1.1 years. 96% graduate from high school, which is not surprising given that our sample is restricted to individuals of European descent who attended an Add Health high school or an associated feeder school in wave 1. The 2-year college completion rate equals $\approx 50\%$.

TABLE 1 – Summary Statistics

N=4,036; Siblings=677; High Schools=72				
	Mean	SD	Min	Max
Educational Attainment				
Years of Education	14.68	2.27	8.00	20.00
High School Degree	0.96	0.20	0.00	1.00
2-year College Degree	0.50	0.50	0.00	1.00
4-year College Degree	0.39	0.49	0.00	1.00
Post-Graduate Degree	0.14	0.35	0.00	1.00
Child and Family Characteristics				
PGI ^{EA}	0.00	1.00	-4.18	3.40
Female	0.54	0.50	0.00	1.00
Firstborn	0.48	0.50	0.00	1.00
Age in Months (Wave 1)	192.41	19.62	144.00	256.00
Maternal Age at Birth	25.33	4.84	16.00	46.08
Christian	0.82	0.38	0.00	1.00
Education Mother (in Years)	13.54	2.48	0.00	19.00
Education Father (in Years)	13.56	2.68	0.00	19.00
Foreign-born Mother	0.03	0.17	0.00	1.00
Foreign-born Father	0.03	0.16	0.00	1.00
Potential Wage/Hour Mother	12.57	1.39	9.40	14.27
Potential Wage/Hour Father	15.40	1.32	11.14	17.11
School Quality Indicators				
Q ^S	0.00	1.00	-2.79	1.82
Teacher w/ MA (%)	51.20	24.10	0.00	95.00
Experienced Teacher (%)	66.66	23.43	0.00	98.00
New Teacher (%)	7.87	7.28	0.00	47.00
Class Size	24.40	4.50	12.00	38.00

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows summary statistics for the core analysis sample. The sample is restricted to genotyped individuals who (i) are of European descent, and (ii) attended an Add Health high school or an associated feeder school in wave 1. Observations with missing information in any of the displayed variables are dropped by list-wise deletion.

To assess sample representativeness, we compare our analysis sample to the 1974–1983 birth cohorts of non-Hispanic Whites in the American Community Survey (ACS) and the Current Population Survey (CPS) (Supplementary Table S.3). This comparison shows a slight over-representation of females and children of young mothers in our sample. Otherwise, our sample is comparable to the corresponding groups in the ACS and CPS. In robustness analyses, we re-weight our analysis sample to match the ACS and CPS concerning gender composition, educational attainment of parents, and the age of mothers at birth. Our results remain unaffected (Supplementary Table S.2).

5 RESULTS

We present our results in four steps. In section 5.1, we discuss the association of educational attainment, genetic endowments, and school quality in light of the identifying assumptions discussed in section 3. In section 5.2, we present our estimates for the gene-environment interaction κ . After a robustness analysis in section 5.3, we conclude with an analysis of mechanisms in section 5.4. In all analyses, we standardize PGI^{EA} , Q^{S} , and the variables in $\mathbf{X}_i(a)$ so that they have a mean of zero ($\mu = 0$) and a standard deviation of one ($\sigma = 1$).

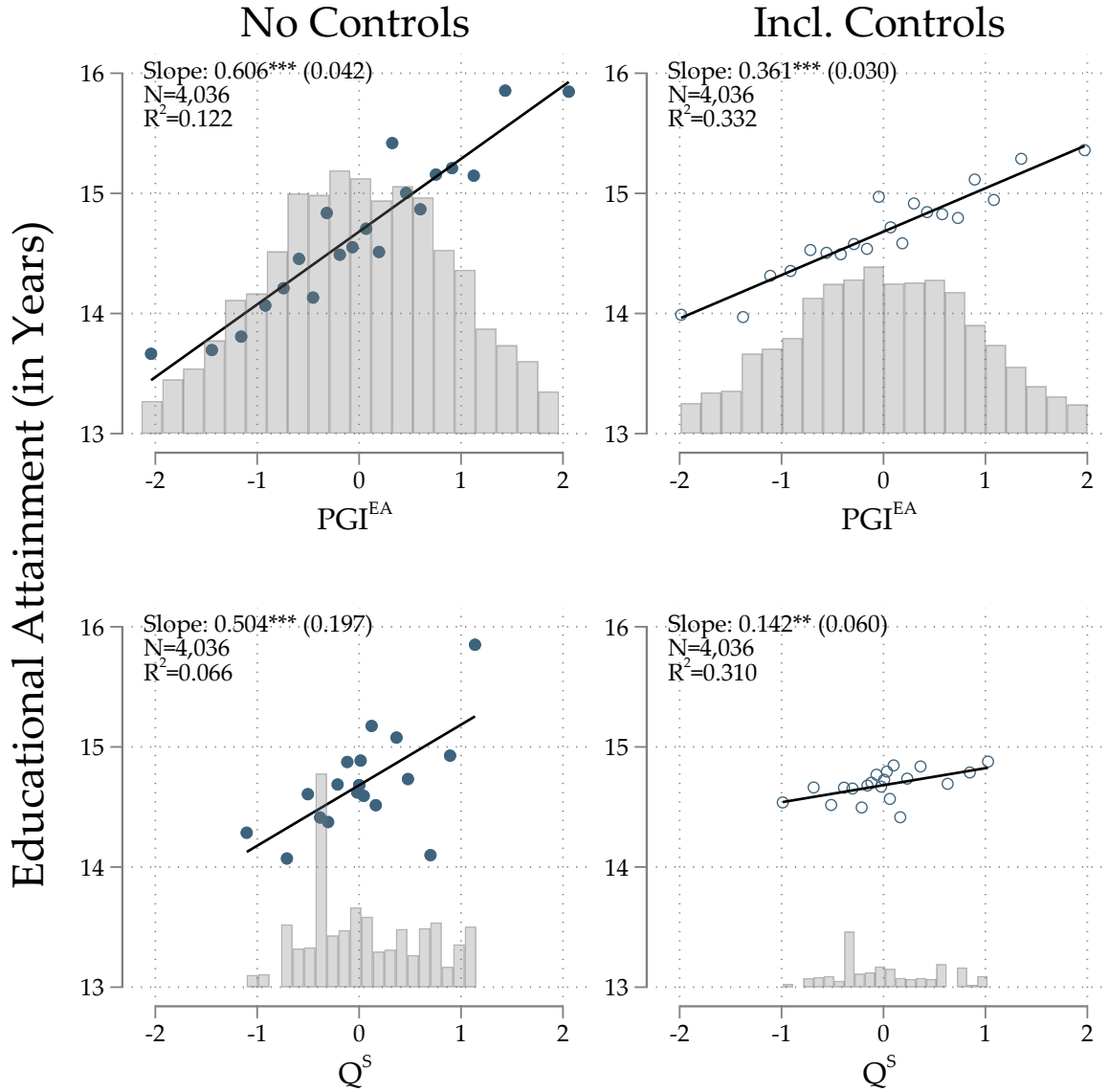
5.1 *The Association of Educational Attainment with Genetic Endowments and School Quality*

Figure 1 displays the association of educational attainment with our measures for genetic endowments PGI^{EA} and school quality Q^{S} . In the left column, we show raw correlations that do not account for the control variables $\mathbf{X}_i(a)$. In the right column, we show associations conditional on $\mathbf{X}_i(a)$.

First, PGI^{EA} is highly predictive of educational attainment. Without controls, an increase in PGI^{EA} by one standard deviation (1 SD) is associated with an increase in educational attainment by 0.606 years. This association does not have a causal interpretation as it may be confounded by genetic nurture effects. When we control for pre-determined child and family characteristics as well as the control function, a 1 SD increase in PGI^{EA} is associated with an increase in educational attainment by 0.361 years. This decrease is consistent with sibling studies showing that genetic nurture effects usually account for 40–50% of the raw association between PGI^{EA} and educational attainment (Kweon et al., 2020; Muslimova et al., 2020; Ronda et al., 2022; Selzam et al., 2019).

Is $\mathbf{X}_i(a)$ sufficient to control for genetic nurture effects? We test whether there remains confounding due to genetic nurture by comparing estimates of the between-family model with a sibling fixed effects model that we estimate on a subsample of our data ($N = 677$). The within-family comparison allows us to perfectly control for genetic nurture effects. Therefore, a strong divergence of between- and within-family estimates would suggest that there is residual genetic nurture that is not picked up by $\mathbf{X}_i(a)$. In Table 2 we show that this is not the case. In the sibling sample, the between-family estimate yields a point estimate of 0.419 after controlling for $\mathbf{X}_i(a)$. The within-family comparison yields a point estimate of 0.445. Both point estimates are precisely estimated and very close to each other. We cannot reject the null hypothesis of their equality at conventional levels of statistical significance ($p = 0.863$). This result suggests that after conditioning on $\mathbf{X}_i(a)$, residual genetic nurture is very low, lending credence to our research design.

FIGURE 1 – Association of Educational Attainment with PGI^{EA} and Q^S



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows the correlation of completed years of education with PGI^{EA} and Q^S , respectively. We bin scatterplots using 20 quantiles of the variable of interest. Gray bars indicate density distributions of the (residualized) variable of interest. Black lines are fitted from linear regressions of educational attainment on the variable of interest. In the left-column, we only control for state fixed effects. In the right column, we introduce the full set of control variables. *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the school level.

Second, Q^S is highly predictive of educational attainment. Without controls, a 1 SD increase in Q^S is associated with an increase in educational attainment by 0.504 years. This association does not have a causal interpretation as it may be confounded by selection effects due to sorting into schools. When we control for pre-determined child and family characteristics as

TABLE 2 – Testing for Genetic Nurture and Selection into Schools

	Years of Education: Between- vs. Within-Family		Predicted Years of Education: w/o vs. w/ Control Function	
	(1)	(2)	(3)	(4)
PGI ^{EA}	0.419*** (0.086)	0.445*** (0.142)	–	–
Q ^S	–	–	0.263** (0.129)	0.005 (0.055)
Equality of coefficients (<i>p</i> -value)	0.863		0.024	
Child Controls	✓	×	×	×
Family Controls	✓	×	×	×
Control Function	✓	×	×	✓
Sibling Fixed Effect	×	✓	×	×
N	677	677	4,036	4,036
R ²	0.416	0.760	0.084	0.181
Outcome Mean	14.722	14.722	14.681	14.681
Outcome SD	2.277	2.277	1.163	1.163

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows the associations of PGI^{EA} and Q^S with years of education (left panel) and predicted years of education (right panel). The left panel shows estimates in the sibling sample: Column (1) displays results from a between-family comparison. Column (2) displays results from a within-family comparison. The right panel shows estimates in the core analysis sample: Column (3) displays results without any controls. Column (4) displays results including the control function variables. Predicted education is calculated from a regression of completed years of education on all *Child Controls* and *Family Controls*. *Child Controls*: Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls*: Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function*: Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. All right-hand side variables are standardized on the estimation sample ($\mu = 0$, $\sigma = 1$). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors (in parentheses) are clustered at the school level.

well as the control function, a 1 SD increase in Q^S is associated with an increase in educational attainment of 0.142 years. This decrease reflects positive selection into schools—a pattern that has been thoroughly documented in the existing literature for the United States (Deming et al., 2014; Rothstein, 2006). Nevertheless, even when accounting for selection, the association of Q^S and educational attainment remains strong and positive. This result is consistent with prior literature showing positive effects of high school quality on students’ educational success (Angrist et al., 2019; Deming, 2014; Hanushek and Rivkin, 2010; Jackson et al., 2020).

Is $X_i(a)$ sufficient to control for selection into schools? To control for sorting into schools, we follow the control function approach suggested in Altonji and Mansfield (2018) and include school-level averages of observed individual characteristics into $X_i(a)$. We test the effectiveness of this control function approach as follows: First, we replace educational attainment with

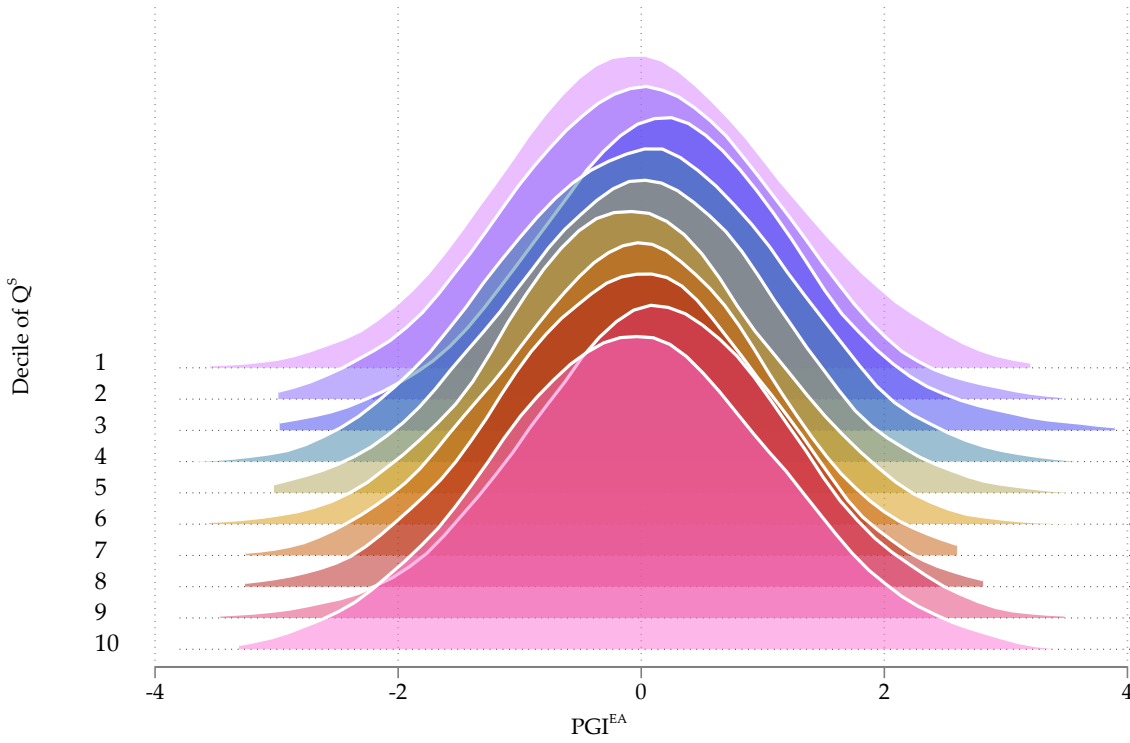
predicted educational attainment. In particular, we predict years of education using the full set of pre-determined child and family characteristics included in $\mathbf{X}_i(a)$. These characteristics are highly predictive of years of education ($R^2 = 0.297$) and other measures of educational attainment and skills (Supplementary Figure S.4). Second, we assess the correlation between predicted educational attainment and Q^S before and after conditioning on the variables of the control function. If the control function approach is effective, any positive correlation between predicted educational attainment and Q^S should be attenuated to zero after conditioning on the variables of the control function. In Table 2, we show that the unconditional association between predicted educational attainment and Q^S is 0.263 indicating significant positive selection into schools. Conditional on the control function, this correlation shrinks to 0.005 and becomes insignificant. This result suggests that the control function approach indeed is effective in controlling for selection into schools. Therefore, conditional on $\mathbf{X}_i(a)$, residual sorting into schools is likely to be low, lending further credence to our research design.

Alternatively, we can assess potential confounding through remaining selection effects by assuming that changes in the coefficient of Q^S due to the introduction of $\mathbf{X}_i(a)$ provide information about the extent of confounding due to unobservables (Altonji et al., 2005; Cinelli and Hazlett, 2020; Oster, 2019). We follow Cinelli and Hazlett (2020) and assess what association unobserved confounders would need to have with both the variable of interest (Q^S) and the outcome of interest (educational attainment) to change our conclusions. In Supplementary Figure S.2, we show that Q^S would remain positive and statistically significant at the 5%-level even if the partial R^2 of unobserved confounders with Q^S and educational attainment were more than ten times higher than the corresponding partial R^2 of paternal education with these variables. Given the decisive role of parental education in school choices, and its strong predictive power for educational outcomes of children, these results lend further confidence that our results are genuine and not a mere reflection of selection into schools based on family background.

In addition to genetic nurture effects and selection effects, a high correlation between PGI^{EA} and Q^S would pose another threat to the identification of the gene-environment interaction. Such a correlation could arise if children were sorted into schools based on their genetic endowments. Figure 2 shows that this concern does not apply in our setting. In this figure, we plot unconditional PGI^{EA} distributions by decile of Q^S . Visual inspection suggests that PGI^{EA} distributions are almost congruent to each other within each decile of the school quality indicator. This conclusion also holds after residualizing PGI^{EA} and Q^S from $\mathbf{X}_i(a)$ (Supplementary Figure S.3). More formally, we compute two-sample Kolmogorov-Smirnov tests for the equality of PGI^{EA} distributions within the deciles of Q^S . Out of 45 pairwise comparisons, only 2 differences are significant at the 10% level. This result is expected by chance and hence we conclude that PGI^{EA} and Q^S are indeed independently distributed.¹⁴

¹⁴See also Supplementary Table S.1, where we show that PGI^{EA} is correlated with the socioeconomic composition of schools. To the contrary, we cannot reject the null hypothesis of zero correlation between PGI^{EA} and Q^S .

FIGURE 2 – Distribution of PGI^{EA} by Q^S



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows unconditional PGI^{EA} distribution by deciles of Q^S . Density distributions are smoothed using the Epanechnikov kernel function with a bandwidth of 0.5.

5.2 The Interplay of Genetic Endowments and School Quality

Table 3 shows our baseline estimates for the interaction of genetic endowments and school quality. In all regressions, we include the vector $X_i(a)$ to control for genetic nurture and selection into schools. As noted previously, $X_i(a)$ comprises an extensive set of pre-determined child and family characteristics as well as the control function using school-level averages of observed individual characteristics (see section 4).

The point estimates in column (1) replicate the findings from Figure 1 and show a strong and positive association of PGI^{EA} and Q^S with educational attainment. A 1 SD increase in PGI^{EA} (Q^S) increases educational attainment by ≈ 0.36 (≈ 0.13) years. The coefficient of the interaction $PGI^{EA} \times Q^S$ is our estimate of κ . The negative interaction coefficient indicates that genetic endowments and school quality are *substitutes* in the production of educational attainment: a 1 SD increase in school quality reduces the positive association of educational attainment with PGI^{EA} by $\approx 17\%$ ($= 0.062/0.360$).

In column (2), we replicate the analysis for an alternative way of constructing the school quality indicator. Instead of using weights from principal component analysis, we aggregate dimen-

TABLE 3 – Association of Years of Education with PGI^{EA} and School Environments

Outcome: Years of Education	Overall		Decomposition of Q^S			
	PCA (1)	Anderson (2008) (2)	(3)	(4)	(5)	(6)
PGI^{EA}	0.360*** (0.028)	0.361*** (0.029)	0.360*** (0.027)	0.361*** (0.029)	0.361*** (0.030)	0.361*** (0.030)
Q^S	0.128** (0.054)	0.095** (0.047)	–	–	–	–
$PGI^{EA} \times Q^S$	-0.062** (0.026)	-0.059** (0.027)	–	–	–	–
Teacher w/ MA	–	–	0.169*** (0.065)	–	–	–
$PGI^{EA} \times$ Teacher w/ MA	–	–	-0.065** (0.026)	–	–	–
Exp. Teacher	–	–	–	0.083* (0.049)	–	–
$PGI^{EA} \times$ Exp. Teacher	–	–	–	-0.043* (0.026)	–	–
New Teacher	–	–	–	–	-0.001 (0.045)	–
$PGI^{EA} \times$ New Teacher	–	–	–	–	0.036 (0.029)	–
Class Size	–	–	–	–	–	-0.004 (0.038)
$PGI^{EA} \times$ Class Size	–	–	–	–	–	-0.006 (0.030)
Child Controls	✓	✓	✓	✓	✓	✓
Family Controls	✓	✓	✓	✓	✓	✓
Control Function	✓	✓	✓	✓	✓	✓
N	4,036	4,036	4,036	4,036	4,036	4,036
R^2	0.334	0.334	0.334	0.333	0.333	0.332
Outcome Mean	14.681	14.681	14.681	14.681	14.681	14.681
Outcome SD	2.268	2.268	2.268	2.268	2.268	2.268

Data: National Longitudinal Study of Adolescent to Adult Health.

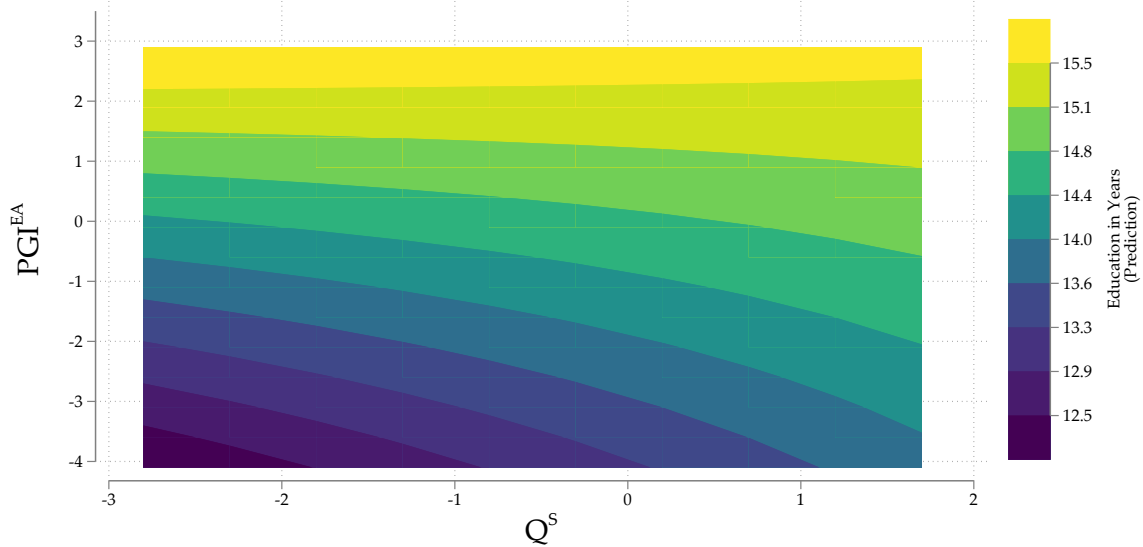
Note: Own calculations. This table shows the joint association of PGI^{EA} and Q^S with completed years of education. *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. All right-hand side variables are standardized on the estimation sample ($\mu = 0, \sigma = 1$). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors (in parentheses) are clustered at the school level.

sions using the method of Anderson (2008) and Kling et al. (2007). Our results are not sensitive to this alternation: a 1 SD increase in school quality reduces the positive association of educational attainment with PGI^{EA} by $\approx 16\%$ ($= 0.059/0.361$).

In columns (3)-(6), we decompose the overall index Q^S into its underlying components. Columns (2) and (3) show that school-specific teacher tenure and teacher education have positive average effects on educational attainment. These results are consistent with previous literature showing that teaching experience and teacher education tend to have positive effects on student learning (Hill and Jones, 2018; Hwang et al., 2021; Jackson and Bruegmann, 2009; Rockoff, 2004). Furthermore, there is evidence for substitutability, i.e., the positive impact of more experienced and more educated teachers is larger for students in the lower parts of the PGI^{EA} distribution. To the contrary, Columns (4) and (5) show that class size and the percentage of new teachers neither increase average educational attainment nor is there evidence for heterogeneity across the PGI^{EA} distribution. Overall, the decomposition suggests that the results for the overall index Q^S are mostly driven by teachers' school-specific tenure and education levels.

In principle, the negative gene-environment interaction shown in Table 3 could be due to low PGI^{EA} students benefiting from high-quality schools, or high PGI^{EA} students losing from high-quality schools. In Figure 3, we provide evidence for the former, but not for the latter. In this figure, we show predictive margins of years of education for different combi-

FIGURE 3 – Association of Years of Education with PGI^{EA} by Q^S



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows predictions of completed years of education by PGI^{EA} and Q^S cell. Predictions are calculated using the model estimated in column (1) of Table 3.

nations of PGI^{EA} and Q^S while controlling for $X_i(a)$. Moving horizontally from left to right at a given PGI^{EA} level, we see that predicted education increases sharply in the lower parts of the PGI^{EA} distribution. On the contrary, in the upper ranges of the PGI^{EA} distribution, predicted education remains unchanged regardless of school quality. This pattern is encouraging

as it suggests that investments in school quality mitigate genetic inequality in educational outcomes without compromising the attainment of genetically advantaged students.¹⁵

It is interesting to contrast this substitutability result with existing evidence on Scarr-Rowe interactions. In this literature, researchers tend to find (weakly) positive gene-environment interaction between PGI^{EA} and parental socio-economic status, i.e., in the family context genetically advantaged children benefit more from environments associated with higher investment into children.¹⁶ To the contrary, we find that sign of interplay between genetic endowments and high-investment environments reverses in the school context, i.e., genetically disadvantaged children benefit more from environments associated with higher investment into children.

5.3 Robustness Analysis

We check the robustness of our results in two steps. First, we investigate whether Q^S picks up the effects of other school characteristics that may correlate with student outcomes. Second, we test whether our estimates of the gene-environment interaction κ are confounded by the heterogeneity of genetic effects in different family environments.

Other school characteristics. In Table 4, we focus on potential confounders at the school-level. For ease of comparison, we replicate our baseline estimates in column (1).

In columns (2)–(4), we address the concern that our index for school quality may be conflated with other school policies and practices that have an impact on educational attainment. To this end, we sequentially introduce indicators for other school policies as well as their interaction with PGI^{EA} . Indicators for other school policies include the average share of retained students across grades 9-11 (column 2), a binary indicator of whether schools group students by English ability (column 3), and an index for the strictness of school sanctions (column 4).¹⁷ Comparing columns (2)–(4) to our baseline estimate, we see that all coefficients of interest are robust to the inclusion of these indicators.

¹⁵One might suspect that this pattern is driven by ceiling effects in educational attainment. To check this potential explanation, we replace educational attainment with the PVT, i.e., an educational outcome that is not artificially censored from above. The pattern is the same (Supplementary Figure S.7).

¹⁶See for example Ronda et al. (2022) and Turkheimer et al. (2003) for positive Scarr-Rowe interactions and Figlio et al. (2017) for a null finding. We are not aware of any studies finding negative effects.

¹⁷The strictness index is based on headmaster questionnaires. Headmasters were asked about the school's policy in the following domains of behavior: cheating, fighting with or injuring another student, alcohol or drug possession, drinking alcohol or using illegal drugs, smoking, verbally or physically abusing a teacher, and stealing school property. Possible measures are (i) no action, (ii) verbal warning, (iii) minor action, (iv) in-school suspension, (v) out-of-school suspension, and (vi) expulsion. Following Anderson (2008) and Kling et al. (2007) we standardize each response dimension to $\mu = 0$ and $\sigma = 1$ and aggregate them linearly to obtain the strictness index. See Supplementary Material D for details.

In columns (5)–(7), we address the concern that our index for school quality may be conflated with the demographic composition of teachers at a school. This possibility arises if teachers sort into particular schools based on student characteristics (Jackson, 2009a). To this end, we sequentially introduce indicators for teacher demographics as well as their interaction with PGI^{EA} . Indicators for teacher demographics include the share of white teachers (column 5), the share of Hispanic teachers (column 6), and the share of female teachers (column 7). Our results remain robust to the inclusion of these variables.

In column (8), we re-estimate the coefficients of interest while accounting for unobserved differences across schools through the introduction of school fixed effects. In this specification, we cannot estimate the effect of Q^{S} on educational attainment. However, it is reassuring that the estimates for PGI^{EA} and the gene-environment interaction are very close to our benchmark estimates.

Overall, these results suggest that our estimates of the gene-environment interaction κ are not confounded by other school environments that are not reflected in our school quality index.

Family environments and behavioral responses. In Table 5 we focus on potential confounders at the family-level. For example, highly educated parents may try to compensate their children for poor school quality by helping with homework, providing additional educational resources, etc. Since parental education correlates with PGI^{EA} , such parental responses could also confound our estimates for the interaction between PGI^{EA} and Q^{S} . We address these concerns in the following.

In column (2), we include additional controls for family investments which we omit from our baseline analysis since they may reflect endogenous family responses to genetic endowments of children and school quality. In particular, we include indexes for breastfeeding, parental time investments, and the log of annual family income in the set of controls.¹⁸ Despite a sizable sample reduction, our results remain unaffected. This result supports our previous conclusion that the set of pre-determined family background characteristics and the control function capture most of the relevant information on family environments that are associated with higher investments in children.

In column (3), we include higher-order polynomials for PGI^{EA} and Q^{S} and allow for all possible interactions between both variables. The expansion to higher-order polynomials is motivated by Biroli et al. (2022) who suggest this procedure to control for heterogeneity in family

¹⁸We collect information on a series of activities that the child has engaged in with either their mother or father over the past four weeks. For both parents, these activities include shopping, playing sports, going to church, talking about dates, going to the movies and similar events, talking about personal problems, talking about school work, working together on school work, and talking about other things at school. Following Anderson (2008) and Kling et al. (2007) we standardize each response dimension to $\mu = 0$ and $\sigma = 1$ and aggregate them linearly to obtain an aggregate index of time investment. See Supplementary Material D for details.

TABLE 4 – Robustness to Additional School Characteristics

	Baseline	+ School Policies			+ Teacher Composition			+ School FE
Outcome: Years of Education	(1)	Retention Policy (2)	Ability Groups (3)	Strict. Index (4)	White Teacher (5)	Hispanic Teacher (6)	Female Teacher (7)	(8)
PGI^{EA}	0.360*** (0.028)	0.360*** (0.028)	0.365*** (0.028)	0.362*** (0.029)	0.360*** (0.028)	0.361*** (0.028)	0.359*** (0.027)	0.352*** (0.029)
Q^S	0.128** (0.054)	0.130** (0.054)	0.119** (0.053)	0.138** (0.056)	0.125** (0.053)	0.127** (0.055)	0.162*** (0.054)	–
$PGI^{EA} \times Q^S$	-0.062** (0.026)	-0.062** (0.025)	-0.061** (0.025)	-0.058** (0.027)	-0.060** (0.026)	-0.062** (0.026)	-0.060** (0.026)	-0.063** (0.027)
School Characteristic	–	0.047 (0.032)	-0.058 (0.054)	0.061* (0.033)	-0.006 (0.065)	0.100*** (0.032)	-0.096*** (0.037)	–
$PGI^{EA} \times$ School Characteristic	–	-0.017 (0.029)	0.030 (0.030)	0.019 (0.024)	-0.021 (0.032)	0.007 (0.022)	0.040 (0.033)	–
Child Controls	✓	✓	✓	✓	✓	✓	✓	✓
Family Controls	✓	✓	✓	✓	✓	✓	✓	✓
Control Function	✓	✓	✓	✓	✓	✓	✓	✓
N	4,036	4,036	3,971	4,036	4,036	4,036	4,036	4,036
R^2	0.334	0.334	0.335	0.334	0.334	0.335	0.335	0.343

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows the joint association of PGI^{EA} and Q^S with completed years of education. We control for additional school characteristics and their interaction with PGI^{EA} . The relevant school characteristics are indicated in the column header. *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. All right-hand side variables are standardized on the estimation sample ($\mu = 0$, $\sigma = 1$). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors (in parentheses) are clustered at the school level.

responses to genetic endowments and school quality. Our results are unaffected by this expansion.

In column (4), we test for potential confounding due to gene-environment interactions with family socio-economic status. In particular, we follow Domingue et al. (2020) and Keller (2014) and extend our estimation model by interacting PGI^{EA} and Q^S with the full control vector $\mathbf{X}_i(a)$. In doing so, we allow for the possibility that family socioeconomic status interacts with both genetic endowments and school quality. In this model, the interpretation for the base coefficients of PGI^{EA} and Q^S changes because the estimated coefficients now reflect effects for the subgroup of individuals that sit at the mean of all control variables in $\mathbf{X}_i(a)$.¹⁹ This change in

¹⁹Effects are estimated at the mean of the control variables since we standardize all variables in $\mathbf{X}_i(a)$ to have a mean of zero and a standard deviation of one.

interpretation explains the lower coefficient for Q^S in comparison to the baseline model. Importantly, however, our estimate for the interaction of genetic endowments and school quality remains unaffected.

TABLE 5 – Robustness to Family Environments and Behavioral Responses

Outcome: Years of Education	Baseline (1)	Endogenous Controls (2)	Higher-Order Polynomials (3)	Full Interaction (4)
PGI^{EA}	0.360*** (0.028)	0.325*** (0.031)	0.386*** (0.036)	0.352*** (0.028)
Q^S	0.128** (0.054)	0.169*** (0.057)	0.142** (0.063)	0.082 (0.063)
$PGI^{EA} \times Q^S$	-0.062** (0.026)	-0.072** (0.031)	-0.069*** (0.025)	-0.065** (0.029)
Child Controls	✓	✓	✓	✓
Family Controls	✓	✓	✓	✓
Control Function	✓	✓	✓	✓
Endogenous Controls	×	✓	×	×
Higher-Order Polynomials	×	×	✓	×
Full Interaction	×	×	×	✓
N	4,036	3,355	4,036	4,036
R^2	0.334	0.348	0.334	0.347

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows the joint association of PGI^{EA} and Q^S with completed years of education. In column (2) we introduce potentially endogenous control variables. Endogenous control variables include an indicator for breastfeeding, an index for maternal time investments, and log family income. In column (3) we control for second-order polynomials of PGI^{EA} and Q^S and allow for all possible interactions of both indicators. In column (4) we control for all possible interactions between PGI^{EA} , Q^S , *Child Controls*, *Family Controls*, and the *Control Function*. *Child Controls*: Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls*: Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function*: Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. All right-hand side variables are standardized on the estimation sample ($\mu = 0, \sigma = 1$). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors (in parentheses) are clustered at the school level.

Overall, these results suggest that our estimates of the gene-environment interaction κ are not confounded by the heterogeneity in genetic effects across family environments.

Further robustness checks. In the Supplementary Material, we show the results of additional robustness analyses. First, we check whether our results are driven by outlier schools. Therefore, we re-run our analysis 72 times, excluding one school from the sample per iteration. Reassuringly, in each iteration, the results are very close to our benchmark estimates (Supple-

mentary Figure S.5). Second, we check whether our findings are driven by ceiling effects in educational attainment. To this end, we re-run our analysis, artificially censoring educational attainment step-wise from above. If ceiling effects were driving our results, we would expect the gene-environment interaction to increase across parts of the censoring interval. However, this is not the case. Instead, the corresponding coefficient decreases monotonically (Supplementary Figure S.6). The absence of ceiling effects is further supported by Supplementary Figure S.7. In this figure, we replicate Figure 3 while replacing educational attainment with the PVT, i.e., an educational outcome that is not artificially censored from above. The figure replicates the data pattern of Figure 3 very well and suggests that our results are indeed driven by relative gains of low PGI^{EA} students. Third, we check whether we pick up the relevant genetic variation. In particular, we sequentially control for six alternative PGIs that target other phenotypes than educational attainment as well as their interaction with Q^{S} . While some PGIs have predictive power over and above PGI^{EA} , our estimates for PGI^{EA} , Q^{S} and the gene-environment interaction κ remain unaffected (Supplementary Table S.4). Fourth, we re-estimate our baseline model while applying the correction method of Becker et al. (2021) to account for measurement error and resulting attenuation bias in PGI^{EA} and the gene-environment interaction. As expected, the measurement-error-corrected estimates for PGI^{EA} and the gene-environment interaction are substantially higher than in our baseline estimates. However, they increase in roughly equal proportions. As a result, our conclusions about the relative strength of the substitutability of PGI^{EA} and Q^{S} remain unaffected. In the measurement-error-corrected version, a 1 SD increase in school quality reduces the positive association of educational attainment with PGI^{EA} by $\approx 14\text{--}15\%$ (Supplementary Table S.5). Lastly, we run a placebo test. That is, we re-estimate our baseline model while permuting the values of Q^{S} 10,000 times and keeping constant all other variables. In Supplementary Figure S.8, we show that the resulting distribution of t -statistics for the interaction of PGI^{EA} and Q^{S} is well-behaved and that roughly 1%, 5%, and 10% of the placebo regressions yield t -statistics at the corresponding critical values. Only slightly more than 1% of the placebo regressions yield a t -statistic more extreme than the one from our baseline estimates, suggesting that our results are not driven by chance.

In summary, this battery of additional checks further supports the conclusion that our main findings are genuine.

5.4 Mechanisms

In this section, we analyze the mechanisms underlying the substitutability of genetic endowments and school quality.

Skill formation. In section 3, we formulated educational attainment Y_i as a function of children's skills θ_i at the end of childhood. The skills that influence educational attainment are multidimensional and comprise a broad set of (non-)cognitive skills and health (Almlund et

al., 2011; Heckman and Mosso, 2014). Furthermore, current literature shows that each of these skill dimensions is shaped, in part, by genetic influence (Buser et al., 2021a; Demange et al., 2020, 2021).

We evaluate these potential channels by analyzing the associations of PGI^{EA} and Q^S with a set of intermediate outcomes. In terms of cognitive skills, we use the Picture Vocabulary Test (PVT) as a measure of verbal intelligence. Furthermore, we focus on personality and preferences as two distinct conceptualizations of non-cognitive skills (Becker et al., 2012; Humphries and Kosse, 2017). In particular, we use measures for self-reported risk aversion and patience, and the Big Five personality traits. In terms of health outcomes, we focus on subjective health, measured by quality-adjusted life years (QALY), and objective health, measured by an index that comprises information on whether the respondent is obese, has first-stage hypertension or has high cholesterol. All measures were collected in waves 3 and 4 of Add Health, that is, after respondents had left high school but potentially before they had completed their highest level of education (see Supplementary Material D for details).

Existing literature shows cognitive skills, risk aversion, patience, and health are strong predictors of educational attainment (Burks et al., 2015; Castillo et al., 2018a,b; Jackson, 2009b). Furthermore, openness and emotional stability—the opposite of neuroticism—are positively associated with educational attainment (Becker et al., 2012; Buser et al., 2021b).²⁰ Based on this evidence, one would expect positive associations of both PGI^{EA} and Q^S with each of these intermediate outcomes. The sign of the gene-environment interaction is a priori unclear. However, given the substitutability of PGI^{EA} and Q^S in the production of educational attainment, we expect similar substitutability patterns for a subset of these intermediate outcomes as well.

Table 6 summarizes the results. In column (1) of Panel (a), we focus on the PVT as a measure of cognitive skills. As expected, our results show positive associations of both PGI^{EA} and Q^S with the PVT. A 1 SD increase in PGI^{EA} (Q^S) is associated with a 0.174 SD (0.123 SD) increase in the PVT. Furthermore, both factors are substitutes for each other. A 1 SD increase in school quality reduces the positive association of PVT and PGI^{EA} by $\approx 28\%$ ($= 0.048/0.174$).

In columns (2)–(3) of Panel (a), we focus on economic preferences. As expected, we find positive associations of PGI^{EA} and Q^S with both risk aversion and patience. 1 SD increases in PGI^{EA} and Q^S are associated with increases in risk aversion by 0.40 SD and 0.68 SD, respectively. The corresponding increases in patience are 0.86 SD and 0.75 SD. Furthermore, PGI^{EA} and Q^S are substitutes for each other. A 1 SD increase in Q^S reduces the positive associations of risk aversion and patience with the PGI^{EA} by $\approx 105\%$ ($= 0.042/0.040$) and $\approx 33\%$ ($= 0.029/0.086$), respectively.

In columns (4)–(5) of Panel (a), we focus on health outcomes. As expected, our results show

²⁰We replicate these findings in our data by showing that each intermediate outcome is highly predictive for educational attainment conditional on our set of controls $X_i(a)$. See Supplementary Figure S.9.

TABLE 6 – Association of Skill Measures with PGI^{EA} by Q^S

<i>Panel (a)</i>	Cognitive	Preferences		Health	
	PVT (1)	Risk (2)	Patience (3)	Subjective (4)	Objective (5)
PGI^{EA}	0.174*** (0.014)	0.040** (0.017)	0.086*** (0.016)	0.077*** (0.016)	0.042*** (0.016)
Q^S	0.123*** (0.045)	0.068** (0.030)	0.075* (0.039)	0.038 (0.040)	-0.013 (0.025)
$PGI^{EA} \times Q^S$	-0.048*** (0.011)	-0.042*** (0.016)	-0.029** (0.012)	-0.036*** (0.013)	-0.000 (0.017)
Child Controls	✓	✓	✓	✓	✓
Family Controls	✓	✓	✓	✓	✓
Control Function	✓	✓	✓	✓	✓
N	3,382	3,467	3,467	3,473	4,036
R ²	0.226	0.111	0.104	0.079	0.053
Personality					
<i>Panel (b)</i>	Open- ness (1)	Conscient- iousness (2)	Extra- version (3)	Agree- ableness (4)	Neuro- ticism (5)
PGI^{EA}	0.075*** (0.014)	-0.008 (0.015)	-0.020 (0.019)	0.042** (0.017)	-0.086*** (0.018)
Q^S	0.005 (0.025)	0.045 (0.031)	-0.061** (0.024)	0.044 (0.029)	-0.037* (0.022)
$PGI^{EA} \times Q^S$	-0.014 (0.011)	-0.006 (0.014)	-0.019 (0.023)	-0.002 (0.014)	0.020 (0.019)
Child Controls	✓	✓	✓	✓	✓
Family Controls	✓	✓	✓	✓	✓
Control Function	✓	✓	✓	✓	✓
N	3,998	4,030	4,029	4,028	4,024
R ²	0.100	0.046	0.035	0.139	0.090

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows the joint association of PGI^{EA} and Q^S with cognitive skills, preferences, health, and personality. *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. All right-hand side variables are standardized on the estimation sample ($\mu = 0$, $\sigma = 1$). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors (in parentheses) are clustered at the school level.

a positive association of PGI^{EA} with both subjective and objective health. A 1 SD increase in PGI^{EA} increases subjective (objective) health by 0.077 SD (0.042 SD). Furthermore, the negative coefficient on the interaction of PGI^{EA} and Q^S suggests that this increase is particularly

pronounced for low PGI^{EA} students: a 1 SD increase in school quality reduces the positive association of subjective health with the PGI^{EA} by $\approx 47\%$ ($= 0.036/0.077$).

In Panel (b), we focus on personality traits. We find positive associations of PGI^{EA} with openness and agreeableness, and a negative association of PGI^{EA} with neuroticism. Q^S is associated with decreases in extraversion and neuroticism. However, we find no evidence of an interaction between PGI^{EA} and Q^S in the production of personality traits.

To summarize: we find negative gene-environment interactions between genetic endowments and school quality in the production of cognitive skills, risk aversion, patience, and subjective health. Given their predictive power for educational attainment, these intermediate outcomes are plausible transmission channels for the substitutability of genetic endowments and school quality in the production of educational attainment.

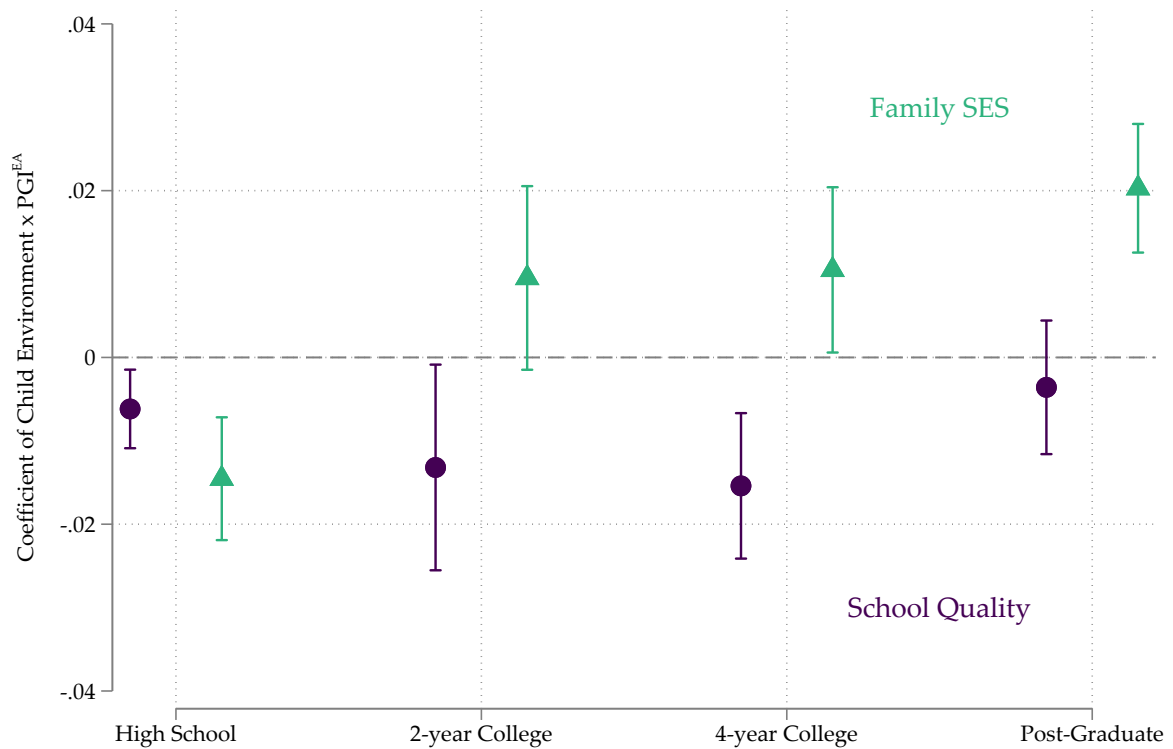
Educational degrees. Total years of education summarizes information from various educational stages, with each stage requiring a different mix of skills θ_i (Cunha et al., 2006, 2010). Therefore, we repeat our analysis by replacing total years of education with binary variables for whether respondents obtained (i) at least a high school degree or GED, (ii) a 2-year college degree, (iii) a 4-year college degree, or (iv) a post-graduate degree.

In Figure 4, we present the resulting point estimates for the gene-environment interaction κ and the associated confidence bands. The circular series suggests that the substitutability of school quality and genetic endowments follows a U-shaped pattern throughout the educational life cycle. For students with low PGI^{EA} , there is a small decrease in the probability of dropping out of high school if they attend high-quality schools, followed by increases in substitutability for 2-year and 4-year college degrees. The substitutability of high-quality schools and genetic endowments levels off at the post-graduate level. This pattern is consistent with the following interpretation: high school graduation is a relatively "inclusive" educational outcome that is accessible to most, including low PGI^{EA} students in low-quality schools. Evidence of this effect is the high school graduation rate of 96% in our sample (Table 1). In contrast, post-graduate education is a relatively "exclusive" educational outcome that is more accessible to students who have advantageous genetic endowments *and* attend high-quality schools. In both cases, there is limited opportunity for high-quality schools to make a difference for low PGI^{EA} students. College education, however, takes a middle ground between these two polar outcomes and therefore offers scope for disadvantageous genetic endowments to be offset by school quality and vice versa. We interpret this pattern as suggestive since the confidence bands are too wide to statistically distinguish among the point estimates for different educational degrees.

The triangular series shows the gene-environment interaction with family socio-economic status (SES) increases over the educational life-cycle of individuals.²¹ This pattern replicates recent

²¹In particular we use the "social origins score" from Belsky et al. (2018) measured in wave 1. Results for alterna-

FIGURE 4 – Association of Education Degrees with PGI^{EA} by Q^S and Family SES



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows point estimates and 90% confidence bands for the interaction of PGI^{EA} with Q^S , the interaction of PGI^{EA} with an indicator for family SES, and their association with education degrees. For each outcome, coefficients are estimated jointly following the specification of equation (7). *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. Standard errors are clustered at the school level.

evidence from the United States and Sweden (Buser et al., 2021a; Papageorge and Thom, 2020) and is consistent with the idea that endowments and investments may be substitutes at the early stages of childhood but that their complementarity increases over the life-cycle (Heckman and Mosso, 2014).

More generally, the contrast between the circular and the triangular series shows the interplay of advantageous genetic endowments and conducive environments varies across different sources of investments in children. After high school and relative to their high PGI^{EA} peers, genetically disadvantaged children consistently gain more from having attended a high-quality school than from having a high SES background. There are different potential explanations for this pattern. First, schools may allocate resources differently than parents. For example, Houmark et al. (2020) show that parents magnify skill inequality by investing more in chil-

tive measures of family SES, such as family income or potential wages of either parent, yield similar results.

dren with higher genetic endowments.²² Hence, our results are consistent with a model where schools allocate resources in a more equalizing way than families. Second, schools may provide different types of investments than families. For example, investments in schools tend to happen in the context of larger groups, whereas investments at home tend to be focused on individuals or smaller sibling groups. Hence, our results are consistent with a model where, relative to their high PGI^{EA} peers, low PGI^{EA} students respond more positively to the type of investments they receive in schools than to the type of investments they receive at home. While we cannot distinguish among these explanations, our results are generally consistent with recent evidence documenting large relative gains of disadvantaged students from attending high-quality schools in the United States (Cohodes et al., 2021; Jackson et al., forthcoming).

6 CONCLUSION

The question of how natural endowments and environmental factors determine life outcomes has a long history of inquiry in philosophy and science (Darwin, 1859; Descartes, 1641; Lamarck, 1838; Locke, 1690). The assumption that life outcomes are the result of genetic and environmental factors initially led to the so-called "nature versus nurture" debate. However, current research has moved beyond this simplistic dichotomy and recognizes that individual life outcomes are the result of a complex interplay between nature and nurture. This insight highlights that the importance of genetic endowments for life outcomes is not immutable. Instead, it opens a path for policy interventions that shape the relevant environment.

In this paper, we contribute to this research agenda by studying the interplay of genetic endowments and school quality. Making use of recent advances in molecular genetics, we link an individual-level index of genetic predispositions for educational success with measures of school quality. In turn, we investigate whether the importance of genetic endowments varies with the quality of high schools.

Our findings suggest that investments in the quality of schools can mitigate the genetic gradient in educational attainment. Furthermore, we show that higher gains in educational attainment for students with lower genetic endowments are mediated by gains in language skills, risk aversion, patience, and subjective health.

The use of genetics in education research has an ugly history. Therefore, many people are wary of the emergence of genetic markers in this context, especially when these markers are used for genetic screening (Martschenko et al., 2019). We emphasize that our results do neither

²²In Supplementary Table S.6, we replicate this result in our data using both between-family and within-family designs. This result is in contrast to Sanz-de-Galdeano and Terskaya (forthcoming) who also use AddHealth and find no significant effects of PGI^{EA} on parental investment once the siblings' PGI^{EA} is taken into account. We note that our analysis differs in several dimensions. Among others, we do not restrict the sample to firstborn children only and we use a more comprehensive index of parental time investments.

presuppose nor endorse the use of genetic screening for educational interventions. Instead, our results suggest that universal policy reform that increases the quality of schools for *all students* may provide an important step to level the playing field regardless of a student's draw in the genetic lottery.

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Genetic Endowments, Educational Outcomes, and the Mediating Influence of School Quality

Benjamin W. Arold, Paul Hufe & Marc Stöckli

Online Appendix

A SUPPLEMENTARY TABLES

TABLE S.1 – Gene-Environment Correlations

	PGI ^{EA}	Q ^S	Educ. Mother (School Av.)	White Student (School Share)	Single Parents (School Share)	GPA English (School Av.)
PGI ^{EA}	1.000 (.)					
Q ^S	0.023 (1.000)	1.000 (.)				
White Student (School Share)	-0.083 (0.000)	-0.027 (1.000)	1.000 (.)			
Single Parents (School Share)	-0.037 (0.382)	-0.017 (1.000)	-0.567 (0.000)			
Educ. Mother (School Av.)	0.129 (0.000)	0.020 (1.000)	-0.336 (0.000)	1.000 (.)		
GPA English (School Av.)	0.058 (0.004)	0.012 (1.000)	-0.279 (0.000)	0.226 (0.000)	1.000 (.)	
GPA Math (School Av.)	0.077 (0.000)	0.010 (1.000)	-0.272 (0.000)	0.447 (0.000)	0.510 (0.000)	1.000 (.)

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows correlations between PGI^{EA} and various school characteristics. *p*-values (in parentheses) are Bonferroni corrected to account for multiple hypothesis testing.

TABLE S.2 – Robustness to Sample Composition

Outcome: Years of Education	Baseline	Alternative Sample Composition		
	(1)	Re- Weighted (2)	Excl. (Potential) Movers before High School (3)	Excl. (Potential) Movers during High School (4)
PGI ^{EA}	0.360*** (0.028)	0.346*** (0.030)	0.352*** (0.035)	0.342*** (0.040)
Q ^S	0.128** (0.054)	0.120** (0.056)	0.162*** (0.062)	0.106 (0.077)
PGI ^{EA} × Q ^S	-0.062** (0.026)	-0.056** (0.026)	-0.082** (0.032)	-0.082** (0.038)
Child Controls	✓	✓	✓	✓
Family Controls	✓	✓	✓	✓
Control Function	✓	✓	✓	✓
N	4,036	3,970	2,964	2,441
R ²	0.334	0.314	0.351	0.344

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows the joint association of PGI^{EA} and Q^S with completed years of education. In column (2) we re-weight our analysis sample to match ACS and CPS with respect to gender composition, educational attainment of parents, and the age of mothers at birth—see also Supplementary Table S.3. In column (3) we exclude respondents that visit feeder schools in wave 1 and for whom we do not have information on subsequent high schools. In column (4) we additionally exclude respondents for whom we do not have information on whether they graduated from an Add Health high school. *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. All right-hand side variables are standardized on the estimation sample ($\mu = 0$, $\sigma = 1$). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors (in parentheses) are clustered at the school level.

TABLE S.3 – Sample Representativeness

	Population (Cohorts 1974-1983)		Analysis Sample	
	All	Non-Hispanic White	Unweighted	Re-Weighted
Gender				
Male	0.498	0.503	0.462	0.503
Female	0.502	0.497	0.538	0.497
Education Mother				
≤ High School	0.536	0.489	0.507	0.489
> High School; < College Degree	0.281	0.302	0.218	0.301
≥ College Degree	0.183	0.209	0.275	0.210
Education Father				
≤ High School	0.472	0.425	0.514	0.425
> High School; < College Degree	0.255	0.271	0.187	0.271
≥ College Degree	0.273	0.304	0.299	0.303
Age Mother at Birth				
< 25 Years	0.353	0.330	0.503	0.330
≥ 25 Years	0.647	0.670	0.497	0.670
Parental Income				
< \$50,000	0.557	0.491	0.536	0.511
≥ \$50,000; < \$100,000	0.352	0.403	0.386	0.406
≥ \$100,000	0.091	0.106	0.078	0.083
Education Respondent				
≤ High School	0.301	0.225	0.196	0.181
> High School; < College Degree	0.327	0.344	0.408	0.406
≥ College Degree	0.372	0.431	0.396	0.413

Data: National Longitudinal Study of Adolescent to Adult Health, American Community Survey (ACS), Current Population Survey (CPS).

Note: Own calculations. This table shows summary statistics of the core analysis sample in comparison to other population samples. It shows respondents' characteristics for the following samples: (i) the US population from birth cohorts 1974–1983, (ii) the Non-Hispanic White US population from birth cohorts 1974–1983, (iii) the core estimation sample, and (iv) the core estimation sample re-weighted to match (ii) with respect to *Gender*, *Education Mother*, *Education Father*, and *Age Mother at Birth*. Population data on *Gender* and *Education Respondent* from IPUMS ACS 2019 (Ruggles et al., 2020). Population data on *Education Mother*, *Education Father*, *Age Mother at Birth*, and *Parental Income* from IPUMS CPS 1994 (Flood et al., 2020).

TABLE S.4 – Alternative Polygenic Scores

Outcome: Years of Education	Baseline	+ Controls for Other Polygenic Scores					
	(1)	Body Mass Index (2)	ADHD (3)	Depressive Symptoms (4)	Intelligence (5)	Ever Smoker (6)	Sleep Duration (7)
PGI ^{EA}	0.360*** (0.028)	0.340*** (0.031)	0.331*** (0.028)	0.358*** (0.028)	0.350*** (0.031)	0.341*** (0.031)	0.359*** (0.028)
Q ^S	0.128** (0.054)	0.124** (0.054)	0.122** (0.053)	0.124** (0.055)	0.128** (0.054)	0.125** (0.053)	0.128** (0.054)
PGI ^{EA} × Q ^S	-0.062** (0.026)	-0.070** (0.029)	-0.065** (0.025)	-0.060** (0.026)	-0.054** (0.028)	-0.061** (0.027)	-0.062** (0.026)
Other PGI	–	-0.080*** (0.026)	-0.129*** (0.028)	-0.039 (0.030)	0.020 (0.030)	-0.095*** (0.036)	0.029 (0.028)
Other PGI × Q ^S	–	-0.029 (0.028)	0.003 (0.028)	0.031 (0.029)	-0.015 (0.027)	0.017 (0.033)	-0.004 (0.029)
Child Controls	✓	✓	✓	✓	✓	✓	✓
Family Controls	✓	✓	✓	✓	✓	✓	✓
Control Function	✓	✓	✓	✓	✓	✓	✓
N	4,036	4,036	4,036	4,036	4,036	4,036	4,036
R ²	0.334	0.335	0.337	0.334	0.334	0.336	0.334

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows the joint association of PGI^{EA} and Q^S with completed years of education. We control for other PGIs and their interaction with Q^S. The relevant PGIs are indicated in the column header. *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. All right-hand side variables are standardized on the estimation sample ($\mu = 0, \sigma = 1$). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors (in parentheses) are clustered at the school level.

TABLE S.5 – Measurement-error-corrected PGI (Becker et al., 2021)

	Coefficient	Standard Error	p-value	Substitutability
Baseline				
PGI^{EA}	0.360	0.031	0.000	
Q^{S}	0.128	0.062	0.040	
$\text{PGI}^{\text{EA}} \times Q^{\text{S}}$	-0.062	0.030	0.040	17%
Add Health ($\rho = 1.968$)				
PGI^{EA}	0.589	0.053	0.000	
Q^{S}	0.110	0.064	0.088	
$\text{PGI}^{\text{EA}} \times Q^{\text{S}}$	-0.088	0.047	0.060	15%
Health and Retirement Study ($\rho = 1.413$)				
PGI^{EA}	0.566	0.053	0.000	
Q^{S}	0.112	0.064	0.079	
$\text{PGI}^{\text{EA}} \times Q^{\text{S}}$	-0.086	0.044	0.049	15%
Wisconsin Longitudinal Study ($\rho = 1.649$)				
PGI^{EA}	0.718	0.069	0.000	
Q^{S}	0.098	0.067	0.143	
$\text{PGI}^{\text{EA}} \times Q^{\text{S}}$	-0.099	0.057	0.084	14%
UK Biobank ($\rho = 1.452$)				
PGI^{EA}	0.589	0.055	0.000	
Q^{S}	0.110	0.065	0.090	
$\text{PGI}^{\text{EA}} \times Q^{\text{S}}$	-0.088	0.045	0.048	15%

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows the joint association of PGI^{EA} and Q^{S} with completed years of education. We apply the correction method of Becker et al. (2021) to account for measurement error in PGI^{EA} . The correction method is based on $\rho = h_{\text{SNP}}^2 / R^2$, where h_{SNP}^2 indicates SNP heritability and R^2 the share of variation in educational attainment explained by PGI^{EA} . For Add Health, we take ρ from Sanz-de-Galdeano and Terskaya (forthcoming), for all other data sets we take ρ from Becker et al. (2021). Standard errors are bootstrapped with 1,000 draws. These standard errors are likely conservative (Becker et al., 2021; Sanz-de-Galdeano and Terskaya, forthcoming).

TABLE S.6 – Association of Parental Investments with PGI^{EA} and Q^S

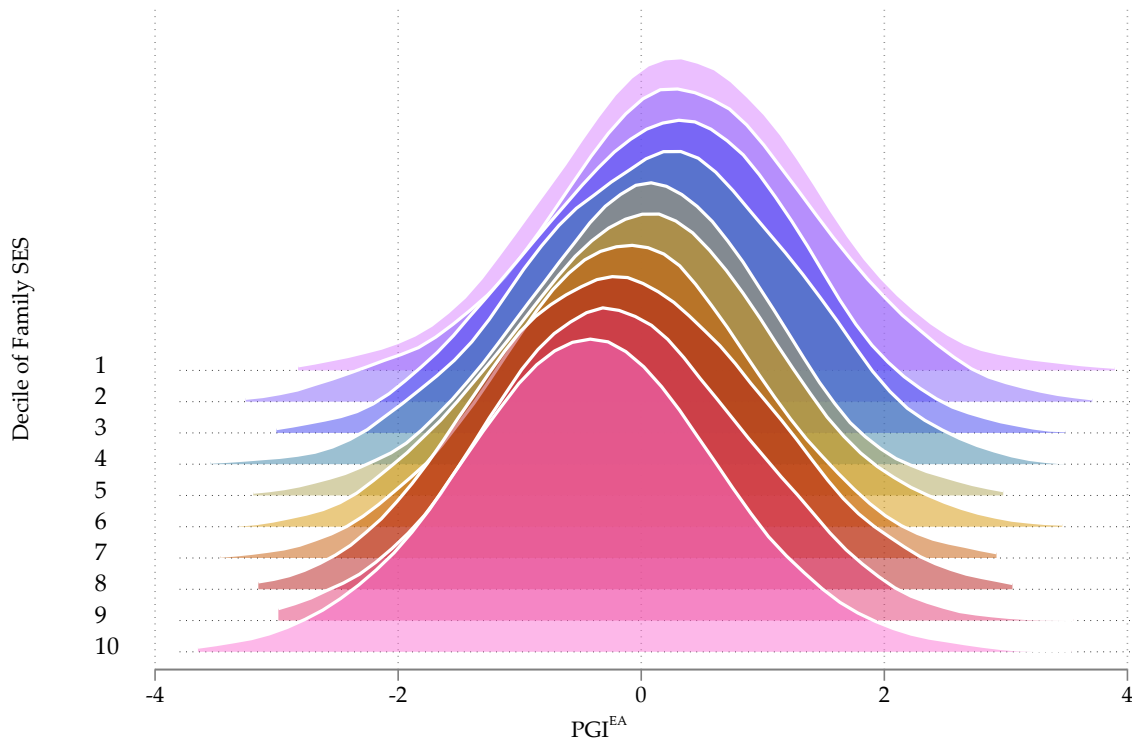
Outcome: Parental Investment Index	Between-Family			Within-Family		
	Both Parents (1)	Mother (2)	Father (3)	Both Parents (4)	Mother (5)	Father (6)
PGI ^{EA}	0.063*** (0.015)	0.049*** (0.015)	0.072*** (0.016)	0.092** (0.043)	0.052 (0.041)	0.141*** (0.048)
Q ^S	-0.022 (0.015)	-0.031* (0.017)	-0.011 (0.017)	-0.042 (0.036)	-0.052 (0.038)	-0.017 (0.047)
Child Controls	✓	✓	✓	✓	✓	✓
Family Controls	✓	✓	✓	✓	✓	✓
Control Function	✓	✓	✓	✓	✓	✓
Sibling PGI ^{EA}	×	×	×	✓	✓	✓
N	4,035	4,035	3,268	670	670	550
R ²	0.086	0.096	0.101	0.119	0.124	0.135
Outcome Mean	0.000	0.000	0.000	0.000	0.000	0.000
Outcome SD	1.000	1.000	1.000	1.000	1.000	1.000

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows the joint association of PGI^{EA} and Q^S with indexes of parental time investments. *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. All right-hand side variables are standardized on the estimation sample ($\mu = 0$, $\sigma = 1$). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors (in parentheses) are clustered at the school level.

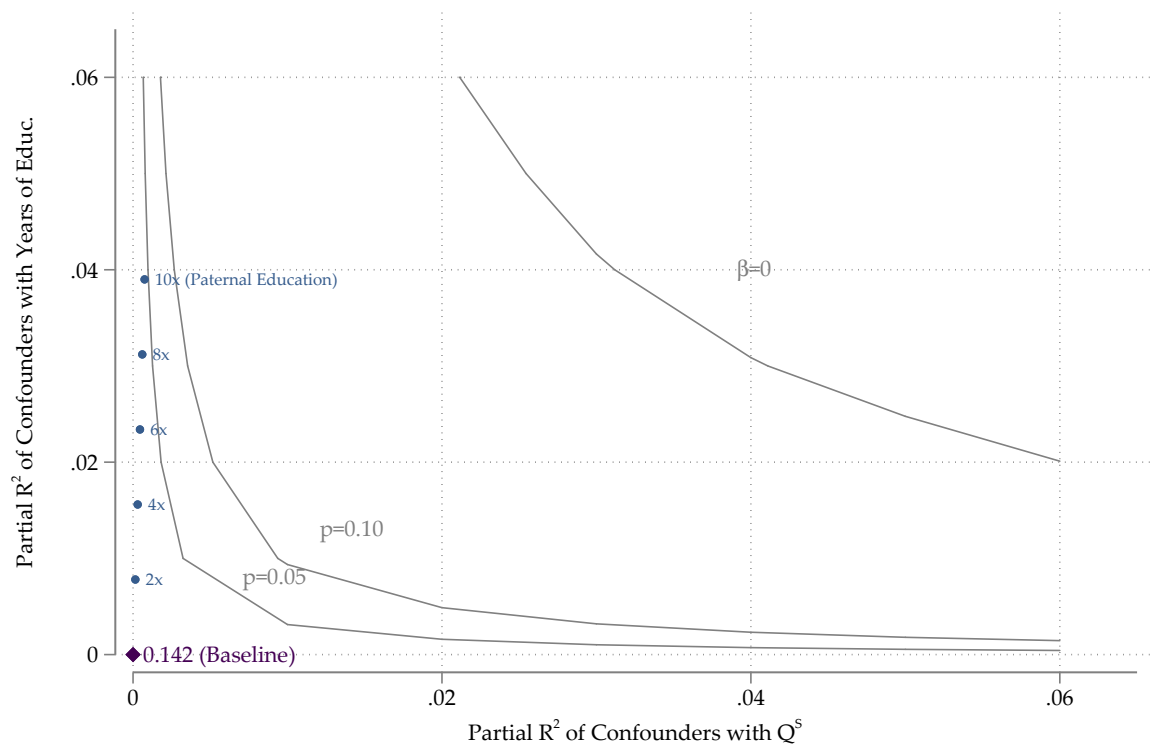
B SUPPLEMENTARY FIGURES

FIGURE S.1 – Distribution of PGI^{EA} by Family SES



Data: National Longitudinal Study of Adolescent to Adult Health.

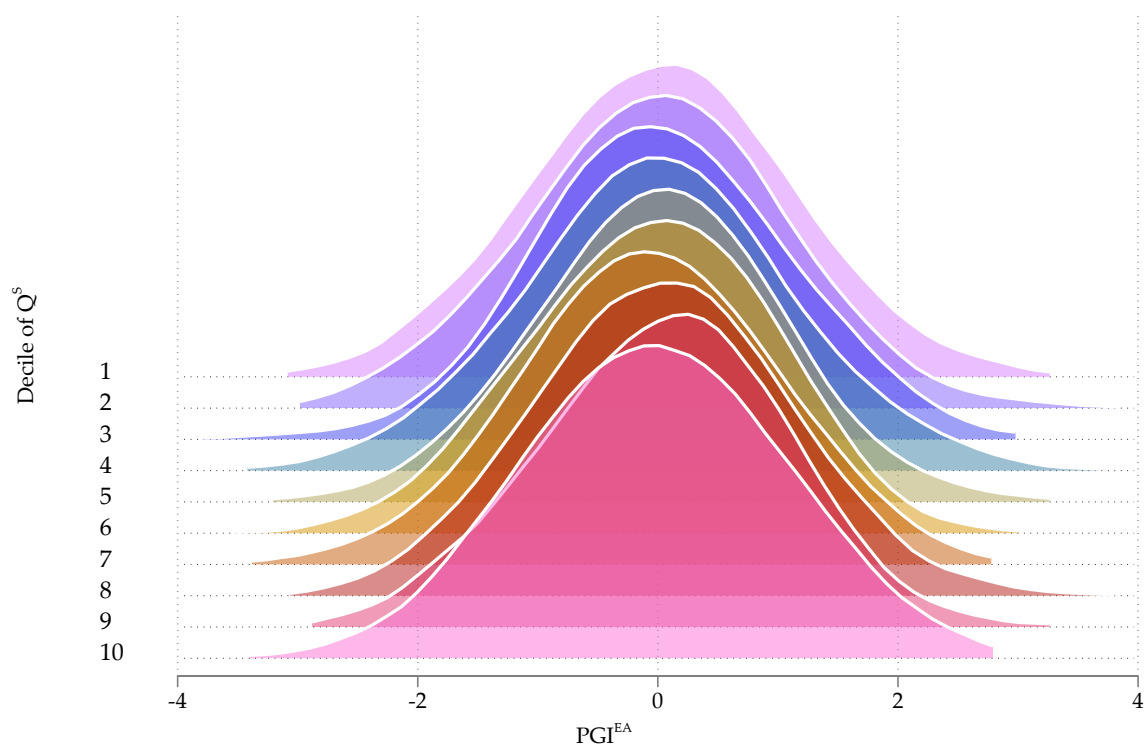
Note: Own calculations. This figure shows unconditional PGI^{EA} distribution by deciles of family SES. Density distributions are smoothed using the Epanechnikov kernel function with a bandwidth of 0.5.



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows the sensitivity of the point estimate for Q^S to unobserved confounding variables. Following the procedure of Cinelli and Hazlett (2020), we calculate the bias-adjusted treatment effect of Q^S under different assumptions about the partial R^2 of confounding variables with the variables of interest and the partial R^2 of confounding variables with years of education. Each contour line shows p -values for the point estimate of Q^S in a regression of years of education on Q^S and controls under different assumptions about the two partial R^2 . Each circle shows analogous values for different multiples of paternal education. The diamond shows baseline estimates from Figure 1. *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. Standard errors are clustered at the school level.

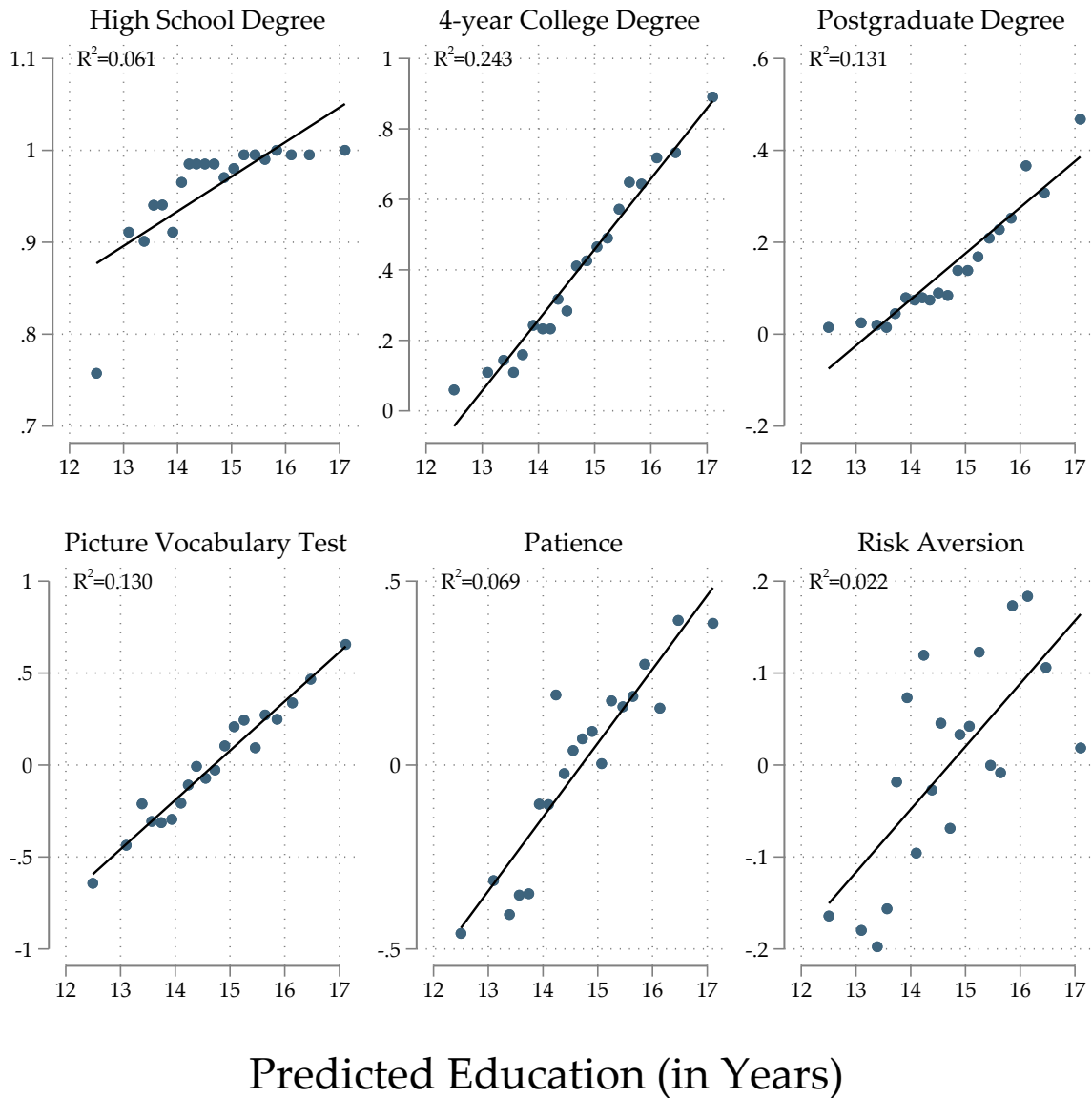
FIGURE S.3 – Distribution of PGI^{EA} by Q^S



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows PGI^{EA} distribution by deciles of Q^S after residualizing PGI^{EA} and Q^S by the full set of control variables. Density distributions are smoothed using the Epanechnikov kernel function with a bandwidth of 0.5.

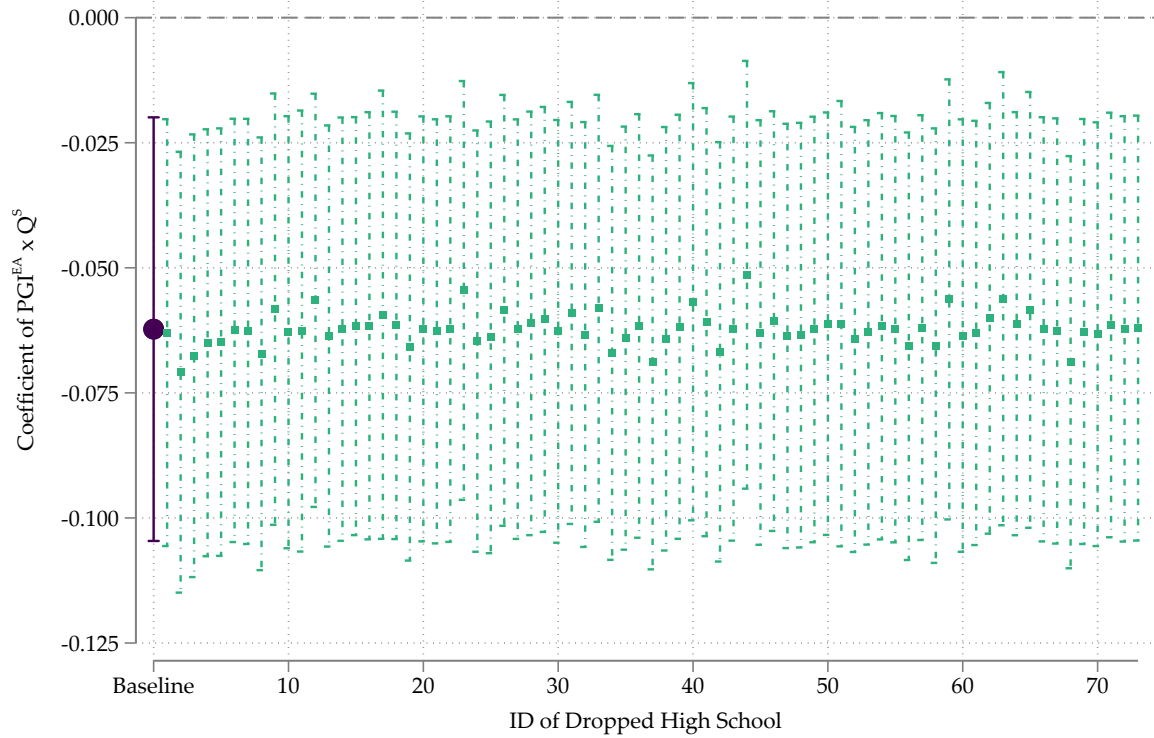
FIGURE S.4 – Association of Educational Attainment and Skills with Predicted Education



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows the association of various measures for educational attainment and (non-)cognitive skills with predicted years of education. We bin scatterplots using 20 quantiles of the variable of interest. Black lines are fitted from linear regressions of the variable of interest on predicted education. Predicted education is calculated from a regression of completed years of education on all *Child Controls* and *Family Controls*. Outcomes are measured in Waves 3, 4 and 5. *Child Controls*: Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls*: Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects.

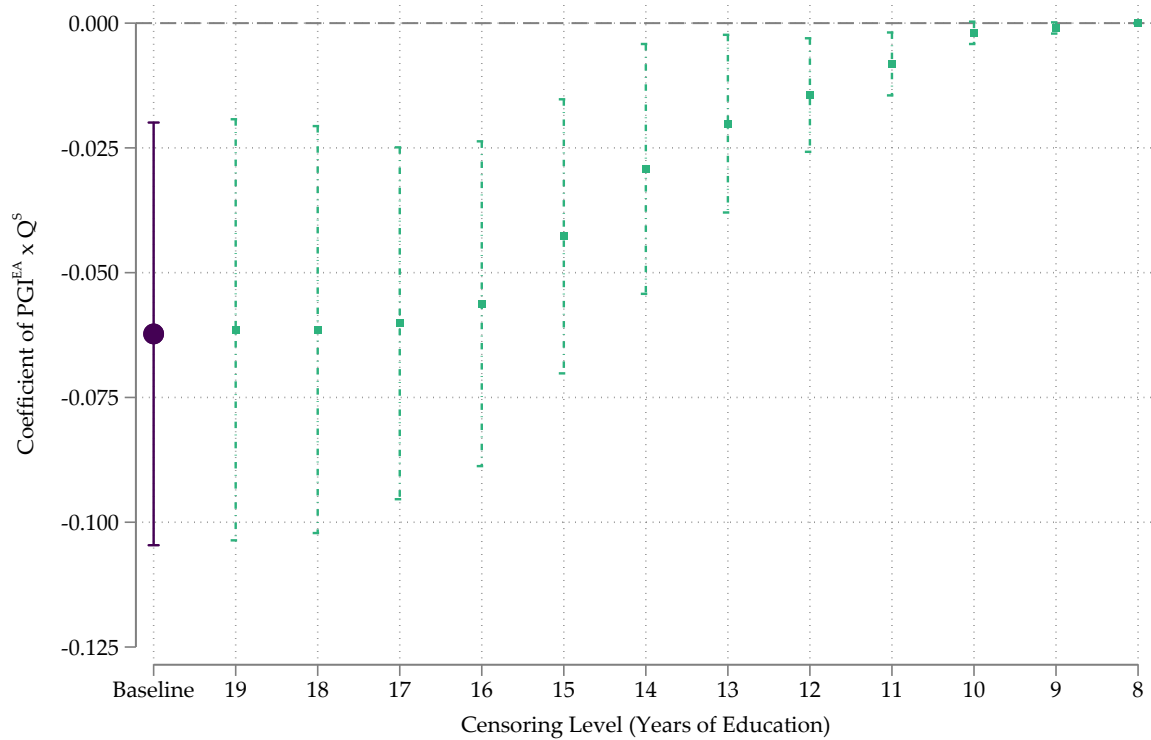
FIGURE S.5 – Sensitivity to Outlier Schools



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows point estimates and 90% confidence bands of the interaction of PGI^{EA} with Q^S , and its association with years of education. Each estimate is derived from a subsample of the data in which we drop one High School, respectively. Estimates follow the specification of equation (7). *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. Standard errors are clustered at the school level.

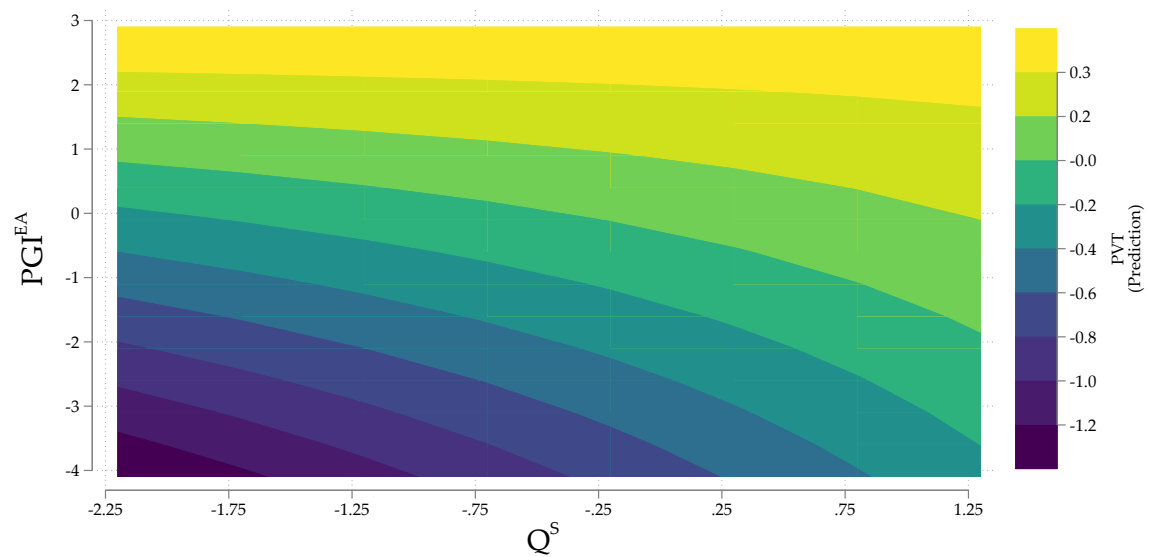
FIGURE S.6 – Sensitivity to Ceiling Effects



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows point estimates and 90% confidence bands of the interaction of PGI^{EA} and Q^S , and its association with years of education. Each estimate is derived from the full sample while censoring the outcome variable at different levels from above. Estimates follow the specification of equation (7). *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. Standard errors are clustered at the school level.

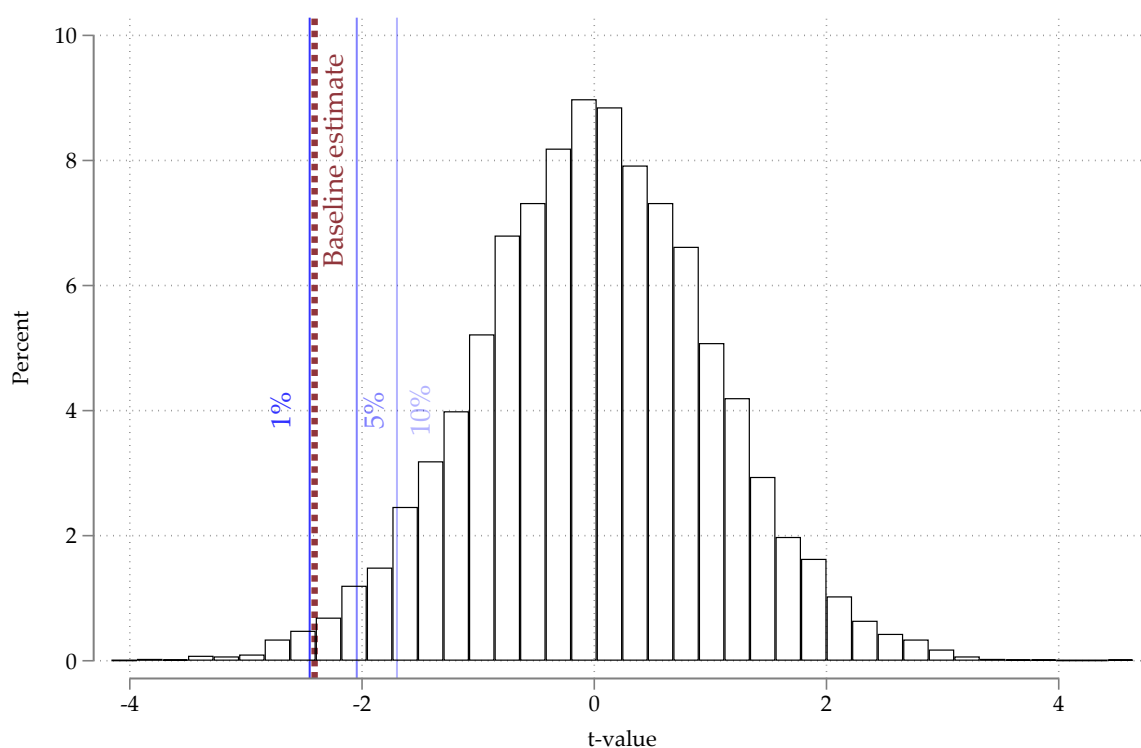
FIGURE S.7 – Association of Picture Vocabulary Test with PGI^{EA} by Q^S



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows predictions of the PVT by PGI^{EA} and Q^S cell. Predictions are calculated using the model estimated in column (1) in Panel (a) of Table 6. *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages. Standard errors are clustered at the school level.

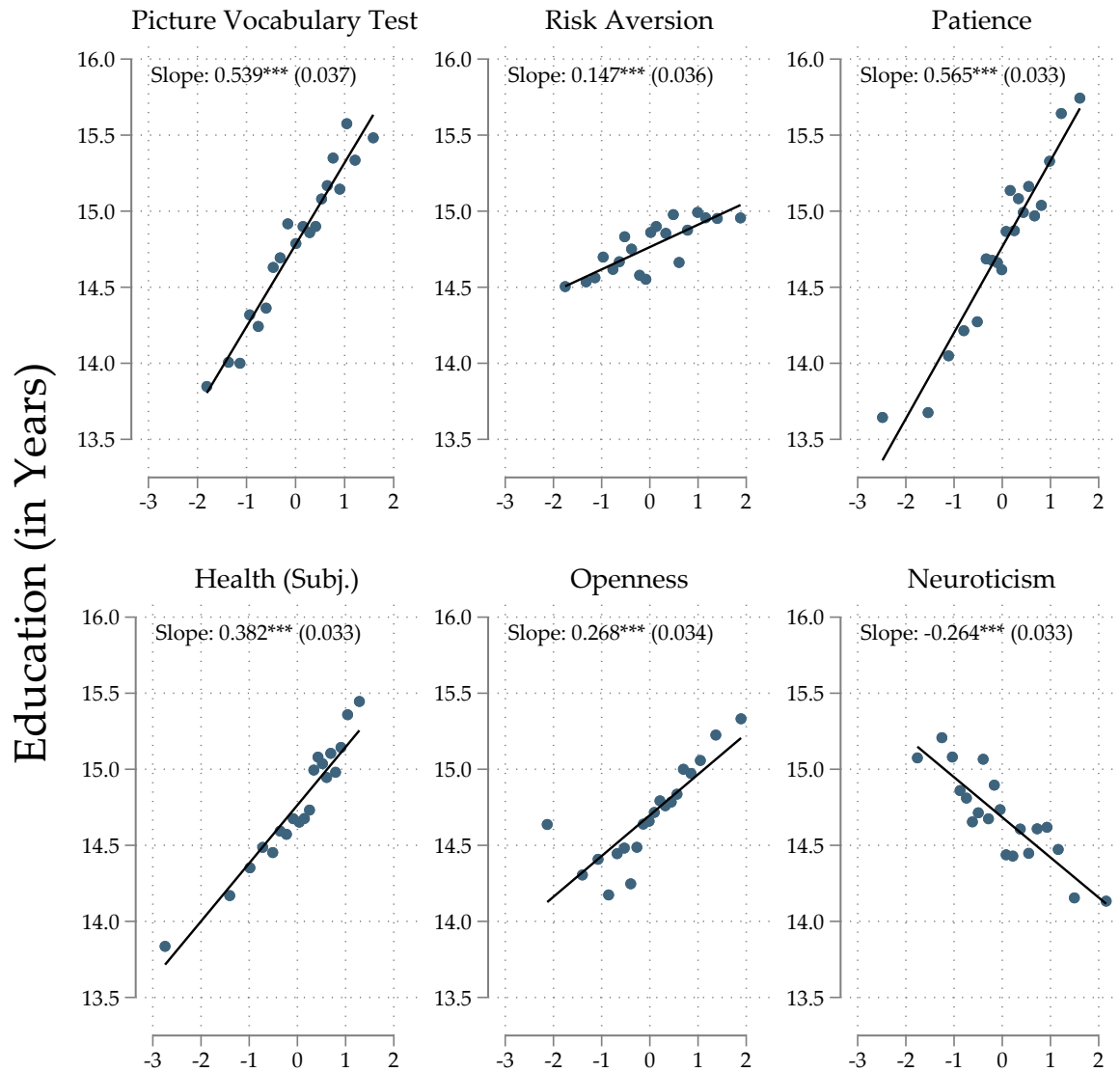
FIGURE S.8 – Permutation test for placebo assignments of Q^S



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows the frequency distribution of t -statistics for the interaction of PGI^{EA} and Q^S under 10,000 permutations of Q^S .

FIGURE S.9 – Association of Educational Attainment with (Non-)Cognitive Skills



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows the association of educational attainment and various measures of (non-)cognitive skills. We bin scatterplots using 20 quantiles of the variable of interest. Black lines are fitted from linear regressions of educational attainment on the variable of interest and the full set of controls. *Child Controls:* Firstborn dummy, linear birth cohort trend (in months) by gender, 20 principal components of the full matrix of genetic data. *Family Controls:* Age of mother at birth, years of education of both mother and father, average potential wages of both mother and father, the standard deviation of potential wages of both mother and father, dummies for non-US born mothers and fathers, a dummy for Christian religion, state fixed effects. *Control Function:* Share white, share single mothers, maternal education (average), peer grades English (average), peer grades Math (average). All control function variables are calculated as leave-cohort-out school averages.

C BIAS IN GENE-ENVIRONMENT INTERACTION

Assume the following population model:

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3(x_{1i} \cdot x_{2i}) + \delta_1 z_i + \delta_2(x_{1i} \cdot z_i) + \delta_3(x_{2i} \cdot z_i) + \epsilon_i,$$

with

$$(x_{1i}, x_{2i}, z_i) \sim N \left(0, \begin{bmatrix} 1 & 0 & \sigma_{x_1 z}^2 \\ 0 & 1 & \sigma_{x_2 z}^2 \\ \sigma_{x_1 z}^2 & \sigma_{x_2 z}^2 & 1 \end{bmatrix} \right), \text{Cov}(\epsilon|z, x_1, x_2) = 0.$$

Note we assume $\sigma_{x_1 x_2}^2 = 0$ based on the evidence presented in Figure 2. Furthermore, we assume $\sigma_{x_1 z}^2 \geq 0, \sigma_{x_2 z}^2 \geq 0$ to reflect concerns about i) genetic nurture and ii) selection into schools based on unobservable family background characteristics z .

Since z is unobserved, we estimate the following model:

$$y_i = \tilde{\alpha} + \tilde{\beta}_1 x_{1i} + \tilde{\beta}_2 x_{2i} + \tilde{\beta}_3(x_{1i} \cdot x_{2i}) + \tilde{\epsilon}_i.$$

What is the bias in the estimated gene-environment interaction $\tilde{\beta}_3$?

By the weak law of large numbers, we know that

$$\hat{\tilde{\beta}} \xrightarrow{p} \beta^* := [\mathbb{E}(X^T X)]^{-1} \mathbb{E}[X^T y].$$

First, under our assumptions about (x_1, x_2) , $[\mathbb{E}(X^T X)]$ simplifies to the identity matrix:

$$\mathbb{E}[X^T X] = \begin{bmatrix} 1 & \mathbb{E}[x_1] & \mathbb{E}[x_2] & \mathbb{E}[x_1 x_2] \\ \mathbb{E}[x_1] & \mathbb{E}[x_1^2] & \mathbb{E}[x_1 x_2] & \mathbb{E}[x_1^2 x_2] \\ \mathbb{E}[x_2] & \mathbb{E}[x_1 x_2] & \mathbb{E}[x_2^2] & \mathbb{E}[x_1 x_2^2] \\ \mathbb{E}[x_1 x_2] & \mathbb{E}[x_1^2 x_2] & \mathbb{E}[x_1 x_2^2] & \mathbb{E}[x_1^2 x_2^2] \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Therefore, bias will be driven by $\mathbb{E}[X^T y]$ only.

Second, under our assumptions about (x_1, x_2, z) , $\mathbb{E}[X^T y]$ reads as follows:

$$\mathbb{E}[X^T y] = \begin{bmatrix} \alpha + \delta_2 \mathbb{E}[x_1 z] + \delta_3 \mathbb{E}[x_2 z] \\ \beta_1 + \delta_1 \mathbb{E}[x_1 z] \\ \beta_2 + \delta_1 \mathbb{E}[x_2 z] \\ \beta_3 + \delta_2 \mathbb{E}[x_1^2 x_2 z] + \delta_3 \mathbb{E}[x_1 x_2^2 z] \end{bmatrix} = \begin{bmatrix} \alpha + \delta_2 \sigma_{x_1 z}^2 + \delta_3 \sigma_{x_2 z}^2 \\ \beta_1 + \delta_1 \sigma_{x_1 z}^2 \\ \beta_2 + \delta_1 \sigma_{x_2 z}^2 \\ \beta_3 + \delta_2 \sigma_{x_2 z}^2 + \delta_3 \sigma_{x_1 z}^2 \end{bmatrix}.$$

Therefore, bias in the gene-environment interaction is captured by:

$$\hat{\beta}_3 - \beta_3 = \delta_2 \sigma_{x_2z}^2 + \delta_3 \sigma_{x_1z}^2.$$

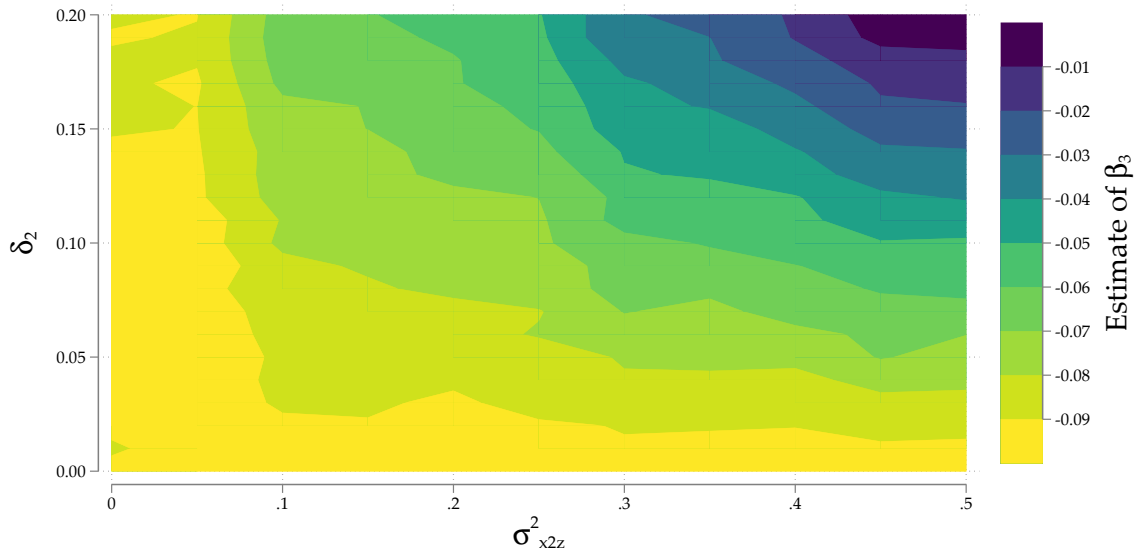
Note our assumptions $\sigma_{x_1z}^2 \geq 0, \sigma_{x_2z}^2 \geq 0$ are insufficient to sign the bias since the signs of δ_2 and δ_3 are unknown. However, we can sign the bias by two additional assumptions that are corroborated by our analysis and existing empirical literature:

$\sigma_{x_1z}^2 = 0 \rightarrow$ We assume the absence of confounding through genetic nurture. This assumption is supported by the similarity of results from between- and within-family models presented in Table 2.

$\delta_2 \geq 0 \rightarrow$ We assume (weak) complementarity between genetic endowments and (unobserved) family background characteristics. This assumption is consistent with the evidence presented in Figure 4 and existing literature confirming (weakly) positive Scarr-Rowe interactions (Figlio et al., 2017; Ronda et al., 2022; Turkheimer et al., 2003).

Under these additional assumptions $\hat{\beta}_3 - \beta_3 \geq 0$. Hence, we would estimate a lower bound of the substitutability between genetic endowments and school quality.

FIGURE S.10 – Simulation for $\beta_3 = -0.10$



Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This figure shows a heatmap of estimates for β_3 under different assumptions about $\sigma_{x_2z}^2$ and δ_2 . The simulation is based on the following data generating process: $y_i = 0.4x_{1i} + 0.2x_{2i} - 0.1(x_{1i} \cdot x_{2i}) + 0.2z_i + \delta_2(x_{1i} \cdot z_i) + 0.2(x_{2i} \cdot z_i) + \epsilon_i$, where $\sigma_{x_1x_2}^2 = \sigma_{x_1z}^2 = 0$, $\sigma_{x_2z}^2 \in [0.0(0.05)0.50]$, and $\delta_2 \in [0.0(0.01)0.20]$. We run 100 iterations for each combination of $\sigma_{x_2z}^2$ and δ_2 .

We confirm this result empirically through a simulation where we estimate β_3 under different assumptions about $\sigma_{x_2z}^2$ and δ_2 . Figure S.10 shows that β_3 increases, i.e., substitutability decreases, with increasing $\sigma_{x_2z}^2$ and δ_2 . This suggests that residual selection into schools that is not accounted for by our control variables would attenuate our estimated gene-environment interaction towards zero.

D DATA APPENDIX

D.1 Outcome Variables

Educational attainment. We measure educational attainment by total *years of education*. In each wave, respondents were asked about their highest level of education at the time of the interview. For each respondent, we use the most recent information and transform education levels into years of education following the mapping suggested by Domingue et al. (2015). Numeric values in parentheses: eighth grade or less (8), some high school (10), high school graduate (12), GED (12), some vocational/technical training (13), some community college (14), some college (14), completed vocational/technical training (14), associate or junior college degree (14), completed college (16), some graduate school (17), completed a Master degree (18), some post-baccalaureate professional education (18), some graduate training beyond a master’s degree (19), completed post-baccalaureate professional education (19), completed a doctoral degree (20).

We use the most recent available information to construct the following measures for educational degrees: *High School* (including GED), *2-year College*, *4-year College*, and *Post-Graduate*. Two-year college degrees include associate and junior college degrees as well as vocational and technical training after high school. Four-year college degrees include bachelor’s degrees. Post-graduate degrees include master’s degrees, doctoral degrees, and post-baccalaureate professional degrees. If available, information is taken from wave 5; otherwise, we take it from wave 4 or 3, respectively. We only include respondents for which we observe educational degrees when they are at least 27 years old at the time of observation. We assume an ordinal ranking of degrees (high school < 2-year college < 4-year college < post-graduate) and assign the possession of a lower-ranked degree if a respondent obtained a higher-ranked degree. For example, we assume that a respondent has finished high school if he or she has obtained a college degree, even if we do not have explicit information about high school graduation status.

Health. We proxy *subjective health* by quality-adjusted life years (QALY) that we derive from self-assessed health (SAH) measures. We use information from waves 3 and 4, where participants were asked “in general, how is your health?” We convert their (categorical) responses into a continuous measure using a mapping proposed by Van Doorslaer and Jones (2003). Using information about objective health—the Health Utility Index Mark III—Van Doorslaer and Jones (2003) scale the intervals of the SAH categories. This approach yields “quality weights” for health between 0 and 1. The values for each health status category are as follows (quality weights in parentheses): “excellent” (0.9833), “very good” (0.9311), “good” (0.841), “fair” (0.707), and “poor” (0.401).¹ We average resulting QALY measures across waves 3 and 4.

¹See Table 4 in Van Doorslaer and Jones (2003).

We construct an index of *objective health* based on information from wave 4. Specifically, we sum the standardized values about whether a respondent (i) is obese, (ii) has stage one hypertension, and (iii) has high cholesterol (as indicated by the respondent). Each item was answered with either “yes” (= 1) or “no” (= 0). We reverse-code our measure of objective health such that higher values indicate better health.

Cognitive skills. The *Picture Vocabulary Test* (PVT) is a test for receptive hearing vocabulary and is a widely-used proxy for verbal ability and scholastic aptitude. To administer the PVT, an examiner presents a series of pictures to the respondent. There are four pictures per page, and the examiner speaks a word describing one of the pictures. The respondent then has to indicate the picture that the word describes. In our analysis, we use age-adjusted PVT percentile ranks from wave 3 (Harris, 2020).

Preferences. We construct two measures of preferences: *risk aversion* and *patience*. In waves 3 and 4, participants were asked (i) whether they like to take risks, and (ii) whether they live their life without much thought for the future. Questions were answered on a five-point Likert scale ranging from “strongly agree” to “strongly disagree.” We reverse-code both measures and use averages from waves 3 and 4 in our analysis.

Personality. The Big Five personality traits are openness to experience, conscientiousness, extraversion, agreeableness, and neuroticism (Almlund et al., 2011). We use information from wave 4 to construct personality measures. Participants were asked a set of questions that each relate to one of the five personality traits. Questions were answered on a five-point Likert scale ranging from “strongly agree” to “strongly disagree.” We use averages of the following questions in our analysis. *Openness*: (i) “I have a vivid imagination,” (ii) “I have difficulty understanding abstract ideas” (reverse-coded), (iii) “I am not interested in abstract ideas” (reverse-coded), (iv) “I do not have a good imagination” (reverse-coded). *Conscientiousness*: (i) “I get chores done right away,” (ii) “I like order,” (iii) “I often forget to put things back in their proper place” (reverse-coded), (iv) “I make a mess of things” (reverse-coded). *Extraversion*: (i) “I am the life of the party,” (ii) “I talk to a lot of different people at parties,” (iii) “I don’t talk a lot” (reverse-coded), (iv) “I keep in the background” (reverse-coded). *Agreeableness*: (i) “I sympathize with others’ feelings,” (ii) “I feel others’ emotions,” (iii) “I am not interested in other people’s problems” (reverse-coded), (iv) “I am not really interested in others” (reverse-coded). *Neuroticism*: (i) “I have frequent mood swings,” (ii) “I get upset easily,” (iii) “I am relaxed most of the time” (reverse-coded), (iv) “I seldom feel blue” (reverse-coded).

TABLE S.7 – Summary Statistics (Outcomes)

	Obs.	Mean	SD	Min	Max
Educational Attainment					
Years of Education	4,036	14.68	2.27	8.00	20.00
High School Degree	4,034	0.96	0.20	0.00	1.00
2-year College Degree	4,036	0.50	0.50	0.00	1.00
4-year College Degree	4,036	0.39	0.49	0.00	1.00
Post-Graduate Degree	4,036	0.14	0.35	0.00	1.00
Health					
Subjective	3,473	0.91	0.07	0.40	0.98
Objective	4,036	0.05	1.94	-6.46	1.62
Cognitive Skills					
Picture Vocabulary Test	3,382	59.26	26.34	0.00	100.00
Preferences					
Risk Aversion	3,467	2.82	0.86	1.00	5.00
Patience	3,467	3.91	0.73	1.00	5.00
Personality					
Openness	3,998	3.63	0.63	1.00	5.00
Conscientiousness	4,030	3.64	0.69	1.25	5.00
Extraversion	4,029	3.33	0.77	1.00	5.00
Agreeableness	4,028	3.86	0.59	1.00	5.00
Neuroticism	4,024	2.57	0.70	1.00	5.00

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows summary statistics for outcome variables in our core analysis sample. The sample is restricted to genotyped individuals who (i) are of European descent, and (ii) attended an Add Health high school or an associated feeder school in wave 1.

D.2 Variables of Interest

Polygenic Indexes. Add Health obtained saliva samples from consenting participants in wave 4. After quality control procedures, genotyped data is available for 9,974 individuals and 609,130 SNPs. Add Health uses this data and calculates a set of different PGI using summary statistics from existing GWAS. Our baseline measure PGI^{EA} is based on statistics from Lee et al. (2018).

School characteristics. In waves 1 and 2, Add Health administered questionnaires to headmasters of Add Health schools. We use this information to construct an indicator for high school quality using principal components analysis (PCA). We extract the first component from a PCA that includes the following school-level information (component loadings in parentheses): (i) the average class size (-0.31), (ii) the share of teachers with a master's degree (+0.63), (iii) the share of new teachers in the current school year (-0.36), (iv) the share of teachers with

school-specific tenure of more than five years (+0.62). We apply a factor rotation for interpretability reasons (oblique oblimin rotation of the Kaiser normalized matrix with $\gamma = 0$; see Gorsuch, 1983). The calculated factor is standardized to $\mu = 0$ and $\sigma = 1$ on the full sample of Add Health respondents in wave 1.

Alternatively, we use the same information and aggregate across dimensions using the procedure suggested in Anderson (2008) and Kling et al. (2007). The calculated factor is standardized to $\mu = 0$ and $\sigma = 1$ on the full sample of Add Health respondents in wave 1.

Family socioeconomic status. We use the *social origins factor score* constructed by Belsky et al. (2018). Their measure uses information about parental education, parental occupation, household income, and household receipt of public assistance in wave 1. The score is standardized to $\mu = 0$ and $\sigma = 1$ on the full sample of Add Health respondents in wave 1.

TABLE S.8 – Summary Statistics (Variables of Interest)

	Obs.	Mean	SD	Min	Max
Polygenic Scores					
PGI ^{EA}	4,036	0.02	0.99	-4.13	3.39
School Characteristics					
Q ^S (PCA)	4,036	0.26	1.23	-3.17	2.49
Q ^S (Anderson, 2008)	4,036	0.25	0.90	-2.59	1.98
Teacher w/ MA (%)	4,036	51.20	24.10	0.00	95.00
Experienced Teacher (%)	4,036	66.66	23.43	0.00	98.00
New Teacher (%)	4,036	7.87	7.28	0.00	47.00
Class Size	4,036	24.40	4.50	12.00	38.00
Family SES					
Social Origins Factor Score	3,960	0.33	1.14	-4.51	3.51

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows summary statistics for variables of interest in our core analysis sample. The sample is restricted to genotyped individuals who (i) are of European descent, and (ii) attended an Add Health high school or an associated feeder school in wave 1.

D.3 Control Variables

Child characteristics. The child's *gender* (female or male, as indicated by the interviewer) is taken from the in-home questionnaire in wave 1.

We calculate the child's *age* (in months) at each wave by subtracting the child's birth date from the date of the interview. Because birth dates have minor inconsistencies across waves, we take

averages across waves 1 to 4.

We use the first 20 *principal components* of the full matrix of the genetic data. The components are obtained from a principal components analysis on the matrix of SNPs in Add Health (see Braudt and Harris, 2020, for a discussion). The principal components are standardized to $\mu = 0$ and $\sigma = 1$ on the full sample of genotyped Add Health respondents.

Family socioeconomic status. We use information from wave 1 to construct measures of *parents' education*. We transform parents' highest degree into years of education following the mapping suggested by Domingue et al. (2015). Numeric values in parentheses: never went to school (0), eighth grade or less (8), some high school (10), completed vocational/technical training instead of high school (10), went to school but level unknown (12), respondent doesn't know (12), high school graduate (12), GED (12), completed vocational/technical training after high school (14), some college (14), completed college (16), professional training beyond a master's degree (19). Where available, mothers' and fathers' education refers to the resident parent. If this information is not available, we use the biological parents' education instead.

Information about *mother's age at birth* (in years) is obtained from wave 1 if available, and wave 2 otherwise. To calculate age at birth, we take information about the mother's age (as indicated by the child) and subtract the age of the child at the respective wave.

Information about religion (*Christian* or not) is obtained from wave 1 (as indicated by the child).

We calculate *potential wages* for population group g in time period t according to the following formula (Shenhav, 2021):

$$\hat{w}_{gt} = \sum_j \frac{E_{jg,1970}}{E_{g,1970}} \times \sum_o \frac{E_{ojg,1970}}{E_{jg,1970}} (\pi_{ojt,-r}) \times w_{ojt,-r},$$

where $\frac{E_{jg,1970}}{E_{g,1970}}$ describes the group-specific employment share of industry j in 1970, $\frac{E_{ojg,1970}}{E_{jg,1970}}$ describes the group- and industry-specific employment share of occupation o in 1970, $\pi_{ojt,-r}$ describes the leave-region-out industry-specific employment growth in occupation o for the year t relative to 1970 (scaled by the overall employment growth in occupation o for the year t relative to 1970), and $w_{ojt,-r}$ describes the leave-region-out average hourly wage paid in year t for each occupation/industry/region cell. We define groups g by individuals that are homogeneous in gender (male, female), educational attainment ($<$ High School, High School, $>$ High School), and ethnicity (Non-Hispanic White, Hispanic, Non-Hispanic Black). We define regions r by census regions (North-East, Midwest, South, West). Employment shares in 1970 are taken from the 1970 decennial census. Employment shares and wages in periods t are taken from the March Supplements of the Current Population Survey (CPS) over the period 1975-2000. We match the time series of $\hat{w}_{gt}$ to the parents of respondents in Add Health based on information about g . Then we calculate (i) mean potential wages across respondents ages 0–14, and (ii) the

standard deviation in potential wages across respondents ages 0–14.

Control function We construct the control function variables from the in-school questionnaires in wave 1. Specifically, for each school, we calculate the (i) *share of white students*, (ii) the *share of students in single-parent households*, (iii) *average years of education of students' mothers* and school averages for (iv) *GPA in Math* and (v) *GPA in English*. We transform mothers' highest degree into years of education following the mapping suggested by Domingue et al. (2015). Numeric values in parentheses: never went to school (0), eighth grade or less (8), some high school (10), went to school but level unknown (12), respondent doesn't know (12), high school graduate (12), GED (12), completed vocational/technical training after high school (14), some college (14), completed college (16), professional training beyond a four-year college (19). Student GPAs are standardized within grade times federal state cells. To prevent mechanical correlation, we calculate all control function variables while excluding the cohort of the respondent (leave-cohort-out).

Other school characteristics. We use information from the school administrator questionnaire in wave 1 to construct for other school policies. Specifically, we construct a binary indicator that assumes a value of 1 if the school uses *ability groups* based on English ability, proxy for the strictness of *retention policy* by calculating the school average of retained students in 1993, and construct a *strictness index* for sanction policies. The strictness index is constructed as follows. School administrators were asked what happens to first-time offenders in the following domains: (i) cheating, (ii) fighting with another student, (iii) injuring another student, (iv) possessing alcohol, (v) possessing an illegal drug, (vi) possessing a weapon, (vii) drinking alcohol at school, (viii) using an illegal drug at school, (ix) smoking at school, (x) verbally abusing a teacher, (xi) physically injuring a teacher, and (xii) stealing school property. Responses are “minor action”, “in-school suspension”, “out-of-school suspension”, and “expulsion.” We standardize answers to $\mu = 0$ and $\sigma = 1$ on the full sample of Add Health respondents and then sum across dimensions (Anderson, 2008; Kling et al., 2007).

We also use information from the school administrator questionnaire in wave 1 to construct measures for teacher composition. Specifically, we calculate the schools' share of full-time classroom teachers that the school administrator identifies as (i) *White*, (ii) *Hispanic*, and (iii) *Female*.

Other PGI. We use the PGI for body mass index (*BMI*) (Yengo et al., 2018), attention deficit hyperactivity disorder (*ADHD*) (Demontis et al., 2019), *depressive symptoms* (Howard et al., 2019), *intelligence* (Savage et al., 2018), *smoking* (Liu et al., 2019), and *sleep duration* (Jansen et al., 2019). All polygenic indexes are standardized to $\mu = 0$ and $\sigma = 1$ on the full sample of genotyped Add Health respondents.

Endogenous controls. To measure *parental time investments*, we use the information on a series of activities that children have done with their mother or father in the last four weeks. Specifically, the child is asked whether he or she has (i) gone shopping, (ii) played a sport, (iii) gone to a religious service or church-related event, (iv) talked about someone he or she is dating, or a party he or she went to, (v) gone to a movie, play, museum, concert, or sports event, (vi) had a talk about a personal problem he or she was having, (vii) talked about his or her school work or grades, (viii) worked on a project for school, (ix) talked about other things he or she is doing in school. Questions were answered with “yes” (= 1) or “no” (= 0). We standardize answers to $\mu = 0$ and $\sigma = 1$ on the full sample of Add Health respondents and then sum across dimensions (Anderson, 2008; Kling et al., 2007).

Information about *breastfeeding* and *family income* is taken from wave 1. Parents were asked about whether the child was breastfed (yes or no) and about their income. We use the logarithm of the latter (replacing zero incomes with a 1 to prevent a loss of observations).

TABLE S.9 – Summary Statistics (Controls)

	Obs.	Mean	SD	Min	Max
Child Controls					
Female	4,036	0.54	0.50	0.00	1.00
Age in Months (Wave 1)	4,036	192.41	19.62	144.00	256.00
Firstborn	4,036	0.48	0.50	0.00	1.00
Family Controls					
Education Mother (in Years)	4,036	13.54	2.48	0.00	19.00
Education Father (in Years)	4,036	13.56	2.68	0.00	19.00
Maternal Age at Birth	4,036	25.33	4.84	16.00	46.08
Christian	4,036	0.82	0.38	0.00	1.00
Foreign-born Father	4,036	0.03	0.16	0.00	1.00
Foreign-born Mother	4,036	0.03	0.17	0.00	1.00
Potential Wage/Hour Mother (Mean)	4,036	12.57	1.39	9.40	14.27
Potential Wage/Hour Father (Mean)	4,036	15.40	1.32	11.14	17.11
Potential Wage/Hour Mother (SD)	4,036	0.35	0.11	0.12	0.51
Potential Wage/Hour Father (SD)	4,036	0.39	0.08	0.20	0.65
Control Function					
White Student (School Share)	4,036	0.81	0.18	0.05	1.00
Single Parents (School Share)	4,036	0.24	0.08	0.03	0.58
Educ. Mother (School Average)	4,036	13.40	0.68	12.03	16.25
GPA English (School Average)	4,036	0.02	0.13	-0.45	0.51
GPA Math (School Average)	4,036	0.02	0.15	-0.45	0.54
Other School Characteristics					
Strictness Index	4,036	0.19	0.45	-3.07	0.77
Ability Groups	4,036	0.66	0.47	0.00	1.00
Retention Policy (%)	3,971	3.55	3.72	0.00	20.75
White Teacher (%)	4,036	93.23	11.49	18.00	100.00
Hispanic Teacher (%)	4,036	1.60	4.25	0.00	52.00
Female Teacher (%)	4,036	57.08	13.85	25.00	96.00
Polygenic Scores					
BMI	4,036	-0.02	1.00	-3.42	3.56
ADHD	4,036	-0.04	1.00	-3.82	3.48
Depressive Symptoms	4,036	-0.02	1.00	-3.79	3.55
Intelligence	4,036	0.02	0.99	-3.57	4.64
Ever Smoker	4,036	-0.02	1.00	-4.25	4.25
Sleep Duration	4,036	0.01	0.99	-3.82	2.97
Endogenous Controls					
Breastfed	3,731	0.49	0.50	0.00	1.00
Time Investment Mother	4,035	0.44	4.17	-7.80	13.48
Family income (log)	3,377	3.74	0.74	0.00	6.91

Data: National Longitudinal Study of Adolescent to Adult Health.

Note: Own calculations. This table shows summary statistics for control variables in our core analysis sample. The sample is restricted to genotyped individuals who (i) are of European descent, and (ii) attended an Add Health high school or an associated feeder school in wave 1.

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Multidimensional Equality of Opportunity in the United States^{*}

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Abstract

Are the United States still a land of opportunity? We provide new insights on this question by invoking a novel approach that allows us to measure unfair inequality in the joint distribution of income and wealth. We show that inequality of opportunity has increased by 56% over the time period 1983-2016. Increases are driven by two distinct forces: (i) a less opportunity-egalitarian distribution of income until 2000, and (ii) a less opportunity-egalitarian distribution of wealth after the financial crisis in 2008. In sum, our findings suggest that the US have consistently moved further away from a level playing field in recent decades.

JEL: D31, D63, J62

Keywords: Fairness, Intergenerational Mobility, Time Trends, Measurement

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1 INTRODUCTION

In a fair economy, people act on a level playing field to acquire monetary resources. This idea—oftentimes labeled as *equality of opportunity*—is widely reflected in fairness conceptions of academic philosophers and the general public (Alesina et al., 2018; Almås et al., 2020; Arneson, 2018; Cappelen et al., 2007; Cohen, 1989; Fong, 2001; Rawls, 1971; Roemer, 1998). As a consequence, there is an active literature in economics that assesses the satisfaction of the opportunity-egalitarian ideal in different countries at different points in time. We contribute to this literature by providing the first analysis of the association between family background characteristics and the joint distribution of income and wealth in the US.

Existing studies on inequality of opportunity and intergenerational mobility focus on income—and to a lesser extent on wealth—to measure monetary resources.¹ By excluding either income or wealth from the analysis, these studies neglect important information on individual consumption possibilities which arguably are the relevant metric to assess the financial well-being of individuals. For example, unidimensional analyses will misrepresent the financial well-being of income-poor heirs who support their lifestyle by selling assets or of asset-poor persons with high incomes. Therefore, if society cares for the financial well-being of individuals more broadly, we should move from unidimensional analyses of monetary resources to analyses of the joint distribution of income and wealth.

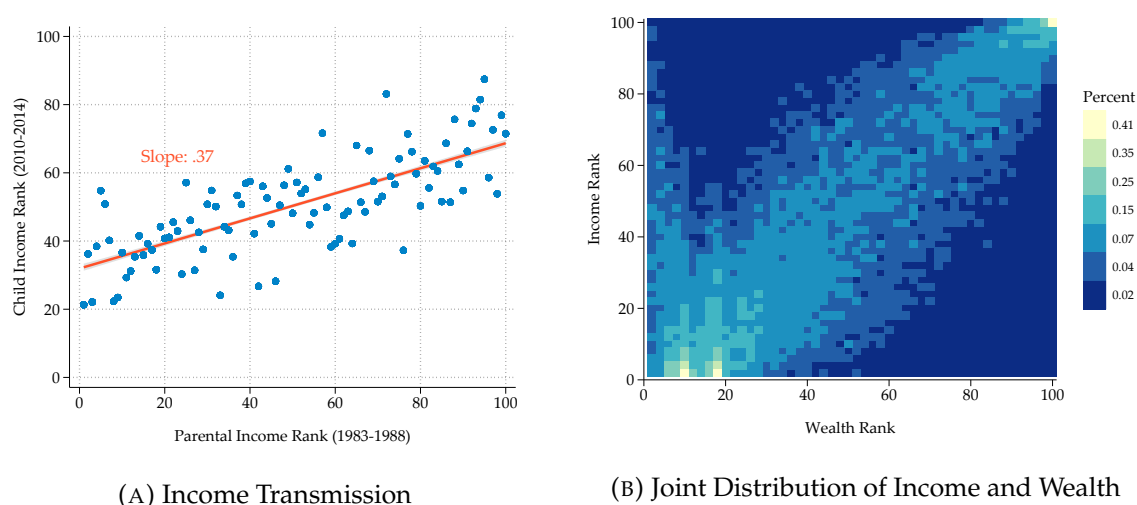
The focus on unidimensional analyses would be innocuous if income and wealth were perfect substitutes as indicators for monetary resources. There are at least two reasons why this is implausible. First, well-off parents transmit monetary resources to the next generation through bequests and inter vivo gifts (Boserup et al., 2016; Elinder et al., 2018; Wolff, 2002). In turn, expected wealth transfers distort the education and labor supply decisions of children (Kindermann et al., 2020; Kopczuk, 2013). Such behavioral responses create a wedge between the relative positions of individuals in income and wealth distributions: individuals that receive a lot of wealth from their parents are not necessarily those who earn high incomes. This observation is particularly relevant for the analysis of time trends as inheritances have grown in many Western societies in recent decades (Piketty and Zucman, 2015). Second, changes in wealth are a function

¹For the US, see Chetty et al. (2014a), Davis and Mazumder (2017), and Solon (1992) for intergenerational income mobility; Charles and Hurst (2003) and Pfeffer and Killewald (2018) for wealth mobility, and Hufe et al. (2022), Niehues and Peichl (2014), and Pistoletti (2009) for inequality of opportunity in incomes.

of savings and asset price changes. While the savings channel depends on income, the price channel depends on portfolio compositions. Therefore, changes in asset prices are another force that drives a wedge between the relative positions of individuals in income and wealth distributions. Again, this observation is particularly relevant for the analysis of time trends as wealth-to-income ratios—and therefore the sensitivity of wealth to asset price fluctuations—has grown over time (Kuhn et al., 2020).

In Figure 1, we use data from the Panel Study of Income Dynamics (PSID) to show that these concerns are relevant for the analysis of equal opportunities in the United States. In Panel (A), we replicate the well-known finding that child incomes increase with the income of their parents during childhood: an increase of parental income by 10 percentile ranks is associated with an average increase of 3.7 percentile ranks in child income. This estimate is very similar to the slope estimate of 0.34 in Chetty et al. (2014a). In Panel (B), a heatmap of income and wealth ranks demonstrates that income and wealth are far from perfect correlates (Rank correlation $\rho = 0.55$).² Taken together,

FIGURE 1. Intergenerational Income Mobility and the Distribution of Monetary Resources in the United States



Data: PSID.

Note: Panel (A) shows a binned scatter plot of average child income ranks in the period 2010-2016 by income rank of their parents in the period 1983-1988. All individuals are aged 25-60. Panel (B) shows a heatmap of year-specific income and wealth ranks for the pooled sample of individuals aged 25-60 in the period 1983-2016. Each data point shows the share of individuals in a fixed two-percentile income (wealth) bin that belong to a particular two-percentile wealth (income) bin. See Section 3 for detailed definitions of income and wealth.

²The moderate rank correlation is not due to idiosyncratic fluctuations in income or wealth. Using 5-year moving averages for income and wealth yields $\rho = 0.59$.

these patterns suggest that unidimensional analyses of equality of opportunity and intergenerational mobility provide a distorted image of the importance of family background for individual consumption possibilities and financial well-being.

In this paper, we address these shortcomings by analyzing the association between family background and the joint distribution of income and wealth. We use the PSID to implement and extend a novel measure of multidimensional equality of opportunity (Kobus et al., 2020). Our analysis proceeds in two steps. First, we construct an *inter-generational sample* in which we measure equality of opportunity in monetary resources by using parental income ranks as the only proxy for socioeconomic background. This practice is consistent with the literature on intergenerational mobility; however, the sparsity of data links across generations prevents meaningful analyses of time trends. Second, we construct an *individual sample* in which we substitute parental income ranks by a vector of alternative socioeconomic background characteristics. These data are available on an annual basis and allow us to assess trends over the period 1983-2016.

Our findings can be summarized as follows. First, multidimensional inequality of opportunity is consistently and substantially higher than inequality of opportunity in income. Hence, unidimensional analyses that focus on income only underestimate the extent to which monetary resources are associated with family background. Second, the playing field in the US has become more tilted in recent decades: inequality of opportunity in 2016 is 56% higher than in 1983. Furthermore, time trends are markedly different when accounting for the multidimensionality of monetary resources. For example, an exclusive focus on income suggests small increases in unequal opportunities after the year 2000. This relative stability, however, is accompanied by strong increases in the wealth dimension leading to an overall increase in unequal opportunities.

The contribution of this paper is threefold. First, we complement recent literature that characterizes the joint distribution of income and wealth in the US (Berman and Milanovic, 2020; Kuhn et al., 2020). This literature focuses on inequalities in outcomes but remains silent on opportunities and intergenerational transmission processes. Second, we provide novel insights regarding the development of equality of opportunity in the United States. While existing literature documented relative stability of equality of opportunity in terms of income after 2000 (Chetty et al., 2014b; Hartley et al., 2022), we show that decreases emerge once we account for the wealth dimension. Third, we provide a novel decomposition of the multidimensional measure into inequality of opportunity in income, inequality of opportunity in wealth, and the association of both outcomes across family background types. Association is a distinctive feature of joint

distributions that cannot be captured by unidimensional analyses. It indicates whether individuals of a given family background are more likely to fare better or worse in both dimensions simultaneously. We use a multidimensional framework to combine these dimensions and to obtain an overall conclusion regarding the extent of unequal opportunities in the US.

2 MEASUREMENT OF MULTIDIMENSIONAL EQUALITY OF OPPORTUNITY

Consider a population $\mathcal{N} := \{1, \dots, N\}$ and a set of outcomes $\mathcal{K} := \{1, \dots, K\}$ that capture monetary resources. Individuals $i \in \mathcal{N}$ receive utility from $q \in \mathcal{K}$. We can summarize the distribution of monetary resources by outcome matrix X of dimension $N \times K$, where an element x_{iq} denotes i 's outcome in dimension q . Outcomes are determined by two sets of factors: a set Ω that captures family background characteristics and a set Θ that captures individual choices. We define $\omega_i \in \Omega$ as a comprehensive description of family background and $\theta_i \in \Theta$ as a comprehensive description of the choices made by $i \in \mathcal{N}$. For each q , there is an outcome-generating function defined as follows:

$$x_{iq} = f_q(\omega_i, \theta_i), \forall i \in \mathcal{N}. \quad (1)$$

In an equal-opportunity society, outcome differences are determined by individual choices θ_i but are invariant to family background ω_i (Roemer, 1998). There are different ways of translating this idea into measures. Most empirical literature relies on an *ex-ante* approach, which broadly consists of two steps. First, one partitions the population into types $T = \{t_1, \dots, t_M\}$. Individuals belong to a type if they share the same set of family background characteristics: $i, j \in t_m \Leftrightarrow \omega_i = \omega_j$. For example, in rank-rank measures of intergenerational mobility, types are defined by parental income ranks. Second, one assesses differences in average outcomes across types by regressing child outcomes on a measure of family background:

$$x_{iq} = \alpha_q + \beta_q \omega_i + \epsilon_{iq}. \quad (2)$$

There are two prominent ways of summarizing the resulting information in measures of inequality of opportunity: (i) β_q , which is the standard statistic in the literature on *intergenerational mobility* (Black and Devereux, 2011). (ii) $I(X) = I(\mathbb{E}[x_{iq}])$, where $I(\cdot)$ is any inequality index, and which is the standard statistic in the literature on *equal-*

ity of opportunity (Roemer and Trannoy, 2016). It defines inequality of opportunity as inequality between types: all within-type variation is removed and inequality reflects only inequality that arises due to family background. Clearly, both measures are isomorphic and capture the opportunity-egalitarian idea: the higher β_q , the more life outcomes x_{iq} are predicted by family background ω_i , and the higher the corresponding measure of inequality of opportunity.³

In this paper, we follow the tradition of the equality of opportunity literature and summarize outcome differences across types with an inequality index. In particular, we use the measure of Kobus et al. (2020), which allows us to account for the multidimensionality of monetary resources. For the sake of simplicity and in line with our empirical application, we focus on the case of two outcome dimensions and set $K = 2$. In this case, the index is given by

$$I(X) = 1 - \left(\sum_{t=1}^M \frac{N_t a_t}{\sum_{t=1}^M N_t a_t} \frac{(\mu_1^t)^{r_1} (\mu_2^t)^{r_2}}{(\mu_1)^{r_1} (\mu_2)^{r_2}} \right)^{\frac{1}{r_1+r_2}} \quad \forall_t a_t < 0, r_1, r_2 < 0, \quad (3)$$

where N_t denotes the number of individuals in type t and μ_q^t (μ_q) the type (population) means in outcome q . In the following, we will describe the roles of r_q and a_t which are weights for outcome dimension q and types t , respectively. However, before doing so we note that if $r_q = 0$ for either outcome, $I(X)$ boils down to a unidimensional measure of inequality of opportunity which is the well-known Atkinson (1970) index applied to types t .⁴

Dimension weights r_q govern the sensitivity of the measure to between-type inequality in outcome q . The more negative r_q , the more convex the measure in q , and the higher its sensitivity to between-type inequality in this dimension. For example, if $r_1 < r_2$, $I(X)$ is more sensitive to inequality in the first than in the second outcome, i.e., the former is then relatively more important in the inequality assessment. r_q is also related to the degree of inequality aversion ϵ_q via $r_q = 1 - \epsilon_q$. As ϵ_q rises, the index becomes more sensitive to inequality at the bottom of the distribution than at the top. Note that ϵ_q is a parameter chosen by the researcher. In his seminal work, Atkinson (1970) arbitrarily set ϵ_q equal to 1, 1.5 and 2. Subsequently, empirical research has tried to infer plausible values of ϵ_q from economic policy design and tax schedules (Aristei

³One advantage of the equality of opportunity approach is that it allows for a flexible accommodation of family background characteristics other than family income or wealth.

⁴Generally, the index in Equation (3) is a multidimensional generalized entropy measure. Similarly, the empirical literature on equality of opportunity often uses unidimensional generalized entropy measures like the mean log deviation to summarize inequality between family background types.

and Perugini, 2016; Gouveia and Strauss, 1994; Young, 1990). These estimates range between 1 and 2 depending on the country and period of interest. In our baseline calculations, we choose $\epsilon_q = 1.2$ ($r_q = -0.2$) for both income and wealth. However, in section 5, we show that our conclusions on time trends do not change for a wide range of plausible choices for ϵ_q .

Type weights a_t determine how much the social planner values respective types. The more negative a_t , the higher the weight attached to type t . To ensure that $I(X)$ measures inequality, higher weight is assigned to types that have lower values of $(\mu_1^t)^{r_1}(\mu_2^t)^{r_2}$. Note that a_t is also a parameter chosen by the researcher. In our benchmark calculations, we choose type weights that decrease linearly with type ranks in the values of $(\mu_1^t)^{r_1}(\mu_2^t)^{r_2}$. However, in section 5, we show that our conclusions on time trends do not change when type weights are concave or convex in these ranks.

The functional form $(\mu_1^t)^{r_1}(\mu_2^t)^{r_2}$ is the same as the Cobb-Douglas utility function but the parameterization of weights differs. In particular, both a_t and r_q are negative ensuring that the index is convex and supermodular. Convexity ensures sensitivity to Pigou-Dalton transfers between types, i.e., that $I(X)$ increases after transfers that increase between-type inequality in dimension q . This is a fundamental property for ex-ante measures of inequality of opportunity. Supermodularity ensures sensitivity to correlation-increasing transfers, i.e., that $I(X)$ increases after transfers that increase the cross-type correlation of income and wealth. This is a fundamental property for multidimensional measures of inequality. It is important to note that the functional form is not adopted arbitrarily but that it is derived from first principles: $I(X)$ is the only index fulfilling the fundamental principles of ex-ante equality of opportunity in a multidimensional setting while satisfying standard properties of inequality measures such as *monotonicity*, *utilitarian aggregation*, and *ratio scale invariance*.⁵

Beyond its normative foundations, the index has several useful properties for empirical analyses. First, it can be decomposed to distinguish between the impact of inequality of opportunity in each outcome and the association of outcomes across types. We will use this property in our empirical analyses to understand the drivers of inequality of opportunity in the US. Second, the index is bounded in the interval $[0, 1]$. If $\mu_q^t = \mu_q$ for every type t and outcome q , the index will be zero. Finally, the index is a welfare-based inequality measure in line with the pioneering work by Atkinson (1970). For example, a value of 0.25 (0.5) means that existing inequality of opportunity imposes a welfare cost of 25% (50%) of the population average of each outcome. In other words, if

⁵See also Supplementary Material A for an illustration of its core properties.

there was perfect equality of opportunity, society would achieve the same level of welfare using only 75% (50%) of the available monetary resources in income and wealth (Atkinson, 1970; Kolm, 1969; Sen, 1973).

3 DATA

Data Source. We assess the evolution of equal opportunities in the US while accounting for the multidimensionality of monetary resources. Therefore, we require data with information on income, wealth, and family background for a long time period. In the US, the Panel Study of Income Dynamics (PSID) is the only publicly available data source that satisfies these criteria. For example, while the Survey of Consumer Finances (SCF) offers a long time series on household income and wealth, it contains limited information on the family background of its respondents.

Since 1968 the PSID collects rich information on income and family background for a nationally representative sample of US households. Since 1984 it also collects data on wealth.⁶ Children who leave the parental household become independent sampling units in the PSID. Therefore, it is possible to link data across generations.

Income information is collected for the year predating the survey year. Hence, we use information from the income reference (survey) period 1983-2016 (1984-2017). We now turn to a description of relevant variables.

Monetary Resources. We consider two dimensions of monetary resources: income and wealth. We measure income as annual disposable household income. It comprises total household income from labor, asset flows, windfall gains, private transfers, public transfers, private retirement income and social security pensions net of total household taxes. We measure wealth as household net worth. It comprises the sum of home equity, other real estate, private businesses, vehicles, transaction accounts, corporate equities, annuities/IRAs and other savings net of any debt.

We scale household incomes and wealth by the modified OECD equivalence scale. Hence, we measure both income and wealth at the household level, whereas the units of analysis are individuals. This choice is consistent with our overarching interest

⁶Until 1999 wealth information was collected every five years. Since then, it is a regular part of every PSID wave.

in consumption possibilities since the application of equivalence scales allows for resource sharing among household members.

Wealth data in the PSID is often considered inferior to wealth data in the SCF. Therefore, we compare PSID and SCF concerning time trends in household net worth in Supplementary Figure S.1. Due to oversampling of wealthy households, the SCF assigns a larger share of total net worth to the top 10% of the wealth distribution. Yet, level differences at the top are the only notable difference between PSID and SCF. Importantly, time trends in household net worth are consistent across both data sources.⁷

Family Background Characteristics and Types. We consider two alternative ways to measure family background. First, we use parental income ranks for the total incomes of mothers and fathers averaged over the years 1983-1988. We residualize parental income from the first and second-order polynomials of parental age to account for life-cycle effects in parental earnings profiles. We then partition the population into 36 types by ranking total parental income. Second, we use a vector of alternative socioeconomic background variables. This vector includes parental education (3 categories), parental occupation (3 categories), race (2 categories), and Census region of upbringing (2 categories).⁸ We partition the population into 36 types based on the combination of these family background variables.

Estimation Samples. We base our estimates on two different samples. First, we construct an *intergenerational sample* of 1,366 individuals. To obtain this sample, we drop all individuals with (i) missing or negative income and wealth, and (ii) missing information on parental education, parental occupation, race and region of upbringing. Then we match all respondents to both of their parents and drop observations with (iii) missing parental income. Lastly, we restrict observations to children (parents) aged 25-60 in the period 2010-2016 (1983-1988).⁹ This sample allows us to proxy ω with parental income rank, which is common practice in the literature on intergenerational mobility. However, it imposes restrictions on the analysis of time trends since one requires information on both parental and child outcomes while allowing for sufficient time between these observations.

⁷See also Pfeffer et al. (2016) for a detailed comparison of wealth definitions in PSID and SCF.

⁸Parental education: low (less than high school), intermediate (high school), high (some college and more); parental occupation based on 1-digit 1990 Census codes: low (4,8,9), intermediate (3,5,6,7), high (1,2); race: white (non-Hispanic), other; Census region of upbringing: South, other.

⁹Supplementary Table S.2 details how the different restrictions affect the final sample size.

Second, to investigate time trends, we construct an *individual sample*.¹⁰ In contrast to the previous sample, we drop requirement (iii). Again, we limit the sample to individuals aged between 25-60. We obtain a sample of at least 4,000 observations in every year of the period 1983-2016 which allows us to monitor the development of equality of opportunity in the US over 33 years.

We are conscious that the PSID is subject to selective survey attrition across waves and that our data restrictions may distort our sample through selective item non-response. Therefore, we follow Meyer et al. (2015) and perform all calculations using sampling weights that match the Current Population Survey (CPS). Results remain unchanged when using standard survey weights provided by the PSID instead. Descriptive statistics for all estimation samples are disclosed in Table S.1.

4 RESULTS

Our analysis proceeds in two steps. First, we measure equality of opportunity in the *intergenerational sample*. Thereby, we either use parental income ranks or the vector of alternative socioeconomic background variables to proxy for family background. We will show that both approaches yield very similar results. Second, having validated the use of alternative socioeconomic background variables, we use the *individual sample* to analyze trends in equality of opportunity in the period 1983–2016.

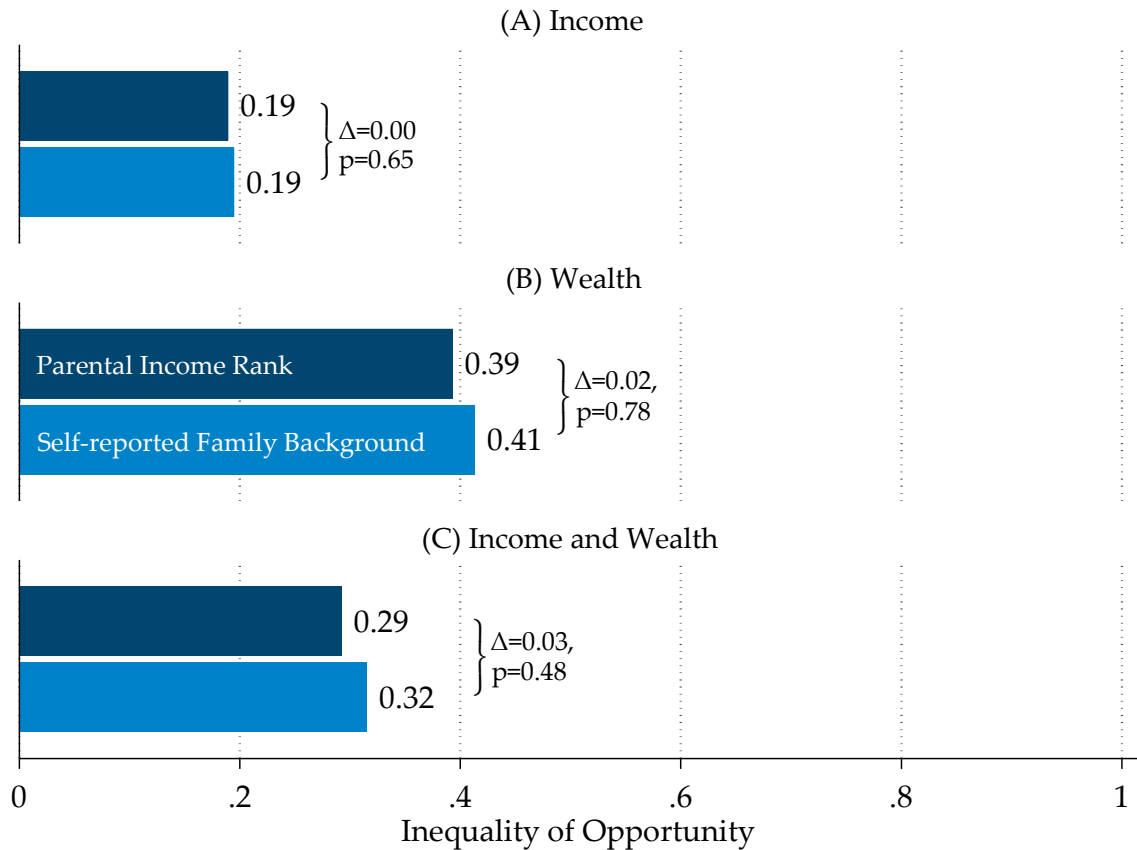
Intergenerational Estimates. Figure 2 shows estimates for inequality of opportunity in the *intergenerational sample* for different combinations of outcomes and family background variables.

First, we focus on the dark-blue bars that show estimates based on parental income ranks. In Panel (A), we measure monetary resources by income only and inequality of opportunity amounts to 0.19. In Panel (B), we measure monetary resources by wealth and inequality of opportunity doubles to a level of 0.39. Finally, in Panel (C) we account for the multidimensionality of monetary resources by considering both income and wealth. Then, inequality of opportunity amounts to 0.29. These results suggest that

¹⁰We note it is possible to analyze time trends in an intergenerational sample when focusing on income and wealth measures in early adulthood (e.g. Hartley et al., 2022). However, such an age restriction would not be adequate in our setting due to lifecycle gradients in both income and wealth.

we tend to underestimate tilt in the playing field when relying on income as the sole proxy for monetary resources.

FIGURE 2. Inequality of Opportunity in the US Intergenerational Sample



Data: PSID.

Note: This figure shows estimates of inequality of opportunity in the US for the *intergenerational sample*. Panel (A) (Panel [B]) shows results for a unidimensional definition of monetary resources based on income (wealth). Panel (C) shows results for a multidimensional definition of monetary resources based on income and wealth. In each panel, inequality of opportunity estimates are based on 36 types according to alternative definitions: parental income rank or self-reported family background. Estimates are computed based on Equation (3) with dimension weights $r_{Income} = r_{Wealth} = -0.2$. Δ indicates the difference in inequality estimates across type definitions. p -values for the null hypothesis that $\Delta = 0$ are bootstrapped using 1,000 draws.

Second, we focus on a comparison between dark-blue bars and light-blue bars. To estimate the latter, we replace parental income ranks with a vector of alternative socioeconomic background characteristics. Point estimates remain virtually unchanged by this alternation and we cannot reject the equality of estimates at conventional levels of significance. This result suggests that parental income ranks and alternative

socioeconomic background characteristics contain similar information about family background. In general, this is an encouraging message as data sets including intergenerational links are much scarcer than data sets including retrospective information on various socioeconomic background variables.¹¹

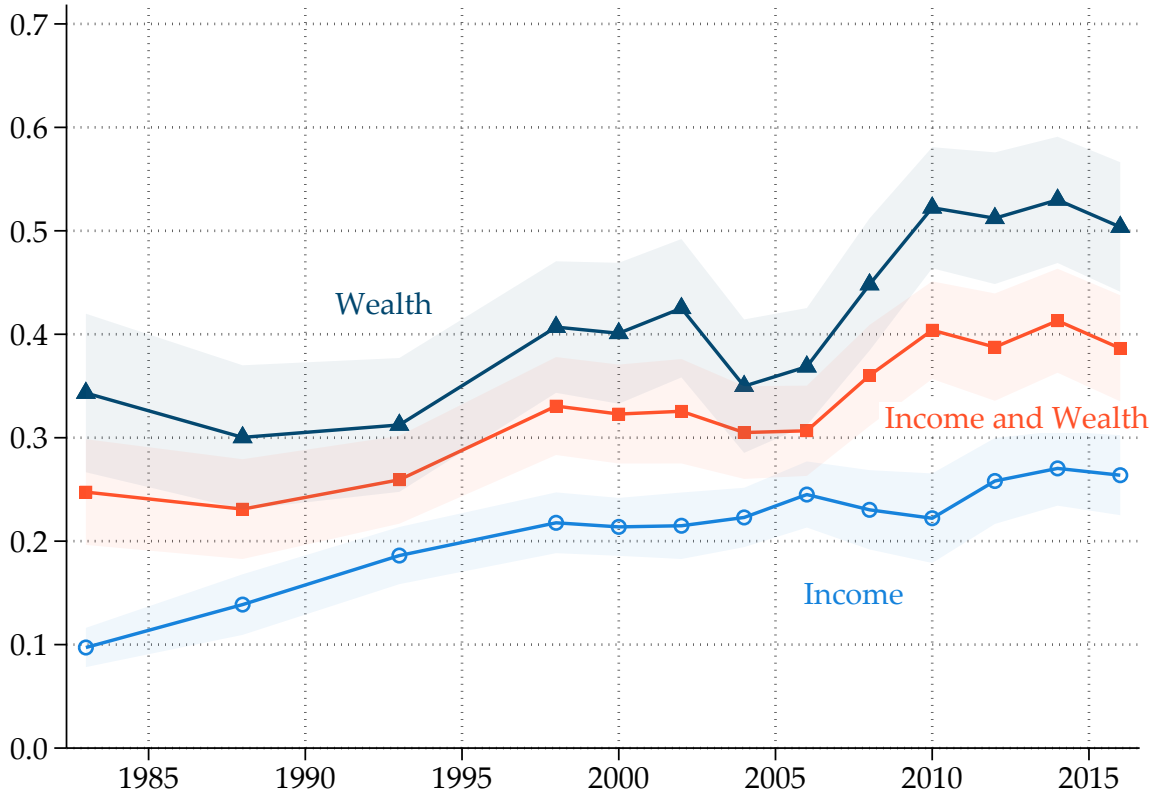
This conclusion is robust to a variety of checks. First, it is well-known that PSID subsamples with intergenerational links are positively selected on their socioeconomic status (Ward, 2021). Therefore, we re-weight the *intergenerational sample* to match the broader population characteristics concerning parental education, parental occupation, race, Census region of upbringing, and age. The re-weighting has little effect on inequality of opportunity estimates (Supplementary Figure S.3). Second, existing literature documents life-cycle bias in intergenerational mobility estimates. Estimates of both income and wealth mobility tend to be downward (upward) biased when children are young (old) (Haider and Solon, 2006; Mazumder, 2018; Nybom and Stuhler, 2016). This bias is usually addressed by measuring income in midlife. There are slight upward corrections of inequality of opportunity in wealth when we restrict our sample to the age range 40-45 (Supplementary Figure S.4). Importantly, however, differences in results based on income ranks and alternative socioeconomic background characteristics remain small for all considered age ranges. Third, we compare estimates based on the alternative background characteristics to expanded sets of family backgrounds where we add parental income and parental wealth ranks. The resulting estimates are very close to our baseline estimates suggesting that the vector of alternative socioeconomic background characteristics captures most of the relevant cross-family variation in socioeconomic status (Supplementary Figure S.5).

We conclude that the vector of alternative socioeconomic background characteristics provides suitable information to capture intergenerational disadvantage. As these data are available on an annual basis, we can use them to assess time trends in inequality of opportunity.

Time Trend (1983-2016). Figure 3 shows the development of inequality of opportunity in the US over the period 1983-2016. The following patterns emerge.

¹¹Jácome et al. (2021) use a similar strategy and approximate parental income with self-reported background characteristics, i.e., retrospective information collected from the respondents of interest and not their parents. In Supplementary Figure S.2, we show that income distributions within family background types are broadly comparable regardless of whether we use types based on parental income ranks or self-reported background characteristics.

FIGURE 3. Inequality of Opportunity in the US, 1983-2016
Baseline Estimates



Data: PSID.

Note: This figure shows estimates of inequality of opportunity in the US for the *individual sample* over the period 1983-2016. Inequality of opportunity estimates are based on 36 types according to the following socioeconomic background characteristics: parental education, parental occupation, race, and region of upbringing. Estimates are computed based on Equation (3) with dimension weights $r_{Income} = r_{Wealth} = -0.2$. 95% confidence intervals (shaded areas) are bootstrapped using 1,000 draws.

First, inequality of opportunity in income increased from 0.10 to 0.26 over time. We can distinguish two distinct periods. On the one hand, we observe marked increases from 1983 to 1998. On the other hand, there are only moderate increases after the year 2000. This two-partite pattern is consistent with findings from the literature on intergenerational income mobility. For example, Davis and Mazumder (2017) show that equality of opportunity decreased for cohorts born in the 1960s and that entered the labor market after 1980. Chetty et al. (2014b) show that this trend flattens for cohorts born in the 1970s that enter labor markets in the 1990s and 2000s. Likewise, Hartley et al. (2022) document a flat time trend in the intergenerational income correlation of mothers and daughters after 2000.

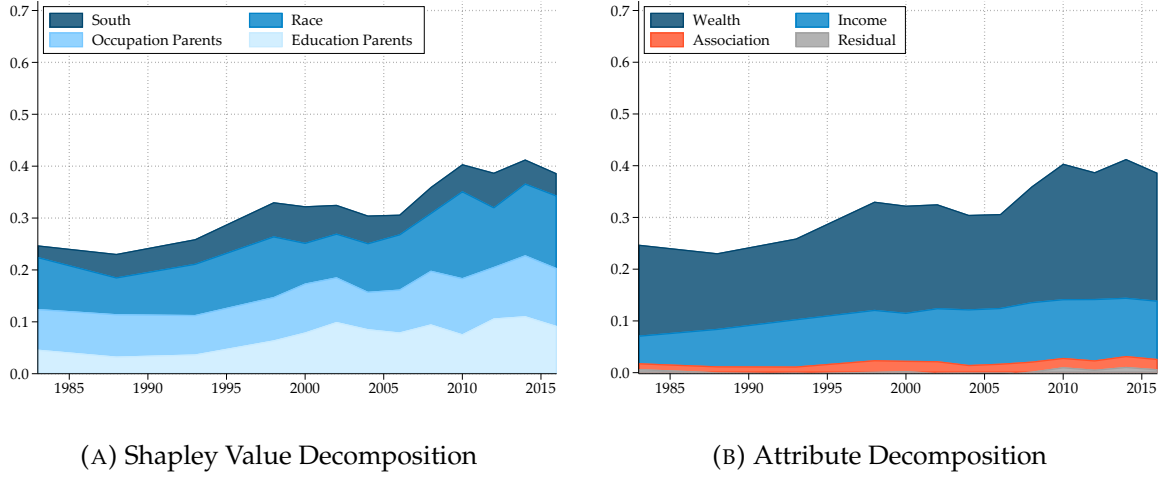
Second, inequality of opportunity in wealth increased from 0.34 to 0.50 over time. Again, we can distinguish two distinct periods. On the one hand, we observe moderate increases from 1983 to 2006. In these years, increases in the stock market were accompanied by a robust housing market (Kuhn et al., 2020; Wolff, 2017). Since owner-occupied housing has a higher weight in the portfolios of individuals from lower socioeconomic backgrounds, increasing house prices attenuated the tendency towards a less opportunity-egalitarian distribution of wealth. On the other hand, differences in portfolio compositions across socioeconomic backgrounds started working in the opposite direction with the financial crisis in 2008. While the stock market experienced a quick recovery, house prices did not catch up to their pre-crisis level. As a consequence, the wealth distribution has become less opportunity-egalitarian with the crisis—a trend that has not reverted ever since.

Taken together, the playing field for the acquisition of monetary resources has become more tilted over time. Starting at a level of 0.25 in 1983, inequality of opportunity in the joint distribution of income and wealth reached a level of 0.39 in the latest period of observation. This shift corresponds to an increase of 56%. Importantly, the trend towards decreasing opportunities to acquire monetary resources continues after the year 2000. This finding can be related to extant literature invoking intergenerational income mobility estimates to conclude relative stability in equality of opportunity in recent years (Chetty et al., 2014b; Hartley et al., 2022). To the extent that these works aim to proxy financial opportunities more generally, they miss important information by focusing on income only. When accounting for the multidimensionality of monetary resources, one cannot reject the claim that opportunities in the US have further declined after the year 2000.

Decomposition. To develop a better understanding of these trends, we conduct a Shapley value decomposition, i.e., we decompose the trend in equality of opportunity into the contributions from different socioeconomic background characteristics: parental education, parental occupation, race, and the region of upbringing (Figure 4, Panel [A]).

First, 57% of the overall increase in inequality of opportunity is accounted for by parental education and occupation. This finding is consistent with Hufe et al. (2022) who identify these components as the strongest drivers of increasing inequality of opportunity in incomes in the US. Second, 29% of the overall increase is accounted for by race. At first glance, this finding appears at odds with the stagnation of racial income gaps since the civil rights era (Bayer and Charles, 2018; Deroncourt and Mon-

**FIGURE 4. Inequality of Opportunity in the US, 1983-2016
Decomposition by Background Characteristic and Outcome Dimension**



Data: PSID.

Note: This figure shows a decomposition of inequality of opportunity in the US for the *individual sample* over the period 1983-2016. Inequality of opportunity estimates are based on 36 types according to self-reported socio-economic background characteristics. Estimates are computed based on Equation (3) with dimension weights $r_{Income} = r_{Wealth} = -0.2$. The decomposition in Panel (A) is based on the Shapley value procedure proposed in Shorrocks (2013). The decomposition in Panel (B) is based on the attribute decomposition derived in Supplementary Material B.

tialoux, 2021). However, the importance of race increases only after the 2008 financial crisis. Therefore, decreased opportunities to acquire monetary resources are most likely driven by the sustained effect of the financial crisis on the housing wealth of Black Americans (Kuhn et al., 2020; Wolff, 2017). Lastly, the contribution of the region of upbringing remains stable over time.

We also conduct an attribute decomposition, i.e., we decompose the time trend into the contributions of (i) inequality of opportunity in income, (ii) inequality of opportunity in wealth, as well as (iii) the cross-type association in both outcomes. The last dimension is of particular interest as it cannot be analyzed in unidimensional measures of inequality of opportunity. In Supplementary Material B, we show that $I(X)$ can be decomposed as follows:

$$\begin{aligned}
I(X) = & \underbrace{\frac{r_1}{r_1+r_2} \left(1 - \left(\sum_{t=1}^M \frac{N_t a_t}{\sum_{t=1}^M N_t a_t} \left(\frac{\mu_1^t}{\mu_1} \right)^{r_1} \right)^{\frac{1}{r_1}} \right)}_{=I_1(\text{Income})} \\
& + \underbrace{\frac{r_2}{r_1+r_2} \left(1 - \left(\sum_{t=1}^M \frac{N_t a_t}{\sum_{t=1}^M N_t a_t} \left(\frac{\mu_2^t}{\mu_2} \right)^{r_2} \right)^{\frac{1}{r_2}} \right)}_{=I_2(\text{Wealth})} \\
& + \underbrace{\frac{1}{r_1+r_2} \left(1 - \frac{\sum_{t=1}^M N_t a_t \sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} (\mu_2^t)^{r_2}}{\sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} \sum_{t=1}^M N_t a_t (\mu_2^t)^{r_2}} \right)}_{=\kappa_I(\text{Association})} \\
& + R,
\end{aligned} \tag{4}$$

where I_q is a unidimensional index of inequality of opportunity in outcome dimension q , κ_I is a measure of cross-type association in outcomes, and R is a residual resulting from linear approximation.¹²

The results of this decomposition are shown in Panel (B) of Figure 4. 43% and 51% of the overall increase in inequality of opportunity can be explained by trends in unidimensional inequality of opportunity in income and wealth, respectively. The cross-type association of income and wealth explains only 6% of the overall increase in unequal opportunities. This finding is somewhat surprising since recent research points to an increased correlation between income and wealth in the US (Berman and Milanovic, 2020; Kuhn and Ríos-Rull, 2016). Our results suggest that these increases at the individual level are mostly driven by increased correlation within family background types while the association of these outcomes across family background types remains rather stable. However, we note that the relative stability of cross-type association κ_I depends on the parameter choices for a_t and r_q . In Supplementary Table S.3, we show the decomposition of time trends under different plausible assumptions for r_q . For example, if we allow for a higher degree of inequality aversion by choosing $r_{\text{Income}} = -0.4$ and $r_{\text{Wealth}} = -0.4$, κ_I explains up to 17% of the overall increase in unequal opportunities. This finding indicates that family background types in the lower tail of the distribution have become more resource-constrained than the rest of the population in both income and wealth simultaneously.

¹² I_q are unidimensional inequality of opportunity measures based on the Atkinson (1970) index of inequality—see our discussion in section 2.

5 SENSITIVITY ANALYSIS

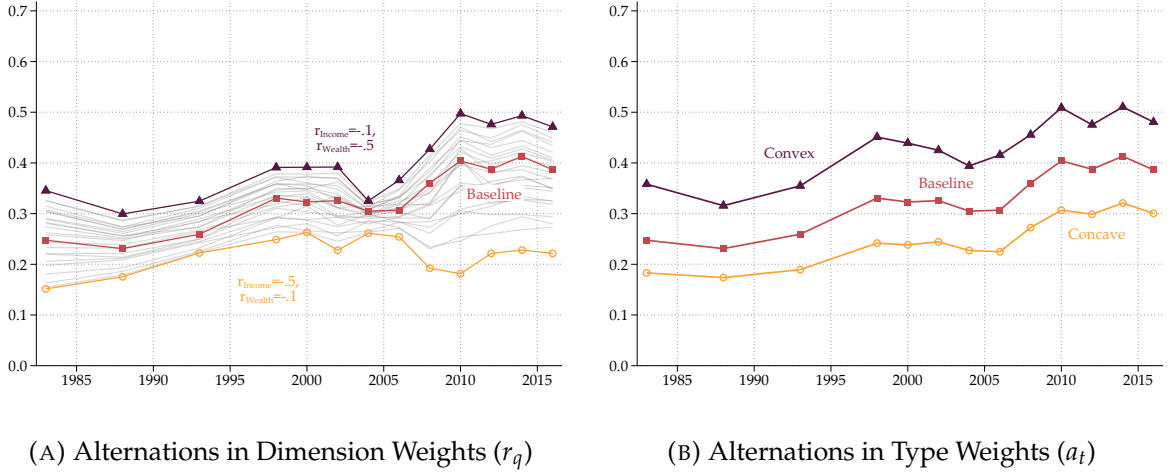
Parameter Choices. We assess the sensitivity of our main conclusions to changes in the measurement parameters, i.e., dimension weights r_q and type weights a_t . Alternative parameter choices correspond to different normative assumptions about inequality aversion. Therefore, they will lead to level shifts in the extent of inequality of opportunity—a property that is well-known in the literature (Atkinson, 1970). However, we are especially concerned with the development of inequality of opportunity over time. In the following discussion, we will therefore abstract from levels and focus on whether changes in unequal opportunities are sensitive to different assumptions about these parameters.

First, dimension weights r_q determine inequality aversion in income and wealth, respectively. In our baseline estimates, we give both dimensions equal weight and choose $r_{Income} = r_{Wealth} = -0.2$. However, there may be good reasons to give different weights to different dimensions of monetary resources. For example, one could argue that wealth should receive a higher weight due to its insurance value. Reversely, one could argue that wealth should receive a lower weight since it is less liquid and might not be available for instantaneous consumption. Panel (A) of Figure 5 shows alternative results for all pairwise combinations over the parameter grid $r_q \in (-0.1, -0.2, -0.3, -0.4, -0.5)$. Lowest estimates of inequality of opportunity are obtained for $r_{Income} = -0.5$ and $r_{Wealth} = -0.1$; that is, in the case where we place little weight on the wealth dimension, and more weight on the income dimension. We note that such income-focused parameterization yields a flat trend after the year 2000. This result is expected and consistent with existing work on intergenerational income mobility (Chetty et al., 2014b; Hartley et al., 2022). However, even small increases in the wealth focus lead to upward corrections in inequality of opportunity estimates and overturn the conclusion of flat time trends after 2000. The highest estimates of inequality of opportunity are obtained for $r_{Income} = -0.1$ and $r_{Wealth} = -0.5$; that is, in the case where we place more weight on the wealth dimension, and little weight on the income dimension.

Second, type weights a_t determine the degree of inequality aversion between types. In our baseline estimates, we choose linear a_t that are inversely related to type ranks in monetary resources. Panel (B) of Figure 5 shows alternative results for convex (a_t^2) and concave type weights ($a_t^{0.5}$). The lowest estimates of inequality of opportunity are obtained for concave type weights, where we place relatively less weight on inequality in the lower tail of the type distribution. Conversely, the highest estimates are ob-

tained for convex type weights, where we place relatively more weight on inequality in the upper tail of the type distribution. Despite changes in levels, our conclusions concerning time trends are insensitive to parameter choices in a_t .

**FIGURE 5. Inequality of Opportunity in the US, 1983-2016
Sensitivity to Parameter Choices**



Data: PSID.

Note: This figure shows the sensitivity of inequality of opportunity in the US for the *individual sample* over the period 1983-2016 under different parameter choices. Panel (A) shows the sensitivity to alternations in r_q . We display all pairwise combinations of $r_{Income} \in (-0.1, -0.2, -0.3, -0.4, -0.5)$ and $r_{Wealth} \in (-0.1, -0.2, -0.3, -0.4, -0.5)$. The central line replicates our baseline estimates from Figure 3 where we use linear $r_{Income} = r_{Wealth} = -0.2$. Panel (B) shows the sensitivity to alternations in a_t . We construct convex (concave) weights as a_t^2 ($a_t^{0.5}$). The central line replicates our baseline estimates from Figure 3 where we use linear a_t .

Data Choices. In Supplementary Figure S.6, we furthermore document that our main conclusions are robust to different data choices.

First, we recompute inequality of opportunity while smoothing transitory changes in income and wealth, i.e., we replace annual values of income and wealth with 5-year averages. As a consequence, outcome variables provide better proxies for the long-term income and wealth potential of individuals (Solon, 1992). Time trends are very close to our baseline estimates.

Second, we recompute inequality of opportunity for different type partitions. To this end, we code three additional variables and add them to the vector of socioeconomic background characteristics: the number of siblings (11 categories), a dummy for foreign-born parents, and a dummy for single-parent families. In turn, we follow

Brunori et al. ([forthcoming](#)) and let a regression tree algorithm decide on the optimal type partition in each year of our analysis. Again, time trends are very similar to our baseline estimates.

Third, we recompute inequality of opportunity for different ways of dealing with non-positive income and wealth. For our baseline, we drop observations with negative income/wealth and set observations with zero income/wealth to 1 USD, respectively. Alternatively, we (i) drop all observations with negative and zero income/wealth, or (ii) retain all observations with negative and zero income/wealth in the sample. Time trends are again very similar, regardless of the chosen specification.

Fourth, we recompute inequality of opportunity using alternative definitions of income and wealth. Our baseline definitions may contain mechanical relationships between income and wealth. Wealth enters household income through capital returns; reversely, savings from household income increase wealth in a given period. Therefore, we divorce both concepts as follows: first, we replace household disposable income with household labor market earnings, i.e., we use an income concept that is not mechanically related to asset returns. Second, we adjust household net worth by deducting active savings in a given year, i.e., we use a wealth concept that is not mechanically related to contemporaneous saving decisions. Our time series are not sensitive to these adjustments, suggesting that mechanical relationships between income and wealth are not the main driver of our results.

We conclude: while the level of inequality of opportunity and the magnitude of its increase varies with different measurement choices, all main conclusions from our baseline estimates remain in place. The only exception arises if we parameterize our index in ways that give little weight to the wealth dimension. In this case, we replicate analyses that focus on the income dimension only and we obtain a flat time trend after the year 2000.

6 CONCLUSION

In this paper, we study inequality of opportunity for the acquisition of monetary resources in the US over the period 1983-2016. In contrast to existing work, we account for the multidimensionality of monetary resources by targeting the joint distribution of income and wealth. Our results show that unidimensional analyses may miss important information when analyzing the playing field in the US: first, we document

a more unequal distribution of opportunities when complementing income with the wealth dimension. Second, there are strong and consistent increases in inequality of opportunity over time. This trend is driven by a less opportunity-egalitarian distribution of income until 2000, and a less opportunity-egalitarian distribution of wealth after the financial crisis in 2008.

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Multidimensional Equality of Opportunity in the United States

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Supplementary Material

April 4, 2023

A MEASUREMENT—SOME SIMPLE EXAMPLES

Consider a society with two types that are of equal size. Both income and wealth are unequally distributed. Since (i) inequality in both dimensions is exactly the same, and (ii) both dimensions are perfectly correlated across types, inequality of opportunity in monetary resources is exactly the same (0.17) regardless of whether we focus on income (I_{Income}) or wealth (I_{Wealth}) in isolation, or whether we focus on the joint distribution of income and wealth ($I_{Income,Wealth}$).

We now consider three alternative societies in which unidimensional and multidimensional measures of inequality of opportunity diverge. As in the main part of the paper, estimates are computed based on Equation (3) with dimension weights $r_{Income} = r_{Wealth} = -0.2$ and linear a_t that are inversely related to type ranks in monetary resources.

			Income		Wealth
			Type 1	50	500
			Type 2	100	1000
			$I_{Income} = 0.17$		
			$I_{Wealth} = 0.17$		
			$I_{Income,Wealth} = 0.17$		
			(a)	(b)	(c)
Income			Income		Wealth
Wealth			Wealth		
Type 1	75	500	Type 1	50	1000
Type 2	75	1000	Type 2	100	500
$I_{Income} = 0.00$			$I_{Income} = 0.17$		$I_{Income} = 0.27$
$I_{Wealth} = 0.17$			$I_{Wealth} = 0.17$		$I_{Wealth} = 0.27$
$I_{Income,Wealth} = 0.09$			$I_{Income,Wealth} = 0.06$		$I_{Income,Wealth} = 0.12$

- (a) We equalize outcomes across types in the income dimension. Therefore, I_{Income} decreases, and I_{Wealth} stays the same. The multidimensional measure $I_{Income,Wealth}$ decreases. This case illustrate the measure's *inequality aversion between types*.

- (b) We maintain inequality across types but reverse the cross-type association of income and wealth. Therefore, I_{Income} stays the same, and I_{Wealth} stays the same. The multidimensional measure $I_{Income,Wealth}$ decreases. This case illustrates the measure's *sensitivity to correlation-increasing transfers*.
- (c) We increase inequality across types in both dimensions and reverse the cross-type association of income and wealth. Therefore, I_{Income} increases, and I_{Wealth} increases. The multidimensional measure $I_{Income,Wealth}$ decreases. This case illustrates the existence of cases where unidimensional and multidimensional measures lead to opposing conclusions. While the former would detect an increase of inequality of opportunity in comparison to the baseline, the latter would detect a decrease in unequal opportunities.

B ATTRIBUTE DECOMPOSITION

In this appendix, we derive and prove the attribute decomposability of $I(X)$ as defined in Equation (3). Our derivation is based on results presented in Abul Naga and Geoffard (2006). For the exposition, we focus on the case of two outcome dimensions with $K = 2$.¹ In this case, X consists of two submatrices X_1 and X_2 that denote outcome matrices for dimensions 1 and 2, respectively. Recall that μ_q^t denotes a type mean in outcome dimension q . Given the notation with two submatrices, an element x_{i1} (x_{i2}) of matrix X_1 (X_2) equals μ_1^t (μ_2^t), i.e., the mean value of dimension 1 (dimension 2) in type t to which individual i belongs. Finally, recall that μ_q denotes the population mean of dimension q .

Attribute Decomposability. In general, $I(X) = 1 - \delta(X)$, where $\delta(X) \in [0, 1]$. $I(X)$ is attribute decomposable if and only if

$$\delta(X) = f_1(\gamma_1(X_1)) + f_2(\gamma_2(X_2)) + f_3(\kappa(X)), \quad (5)$$

where f_1, f_2, f_3 are increasing functions ($\mathbb{R}_+ \mapsto \mathbb{R}_+$), γ_1 and γ_2 are unidimensional equality indices, and κ is a measure of association between X_1 and X_2 .

Proposition 1. $\delta(X)$ is attribute decomposable as follows:

$$\ln \delta(X) = \frac{r_1}{r_1 + r_2} \ln \gamma_1(X_1) + \frac{r_2}{r_1 + r_2} \ln \gamma_2(X_2) + \frac{1}{r_1 + r_2} \ln \kappa(X), \quad (6)$$

¹We note this restriction can be easily relaxed.

where

$$\begin{aligned}\gamma_1(X_1) &= \left(\sum_{t=1}^M \frac{N_t a_t}{\sum_{t=1}^M N_t a_t} \left(\frac{\mu_1^t}{\mu_1} \right)^{r_1} \right)^{\frac{1}{r_1}}, \\ \gamma_2(X_2) &= \left(\sum_{t=1}^M \frac{N_t a_t}{\sum_{t=1}^M N_t a_t} \left(\frac{\mu_2^t}{\mu_2} \right)^{r_2} \right)^{\frac{1}{r_2}}, \\ \kappa(X) &= \frac{\sum_{t=1}^M N_t a_t \sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} (\mu_2^t)^{r_2}}{\sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} \sum_{t=1}^M N_t a_t (\mu_2^t)^{r_2}}.\end{aligned}$$

Proof. First, $\delta(X)$ is the proportion of μ_q that is necessary to achieve the same level of welfare if all attributes were distributed equally across types, see Kobus et al. (2020). Formally, let $w_0 = \sum_{t=1}^M N_t U^t(\delta\mu_1, \delta\mu_2)$ denote the welfare level associated with X . Second, let ρ_1 be the proportion of μ_1 that is necessary to attain w_0 , if (i) the first attribute was equally distributed across types, and (ii) the distribution of the second attribute across types remained as is. Formally, $w_0 = \sum_{t=1}^M N_t U^t(\rho_1\mu_1, \mu_2^t)$. Third, let γ_1 be the proportion of μ_1 that is necessary to attain w_0 , if (i) the first attribute was equally distributed across types, and (ii) the second attribute was equally distributed across types. Formally, $w_0 = \sum_{t=1}^M N_t U^t(\gamma_1\mu_1, \rho_2\mu_2)$.

It follows that

$$w_0 = \sum_{t=1}^M N_t a_t (\delta\mu_1)^{r_1} (\delta\mu_2)^{r_2} = \sum_{t=1}^M N_t a_t (\gamma_1\mu_1)^{r_1} (\rho_2\mu_2)^{r_2}.$$

After modification, we get $\delta^{r_1+r_2} = (\gamma_1)^{r_1} (\rho_2)^{r_2}$, and we obtain

$$\ln(\delta) = \frac{r_1}{r_1 + r_2} \ln(\gamma_1) + \frac{r_2}{r_1 + r_2} \ln(\gamma_2) + \frac{1}{r_1 + r_2} \ln(\rho_2/\gamma_2)^{r_2}, \quad (7)$$

which is the desired decomposition with $\kappa := (\rho_2/\gamma_2)^{r_2}$.

We now need to derive functional forms of γ_1 , γ_2 and κ .

Note that $w_0 = \sum_{t=1}^M N_t a_t (\gamma_1\mu_1)^{r_1} (\rho_2\mu_2)^{r_2} = \sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} (\rho_2\mu_2)^{r_2}$. Solving for γ_1 yields:

$$\gamma_1 = \left(\sum_{t=1}^M \frac{N_t a_t}{\sum_{t=1}^M N_t a_t} \left(\frac{\mu_1^t}{\mu_1} \right)^{r_1} \right)^{\frac{1}{r_1}}.$$

Proceeding in analogy, for γ_2 we get:

$$\gamma_2 = \left(\sum_{t=1}^M \frac{N_t a_t}{\sum_{t=1}^M N_t a_t} \left(\frac{\mu_2^t}{\mu_2} \right)^{r_2} \right)^{\frac{1}{r_2}}.$$

Furthermore, we use $w_0 = \sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} (\rho_2 \mu_2)^{r_2} = \sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} (\mu_2^t)^{r_2}$ to obtain

$$\rho_2 = \left(\frac{\sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} (\mu_2^t)^{r_2}}{\sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} (\mu_2)^{r_2}} \right)^{\frac{1}{r_2}}.$$

Finally, substituting the expressions for γ_2 and ρ_2 into $\kappa := (\rho_2 / \gamma_2)^{r_2}$ we get:

$$\kappa = \frac{\sum_{t=1}^M N_t a_t \sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} (\mu_2^t)^{r_2}}{\sum_{t=1}^M N_t a_t (\mu_1^t)^{r_1} \sum_{t=1}^M N_t a_t (\mu_2^t)^{r_2}}.$$

□

Linear Approximation. Collecting terms and reversing the log-linearization of $\delta(X)$, we obtain the attribute decomposition of $I(X)$ displayed in Equation (5):

$$I(X) = 1 - (\gamma_1)^{\frac{r_1}{r_1+r_2}} (\gamma_2)^{\frac{r_2}{r_1+r_2}} (\kappa)^{\frac{1}{r_1+r_2}}. \quad (8)$$

Applying a linear approximation around the point of perfect equality (i.e., $\gamma_1 = \gamma_2 = \kappa = 1$), we get the linear decomposition displayed in Equation (4):

$$\begin{aligned} I(X) &= \frac{r_1}{r_1+r_2} (1 - \gamma_1) + \frac{r_2}{r_1+r_2} (1 - \gamma_2) + \frac{1}{r_1+r_2} (1 - \kappa) + R, \\ &= \frac{r_1}{r_1+r_2} I_1 + \frac{r_2}{r_1+r_2} I_2 + \frac{1}{r_1+r_2} \kappa_I + R. \end{aligned} \quad (9)$$

C ADDITIONAL FIGURES AND TABLES

TABLE S.1. Descriptive Statistics

	Income	Wealth	Family Background				Age	N
			Educ.	Occ.	Race	Region		
<i>Panel (A): Intergenerational Sample</i>								
	55,745	279,508	2.26	2.32	0.87	0.26	50	1,366
<i>Panel (B): Re-weighted Intergenerational Sample</i>								
	49,150	205,851	2.20	2.28	0.75	0.34	46	1,366
<i>Panel (C): Individual Sample</i>								
1983	34,763	160,272	1.75	1.87	0.84	0.32	41	5,368
1988	42,258	171,767	1.87	1.94	0.82	0.31	40	5,357
1993	40,562	166,160	1.95	2.00	0.81	0.31	41	5,070
1998	44,563	183,894	2.04	2.09	0.79	0.37	42	4,213
2000	46,136	200,777	2.06	2.12	0.78	0.37	43	4,106
2002	46,414	199,987	2.05	2.13	0.77	0.38	43	4,238
2004	48,897	228,384	2.05	2.14	0.77	0.36	43	5,197
2006	49,406	254,948	2.07	2.15	0.76	0.36	43	5,250
2008	48,349	214,078	2.08	2.16	0.76	0.36	44	5,079
2010	45,490	189,976	2.09	2.18	0.73	0.35	44	5,039
2012	46,377	167,962	2.10	2.19	0.72	0.36	44	5,047
2014	46,373	178,185	2.12	2.20	0.71	0.35	44	5,013
2016	46,837	188,240	2.12	2.21	0.70	0.35	44	4,957

Data: PSID.

Note: This table displays summary statistics for the *intergenerational sample* (Panel [A]), the *re-weighted intergenerational sample* (Panel [B]) and the *individual sample* (Panel [C]). Income is defined as annual disposable household income, wealth as household net worth. Both income and wealth are scaled by the modified OECD equivalence scale and expressed in constant 2015 USD. We furthermore drop observations with negative income/wealth and set zero amounts to 1 USD. The family background variables Educ. (Occ.) show the average education (occupation) level of the parent with the highest education (occupation) status measured on a 3-point scale. Race displays the share of whites; region the share of respondents who grew up in the US Census region South. Age refers to the average age in the sample. The last column shows the number of observations.

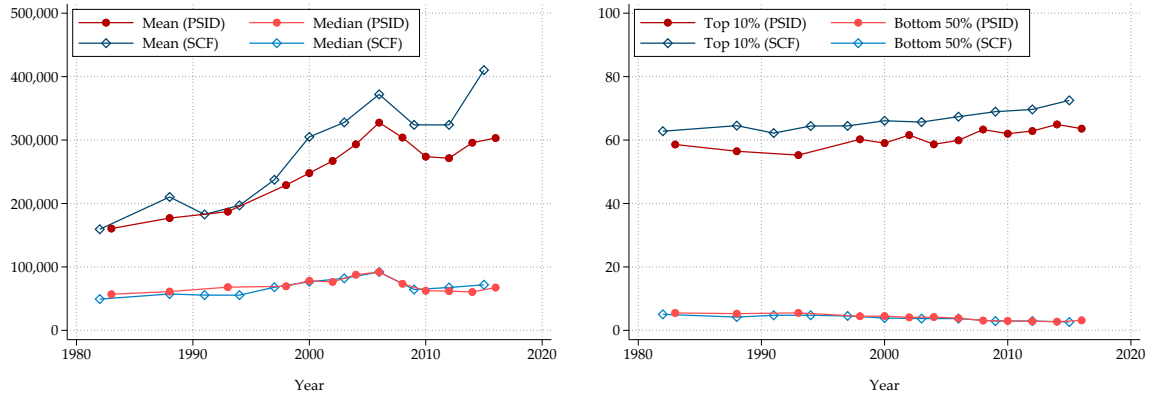
TABLE S.2. Observation Loss due to Sample Restrictions

Time Period	Sample Restriction	Observations	Share of Original Sample
<i>Panel (A): Intergenerational Sample</i>			
2010-2016		8,824	100%
2010-2016	(i)	8,061	91.4%
2010-2016	(ii)	7,187	89.3%
2010-2016	(iii)	1,366	15.5%
<i>Panel (B): Individual Sample</i>			
1983-2016		80,918	100%
1983-2016	(i)	69,267	81.4%
1983-2016	(ii)	63,934	79.0%
1983		6,257	100%
1983	(i)	5,732	91.6%
1983	(ii)	5,368	85.8%
1993		6,012	100%
1993	(i)	5,384	89.6%
1993	(ii)	5,070	84.3%
2004		6,392	100%
2004	(i)	5,610	87.8%
2004	(ii)	5,197	81.3%
2014		6,896	100%
2014	(i)	5,569	80.8%
2014	(ii)	5,013	72.7%

Data: PSID.

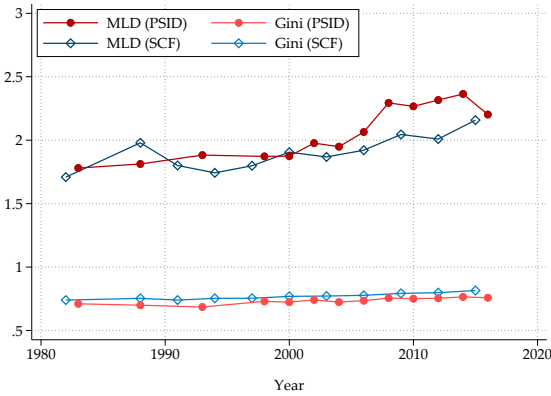
Note: This table shows the loss in observations when defining our analysis samples. For the *intergenerational sample*, we require (i) non-missing and non-negative information on income and wealth, and (ii) non-missing information on parental education, parental occupation, race, and region of upbringing. We further require that (iii) information on the income of both parents is non-missing. For the *individual sample*, we require (i) and (ii) only.

FIGURE S.1. Wealth in PSID and SCF, 1983-2016



(A) Mean and Median (in USD)

(B) Income Shares at Bottom and Top (in %)

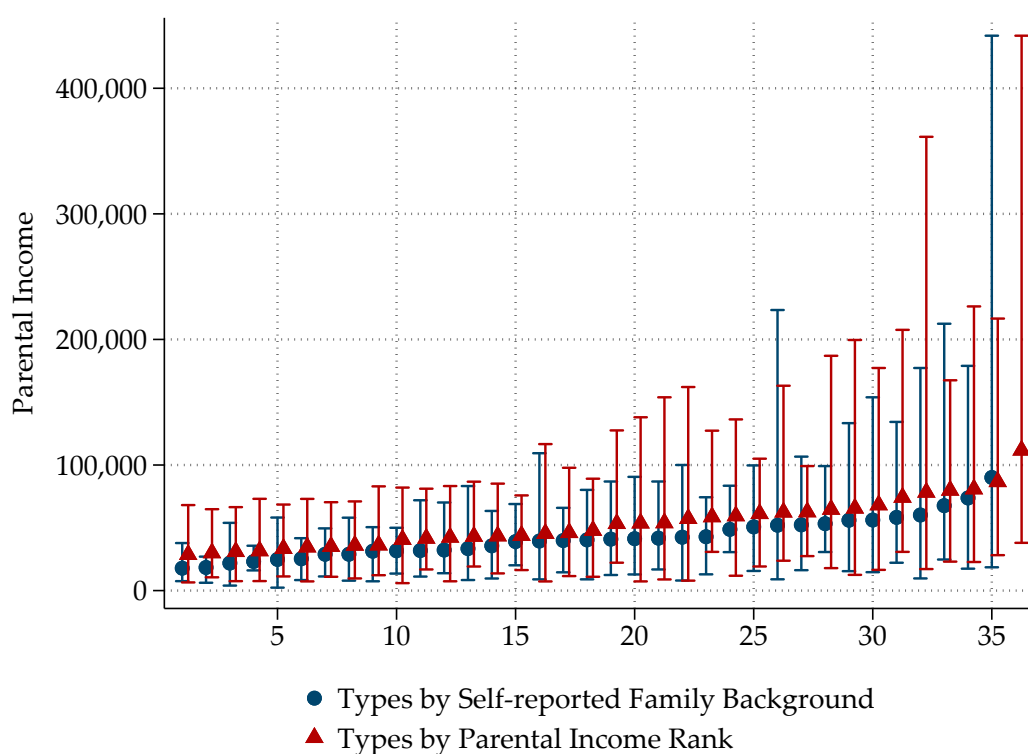


(C) Inequality

Data: PSID, SCF+ (Kuhn et al., 2020).

Note: This figure compares wealth distributions between the PSID and the Survey of Consumer Finances (SCF). In both data sources, wealth is defined as equivalized household net worth (see Section 3); we drop negative values, replace zero values with 1 USD, and winsorize from above at the 99.9 percentile. Samples are restricted to household heads. All figures are expressed in constant 2015 USD.

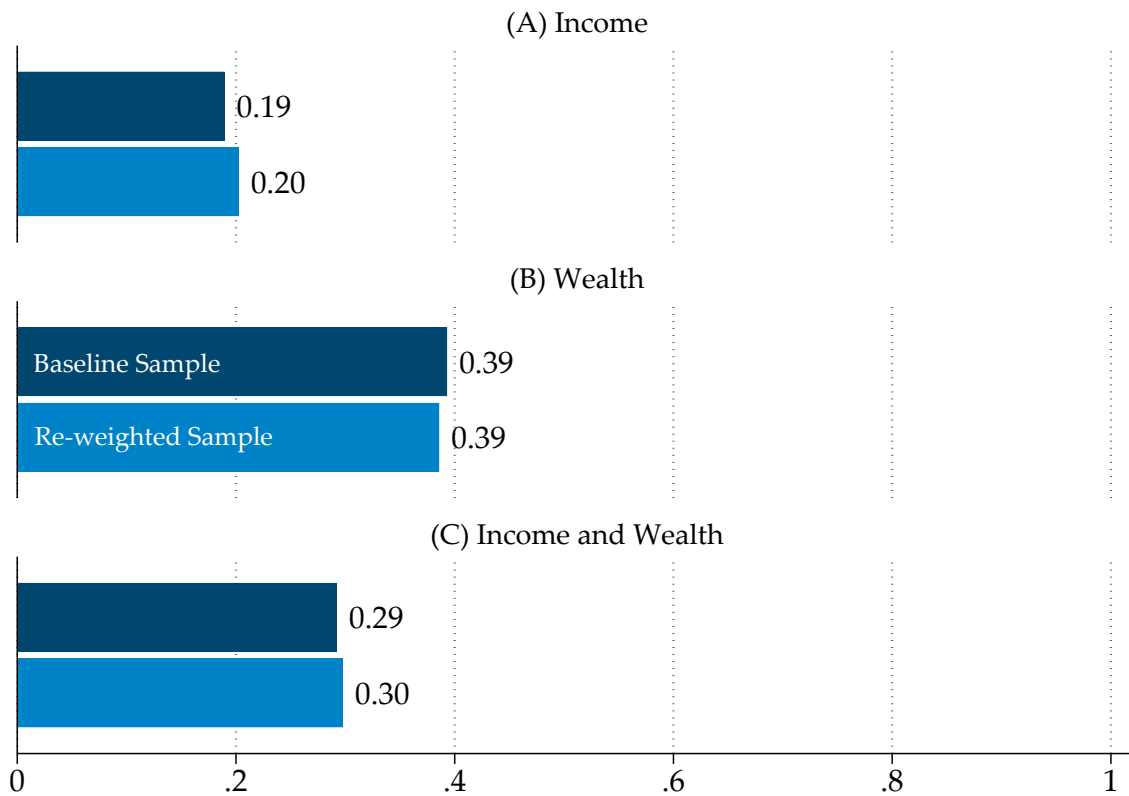
FIGURE S.2. Parental Income by Family Background Type



Data: PSID.

Note: This figure shows the mean, 5th-percentile, and 95th-percentile of income distributions within different types of family background characteristics in the *intergenerational sample*. Blue circles and whiskers refer to self-reported family background characteristics. Red triangles and whiskers refer to parental income ranks. Note that bins of family background characteristics tend to be of unequal size, whereas bins of parental income ranks tend to be of equal size (in absence of ties). This feature and resulting differences in the weighting of types explain that averages based on parental income ranks tend to be slightly higher than corresponding averages based on self-reported family background characteristics although the underlying income distributions are the same.

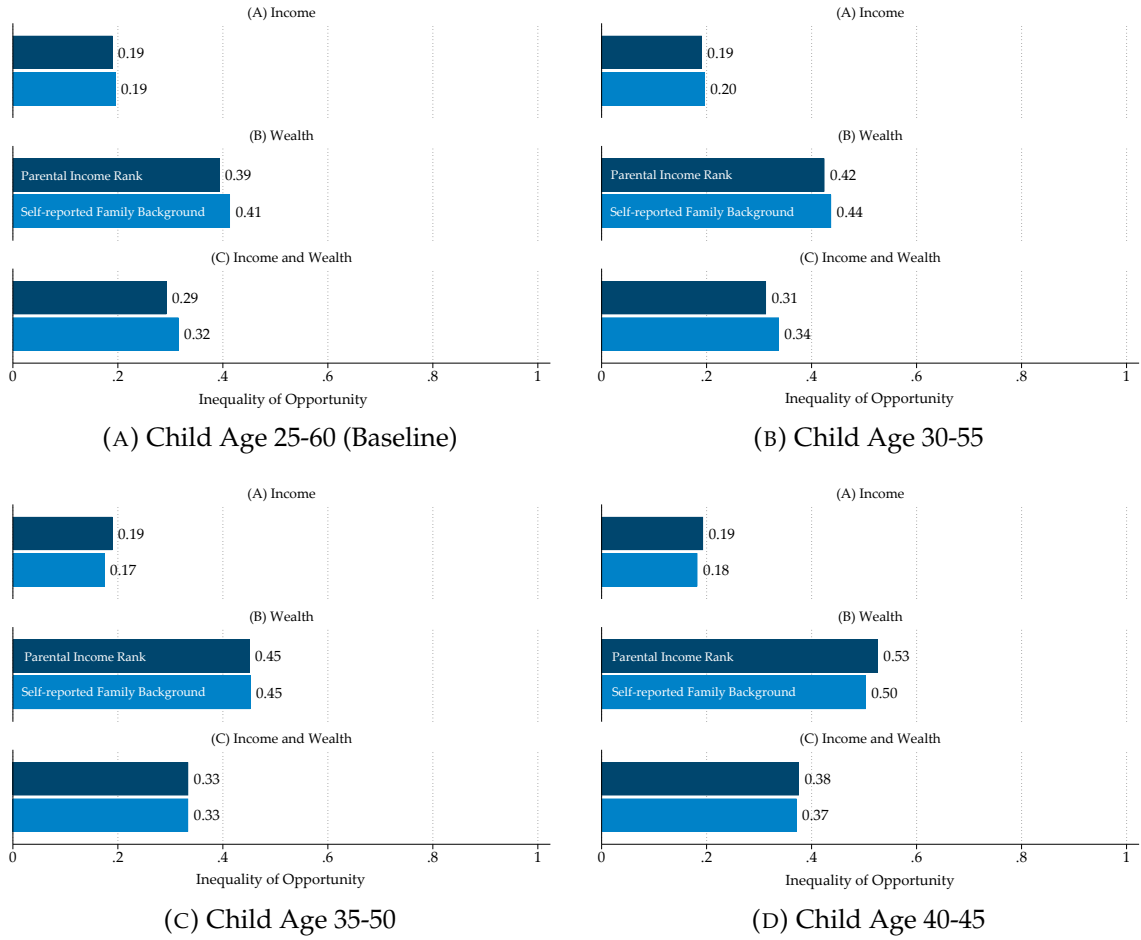
**FIGURE S.3. Inequality of Opportunity in the US
Re-weighted Sample**



Data: PSID.

Note: This figure shows the sensitivity of inequality of opportunity in the US when accounting for selective sample attrition in the *intergenerational sample*. In particular, we re-weight the *intergenerational sample* to match the *individual sample* in the observation period 2010-2016 concerning age, parental education, parental occupation, race, and region of upbringing. All estimates are computed based on Equation (3) with dimension weights $r_{Income} = r_{Wealth} = -0.2$ and use parental income rank as a proxy for family background.

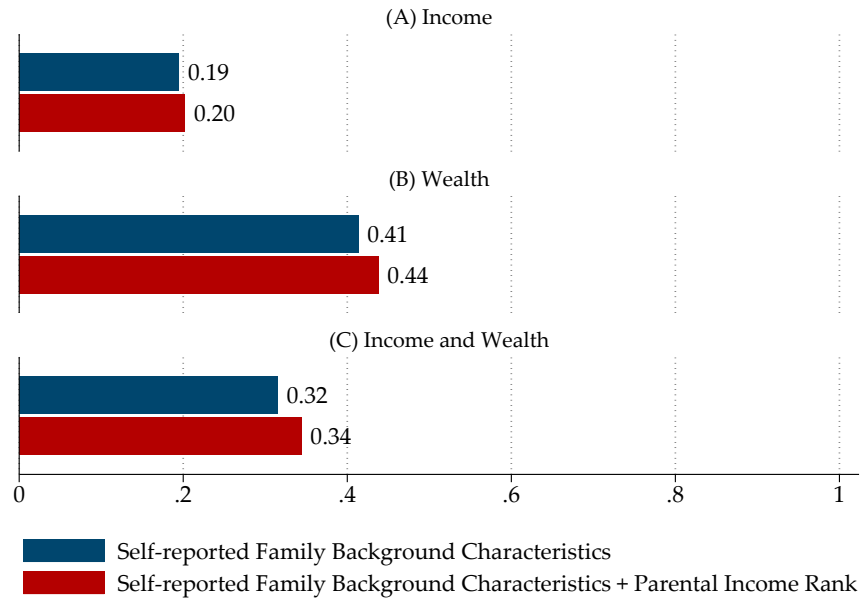
**FIGURE S.4. Equality of Opportunity in the US
Varying Age Restrictions**



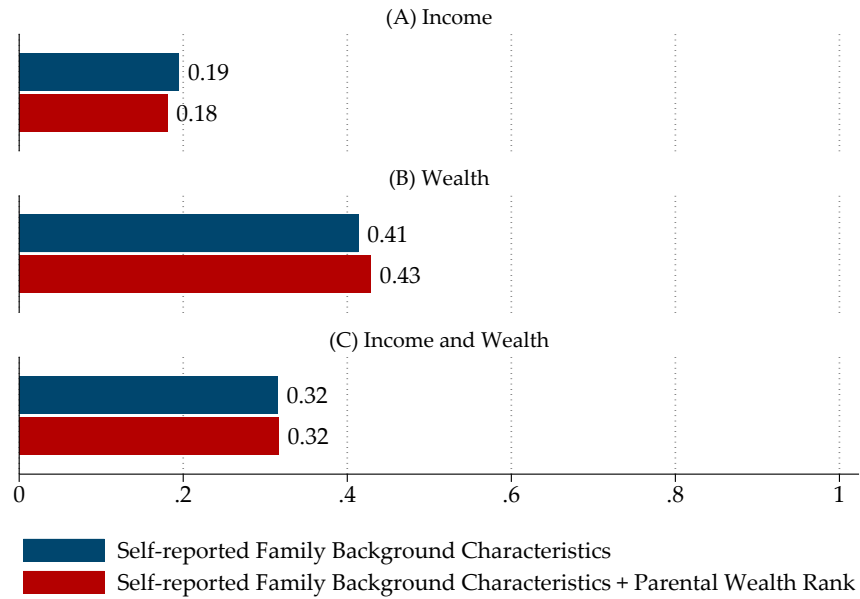
Data: PSID.

Note: This figure shows the sensitivity of inequality of opportunity in the US to different sample restrictions regarding the age of children. Panel (A) replicates our baseline estimates from Figure 2. In Panels (B)-(D) we sequentially narrow the age restriction to 30-55, 35-50, and 40-45. All estimates are computed based on Equation (3) with dimension weights $r_{Income} = r_{Wealth} = -0.2$.

**FIGURE S.5. Equality of Opportunity in the US
Extended Family Background Characteristics**



(A) Adding Parental Income Ranks

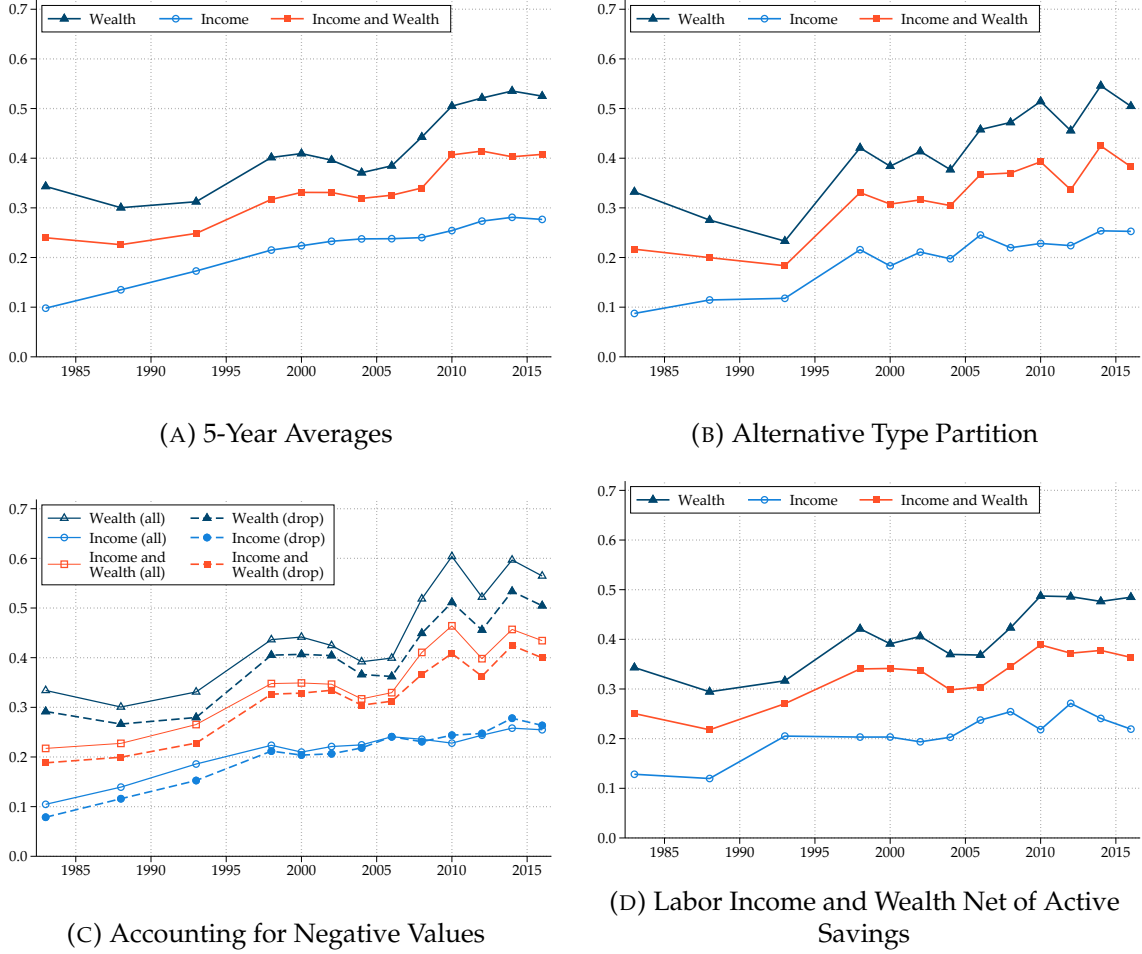


(B) Adding Parental Wealth Ranks

Data: PSID.

Note: This figure shows the sensitivity of inequality of opportunity in the US to extended sets of eligible family background characteristics. The blue bars replicate our baseline estimates from Figure 2 based on self-reported family background characteristics (parental education, parental occupation, race, region of upbringing). The red bars show estimates when adding 100 parental income ranks (Panel A) or 100 parental wealth ranks (Panel B) to self-reported family background characteristics and selecting types via a regression tree algorithm. All estimates are computed based on Equation (3) with dimension weights $r_{Income} = r_{Wealth} = -0.2$.

FIGURE S.6. Inequality of Opportunity in the US, 1983-2016
Sensitivity to Data Choices



Data: PSID.

Note: This figure shows the sensitivity of inequality of opportunity in the US for the *individual sample* over the period 1983-2016. In Panel (A), we take a 5-year moving average of income and wealth. In Panel (B), we let a regression tree determine the underlying type partition. In Panel (C), we keep zero income and wealth without adjustment (solid line) or drop individuals with zero income or wealth (dashed line). Panel (D) displays our estimates for the sub-components of labor income and wealth net of active savings in the period of interest. Estimates are computed based on Equation (3) with dimension weights $r_{Income} = r_{Wealth} = -0.2$.

**TABLE S.3. Attribute Decomposition
Alternative Parameterization**

r_{Income}	r_{Wealth}	Contribution of		
		Income	Wealth	Association
-0.1	-0.1	42%	54%	2%
-0.2	-0.2	43%	51%	6%
-0.3	-0.3	43%	49%	12%
-0.4	-0.4	44%	46%	17%

Data: PSID.

Note: This table displays the relative contribution of I_{Income} , I_{Wealth} , and κ_I to the increase in multidimensional inequality of opportunity over the time period 1983-2016. The decomposition is based on the attribute decomposition derived in Supplementary Material [B](#). Due to the linear approximation error, the percentages do not exactly add up to 100.

References

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- KUHN, M., M. SCHULARICK, and U. I. STEINS (2020). "Income and Wealth Inequality in America, 1949–2016". *Journal of Political Economy* 128 (9), pp. 3469–3519.